

Beyond One-Size-Fits-All: Personalized Delivery and Fulfillment Optimization

Quan Zhou

Desautels Faculty of Management, McGill University, Montreal, Quebec H3A 1G5, Canada, quan.zhou3@mail.mcgill.ca

Mehmet Gümüş

Desautels Faculty of Management, McGill University, Montreal, Quebec H3A 1G5, Canada, mehmet.gumus@mcgill.ca

Sentao Miao

Leeds School of Business, University of Colorado Boulder, Boulder, Colorado 80309, USA, sentao.miao@colorado.edu

Motivated by our collaboration with an online platform operating in North America, we explore the joint optimization of the order fulfillment process with personalized delivery options in the context of e-commerce. Customers can choose from personalized fulfillment options to proceed with the purchase or leave with no purchase. The retailer periodically makes fulfillment decisions and relies on multiple logistic providers to perform the fulfillment operations. We model customer behavior with a general discrete choice model and formulate the joint optimization as a stochastic dynamic program. We propose a tractable deterministic approximation and develop a computationally efficient heuristic with a provable performance guarantee. Using real datasets collected from our industrial partner, we demonstrate the value of personalizing fulfillment options for the customers and jointly optimizing the options with fulfillment assignments. Our results show that demand management via personalized fulfillment options is prominent when customers favor quicker fulfillment and when the fulfillment capacity is limited. However, an optimized fulfillment operation becomes more critical when customers are more willing to wait.

Key words: fulfillment optimization, personalized delivery, e-commerce

History:

1. Introduction

Customer shopping behaviors have been significantly reshaped since the COVID-19 pandemic. There is a persistent trend of customers favoring online shopping, leading to e-commerce accounting for nearly 19% of global retail sales, reaching approximately \$5.2 trillion (eMarketer 2023). This trend also catalyzes the adoption of online shopping in previously unexpected sectors, such as grocery shopping. A recent industry report valued the global online grocery market at \$50.28 billion in 2022, estimating it to reach a value of \$305.13 billion by 2030, with an annual growth rate of 26.8% throughout the forecast period (Grand View Research 2022).

As customers increasingly gravitate towards online shopping, they are drawn to retailers providing personalized delivery services. While fast and cheap fulfillment remains preferable for most consumers, a rising number of shoppers now express heterogeneous preferences for receiving their

orders. Speed, cost, convenience, and eco-friendliness are among the key factors that impact customer preference for fulfillment options (Digital Commerce 360 2023). Thus, offering diversified options such as standard shipping, express shipping, attended home delivery, and out-of-home delivery has become increasingly popular among practitioners. However, although various options can help increase conversions, they complicate the fulfillment operations and require seamless integration of multiple fulfillment methods to cater to the varied needs of the customers (Wolfe 2024). As a result, online retailers are urged to pair diversified customer expectations with complex and expensive fulfillment operations. It is crucial for online retailers to ensure that they not only meet these expectations but also optimize the delivery process. The primary objective of this paper is to address these challenges by jointly optimizing the personalized fulfillment options and the fulfillment process within an e-commerce context.

The setting of this paper is motivated by our collaboration with one of the online grocery platforms in North America. This company enables online shoppers to purchase groceries from various suppliers, offering them personalized fulfillment options at checkout. As depicted in Figure 1, customers are presented with a three-level choice. They have to choose either home or out-of-home delivery options, then specify a delivery date and (if needed) a pickup location such as a convenience store, a restaurant, or a warehouse. In addition, they also need to indicate a time slot for the pickup. To facilitate these services, the company partners with third-party trucking companies and local convenience stores, restaurants, etc.

The screenshot shows a checkout page for a grocery delivery service. The main form is titled "Confirm your delivery method" and is divided into three levels of choice, indicated by red boxes and labels:

- Level 1:** A red box highlights the "Confirm your delivery method" section, which contains two radio buttons: "Home delivery" and "Pick-up point". The "Pick-up point" option is selected.
- Level 2:** A red box highlights the "Pick-up location" section, which includes a dropdown menu showing "Au Jardin Noir - 523 Rang Séraphine, Ange-Gardien, QC, CA, J0E1E0". Below this, there is a section for "Opening time" showing "Mardi : 15h00 à 17h00" and a text input field for "Tell us when you plan to come:" with the date "mardi 18 juin 2024".
- Level 3:** A red box highlights the "Tell us what time you plan to come:" section, which contains four time slot buttons: "15h00", "3:30 p.m.", "16h00", and "16h30". The "16h30" button is highlighted in green.

On the right side of the page, there is a summary section titled "Order summary:" which includes a table of costs:

Order summary :	
Articles (1) :	43.48\$
Estimated delivery	9.02\$
TPS	0.45\$
QST	0.90\$
Grand total	55.00 \$

Below the table, there is a link "How are shipping costs calculated?". At the top right, a green box says "Only \$56.52 left to get free shipping!". Below that, there is a green button "Send to this address" and a text prompt "Choose an address or pickup point to calculate shipping."

Figure 1 The personalized delivery options page from our industrial partner.

The core challenge faced by our partner involves efficiently managing these varied fulfillment options to optimize customer satisfaction and operational efficiency. They encounter difficulties

in predicting customer preferences for different fulfillment options and coordinating the logistic capacities of various partners. The complexity is further amplified by fluctuating demand patterns and the need to minimize costs while maintaining service quality.

To address these issues, we adopt a general discrete choice model under mild assumptions, which allows us to incorporate various customer behaviors when they shop online. Additionally, we develop a stochastic model that jointly optimizes the display of fulfillment options and the fulfillment allocations. This model aims to balance the trade-offs between customer satisfaction, characterized by the availability of preferred fulfillment options, and operational efficiency, which involves minimizing logistics costs and optimizing resource allocation. The integration of these models provides a comprehensive approach to enhance the decision-making process in e-commerce fulfillment, contributing to both theoretical understanding and practical improvements in e-commerce supply chain management.

1.1. Main Contributions

We consider a multi-period joint optimization of personalized fulfillment options and allocations for companies involved in delivery and fulfillment services. Such companies include e-commerce platforms like Amazon or delivery services like Instacart, Shipt, and our industrial partner, specializing in local grocery delivery. The decisions include what fulfillment options to show customers and how to fulfill orders, aiming to maximize total expected profits. To the best of our knowledge, this is among the first studies in the literature to address the dynamic aspect of fulfillment flexibility and assignment problems. This focus is notable, given the significant impact of the problem in enabling companies to satisfy customers and maximize profits in e-commerce contexts. Our results and contributions are summarized as follows:

1. We formulate the problem as a joint stochastic program that integrates a general discrete choice model, where each fulfillment option specifies the latest fulfillment time and a fixed surcharge. To analyze this problem, we introduce a deterministic (fluid) approximation and show that it serves as an upper bound for the original stochastic problem. Furthermore, we establish a key structural property of the optimal option list, proving that an optimal list must be *efficient*—maximizing expected revenue while optimizing the utilization of fulfillment resources. This insight enables us to decompose the deterministic approximation into two steps: identifying efficient option lists for each customer and solving a convex optimization problem. Additionally, it reveals that the dimensionality of the problem can be significantly reduced. These simplifications yield substantial computational benefits in solving the deterministic approximation and designing effective heuristics for the original problem.

2. We proposed several algorithms for identifying the efficient option lists under various discrete choice models and demonstrated their practical effectiveness. Specifically, we show that the number of efficient option lists is bounded by $K + 1$ under a Markov Chain choice model and a mild assumption. Our parametric dual sweep algorithm enumerates them all in polynomial time. We also show that both the number of efficient option lists and the time complexity are polynomially bounded with a Nested Logit choice model and mild consumptions. For a general discrete choice model, such as a mixture of Multinomial Logit, we modified the Largest Marginal Revenue algorithm (Talluri and Van Ryzin 2004) and provide simulation evidence showing that the algorithm remains computationally tractable for problem sizes of practical interest, and that the number of efficient option lists grows linearly in K . These results support the practical applicability and reinforce the computational efficiency of solving the deterministic approximation via efficient option list enumeration.

3. We develop a two-phase algorithm called the “Economic option, Rounded and Threshold fulfillment” policy (**ERT**) as depicted in Figure 2. In the initial (offline) phase, **ERT** generates a set of efficient option lists and solves the deterministic relaxation to form referenced decisions for options and fulfillment allocations. In the subsequent (online) phase, **ERT** iteratively engages in personalization and fulfillment stages throughout the planning cycle. During the delivery personalization stage, the algorithm starts with efficient option lists. It dynamically excludes *economically nonviable* fulfillment options based on customer location and the current state of orders to be fulfilled. In the fulfillment stage, the algorithm consolidates all orders. Then, it allocates them to the appropriate warehouses according to a threshold-based policy determined based on a fulfillment cost function.

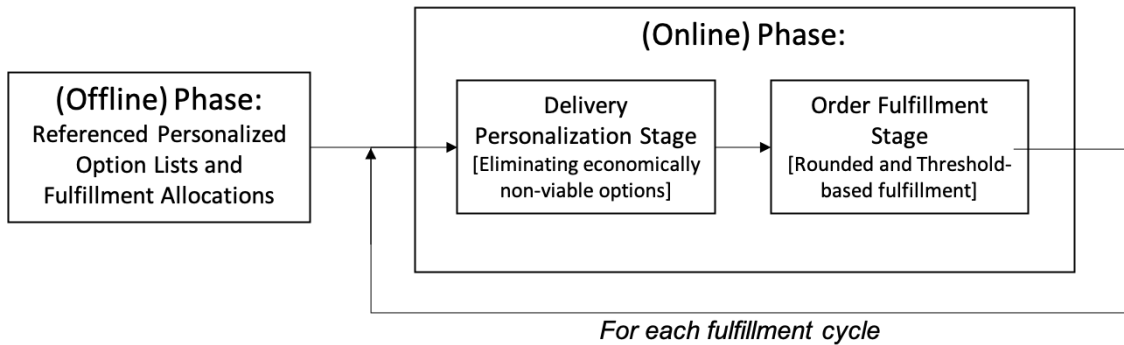


Figure 2 The overall diagram of our proposed (ERT) algorithm.

4. We provide a performance guarantee for the **ERT** algorithm. Our analysis shows that the optimality gap of **ERT** increases at a rate of $\mathcal{O}(T\sqrt{MNK\Lambda})$, with T representing the number

of fulfillment cycles, M the number of warehouses, N the number of customer locations, K the maximum length of a fulfillment window offered to customers, and Λ the largest customer arrival rate per demand location. Specifically, the *sub-linear* growth in N and Λ suggests that **ERT** achieves asymptotic optimality as the number of customer locations or the number of arriving customers per location increases, a scenario commonly encountered by numerous online retailers, including our industrial partner.

5. Beyond these methodological contributions, our study generates managerial insights for an effective joint strategy in managing demand (through personalized fulfillment options) and capacity (via fulfillment allocations), substantiated with real datasets collected from our industrial partner. Our numerical study indicates that while optimizing personalized fulfillment options and allocations are crucial, their roles vary in different scenarios. More specifically, when customers favor quicker fulfillment options, demand management is prominent. Conversely, capacity management becomes more critical when customers prefer more cost-effective (but slower) fulfillment options. Furthermore, demand management is more pivotal when fulfillment capacity is limited, but as capacity increases, the focus shifts toward fulfillment allocations.

1.2. Organization

The remainder of this paper is organized as follows. In §2, we review the related literature. We present the detailed model formulation in §3 and introduce choice-based deterministic relaxation in §4. Utilizing the solution to the relaxation, we develop a heuristic and show its theoretical performance guarantee in §5. We conduct extensive numerical studies of our proposed heuristic policy and present managerial insights in §6. §7 concludes the paper.

2. Literature Review

Our paper falls within the general theme of fulfillment optimization (refer to reviews by Acimovic and Farias 2019 and Qi et al. 2020). However, to the best of our knowledge, ours is among the first to explore fulfillment optimization with personalized delivery options. There are three streams of research closely related to our paper.

E-commerce order fulfillment. The first stream investigates e-commerce fulfillment. Prior work in this stream has examined order consolidation (Xu et al. 2009), single- and multi-item fulfillment under limited inventory conditions (Acimovic and Graves 2015, Jasin and Sinha 2015), and omnichannel retailing (Andrews et al. 2019).

Two specific sub-streams are particularly relevant to our study. The first involves the integration of pricing and fulfillment issues. In this context, Lei et al. (2018) and Harsha et al. (2019) examined the joint pricing and fulfillment problem, while Lei et al. (2022) expanded the scope to include pricing, display, and fulfillment. Notably, Lei et al. (2018) approached the problem as a stochastic

program, devising a two-tier approximation method: Initially, they utilized a deterministic approximation, which, due to its complexity and non-linearity, was further approximated by a solvable linear program through discrete price adjustments. While both study a joint demand-management and fulfillment problem using a fluid approximation, our paper differs from Lei et al. (2018) in several fundamental dimensions. First, they use *dynamic pricing* as the demand lever, a uniform, aggregate instrument applied identically to all consumers. In contrast, our demand lever is *personalized option display*, a customer-specific instrument that exploits heterogeneity in preferences and geographic proximity to warehouses. Second, they assume immediate fulfillment upon order placement with linear shipping costs; our model features multi-period fulfillment windows and convex cost functions that capture congestion effects.

Most closely related to our work is Zhou et al. (2025), which studies the same operational setting of multi-warehouse fulfillment with limited time windows and convex shipping costs. Their model treats demand as entirely *exogenous* and develops a Lagrangian relaxation-based heuristic for the fulfillment-only problem, achieving a performance bound of $\mathcal{O}(TMK\sqrt{N})$. In contrast, our paper complements theirs by introducing a demand-side personalization lever: the option list displayed to each customer at checkout, giving rise to a joint optimization over display decisions and fulfillment allocations. The key technical distinction is that our deterministic relaxation incorporates assortment variables that govern which options each customer sees, thereby introducing a new source of intractability. We exploit structural properties of the problem to reduce its dimension and develop new algorithms to address this challenge under a Markov chain choice model. A further modeling difference is that Zhou et al. (2025) operates at the continuous level, whereas our formulation tracks individual customer arrivals and choices, resulting in integer-valued fulfillment decisions. This individual-level modeling is necessary to support personalized option display but introduces additional complexity. Despite solving a strictly harder problem, our performance bound $\mathcal{O}(T\sqrt{MKNA})$ strictly improves theirs.

The second sub-stream focuses on the strategic benefits of delaying fulfillment decisions. For instance, Mahar and Wright (2009) investigated a quasi-dynamic assignment policy in an omnichannel retail setting with a linear cost structure. Wei et al. (2021) analyzed optimal order consolidation policies, providing heuristic solutions based on structural properties. Finally, Xie et al. (2023) studied online resource allocation with delayed decision-making to leverage real-time demand data. They demonstrated that the performance gap between their online algorithm with delays and the offline optimal policy reduces exponentially as the delay period increases. We add to this literature stream by incorporating personalized delivery options.

Choice-based revenue management. The second stream is related to revenue management (RM) and assortment optimization. Comprehensive reviews are available in Kök et al. (2009)

and Gallego and Topaloglu (2019). Our focus is on the subset of research utilizing choice-based demand models in RM, as our paper explores demand realization through such models. Talluri and Van Ryzin (2004) examined a single-resource RM issue employing a general discrete choice model. They introduced two pivotal concepts: efficient sets and nested-by-fare order. By formulating the problem as a dynamic program (DP), they demonstrated that optimal assortments are efficient sets. Furthermore, they established that optimal assortments adhere to a nested-by-fare order under several specific choice models. Liu and Van Ryzin (2008) expanded the efficient set concept to network revenue management (NRM) within a general discrete choice model framework. They proved that only efficient sets form optimal assortments in the choice-based deterministic relaxation formulation. Unlike in the single-resource scenario, this principle does not universally apply to our DP formulation. Specific choice models provide insights into efficient sets, as discussed in Gallego and Li (2017) and Cao et al. (2022). Our paper focuses on developing an implementable heuristic with a provable performance guarantee based on solving the deterministic relaxation with efficient sets. We extend this efficient-set theory to the Markov chain choice model (Blanchet et al. 2016, Feldman and Topaloglu 2017) by showing that the number of efficient option lists can be identified in polynomial time, which broadens the theoretical foundation of our approach.

Flexible products. The third stream is related to network revenue management (NRM) with flexible products. Gallego and Phillips (2004) describe a *flexible product* as a set of alternatives offered to a market, where the seller later assigns a specific alternative to the purchaser. In our context, an e-commerce company accepts orders encompassing various fulfillment options (akin to products) and later fulfills these orders from a selection of warehouses (comparable to alternatives). Gallego et al. (2004) explored NRM with flexible products under both independent and dependent demand scenarios, employing a general choice model. They demonstrated the asymptotic optimality of deterministic relaxation for both cases. Notably, they used a column generation algorithm to resolve the deterministic relaxation for demand influenced by customer choices. Gönsch et al. (2014) extended the work and introduced heuristic policies derived from linear programming approximations. Subsequently, Cheung and Simchi-Levi (2016) concentrated on an approximate yet efficient solution for the choice-based deterministic relaxation, proposing a Potential-based algorithm with proven efficiency: a polynomial-time method for solving the deterministic relaxation to any desired level of accuracy.

More recently, Ma et al. (2020) developed an approximate dynamic programming (DP) algorithm for solving the choice-based network RM problem. They utilized an *availability tracking basis* for value function approximation, showing that their heuristic’s total expected revenue reaches at least a $1/(1+L)$ fraction of the optimal total expected revenue, where L represents the maximum number of resources a product consumes. Zhu and Topaloglu (2023) expanded this methodology to flexible

product scenarios. The primary challenge in such settings is the availability of remaining capacity information. To overcome this, they introduced an auxiliary variable to track remaining resource capacities, optimistically approximating the value function with this variable. They established that the approximate DP method retains the same constant performance bound even in the context of flexible products. Our paper differs from the previous research in this stream by proposing an asymptotically optimal heuristic based on deterministic relaxation that can be solved efficiently with a commercial convex optimization solver.

3. The Model

We consider an online retailing company, referred to as the seller, that aims to improve its fulfillment system over a finite horizon. In particular, the seller utilizes a combination of logistic carriers to perform fulfillment operations. To cover the total fulfillment cost, as is typical in the industry, the seller relies on two revenue sources. The first one is a fixed percentage of the basket value (i.e., the *fulfillment reserve*), and the second is an additional fare customers pay for shipping (i.e., the *fulfillment surcharge*). The seller’s objective is to balance the revenue and cost terms associated with the delivery process.

Throughout this paper, we denote $[n]$ as the set $\{1, 2, \dots, n\}$. The seller owns M warehouses, where each warehouse $i \in [M]$ stores all products without inventory constraints, as our primary focus is on improving the fulfillment system rather than inventory management. Customers place orders that vary in size and weight, reflecting the diversity of purchase needs. We define a *demand location* as a combination of a specific geographic region and a specific order type, where the order type is determined by the cart’s value and weight. This can be operationalized by discretizing order sizes and weights into representative groups. We assume the seller serves customers from N demand locations, each indexed by $j \in [N]$. This modeling approach is natural, as the seller’s profit depends on both revenue and fulfillment cost, which typically is influenced by the shipping distance and the weight of the order.

We assume that the seller dispatches orders to logistic carriers in discrete time. Once an order is dispatched, it requires a deterministic amount of time for a carrier to deliver the order to the customer. Since the delivery time does not affect the seller’s objective as long as it is dispatched in time, we assume, without loss of generality, that fulfillment is completed instantly upon dispatch. This simplification can be relaxed if needed. For example, if the seller must fulfill an order within three days and it takes one day for a carrier to transport the order from a warehouse to the demand location, the seller can delay dispatching the order by up to two days. Throughout the paper, we define *fulfillment cycle* as the time interval between two consecutive dispatches. The planning horizon is divided into T fulfillment cycles, indexed by $\tau \in [T]$.

Each order specifies the maximum number of fulfillment cycles a customer is willing to wait before receiving her order. We refer to this as the *fulfillment window* k , where $k \in [K]$, and K is the longest number of fulfillment cycles a customer may need to wait. The fulfillment window also reflects the speed of the fulfillment service: a smaller value of k corresponds to faster fulfillment service. At checkout, customers are presented with a menu of fulfillment options (formally defined in the following section). By selecting an option, the customer agrees to wait up to $k \in [K]$ cycles and pay the corresponding fulfillment surcharge. If none of the available options meets their preferences, customers may choose to abandon their purchase.

The sequence of decisions and events are as follows. When a customer lands on the checkout page, the seller decides which fulfillment options to display. The customer decides whether to complete her sales and if yes, she picks one of the fulfillment options displayed on the checkout page. Once a customer proceeds with the purchase, the seller then makes two decisions: (i) to select one of the M warehouses to fulfill the order and (ii) to determine when to fulfill the order so that the order can arrive within the specified window. The objective is to maximize the expected profit associated with the fulfillment process over the planning horizon. In what follows, we explain each component of the fulfillment process.

3.1. Fulfillment Options

Each fulfillment option is characterized by a fulfillment window k . Without loss of generality, we assume that the fulfillment windows are ordered in increasing fashion from 1 to K . We denote a list of fulfillment options (or simply, an option list) presented to a customer by $S \subseteq [K]$. We use the notation $k \in S$ to refer to a specific option in the list, characterized by a fulfillment window of k cycles. Let $\mathcal{N}$ denote the set of all possible option lists that can be presented to a customer. Thus, there are $|\mathcal{N}| = 2^K$ possible option lists, where $|\cdot|$ denotes the number of elements in a set.

A customer from a demand location j (or simply, the customer j) decides whether to proceed with a purchase by selecting one option from the available list based on a known discrete choice model denoted $\{\pi_k^j(\cdot) : k \in [K]\}$. Specifically, if the seller presents an option list $S \subseteq [K]$, customer j will choose a fulfillment option $k \in S$ and proceed with a probability $\pi_k^j(S)$. Additionally, the customer may opt not to proceed with the checkout with probability $\pi_0^j(S)$. This scenario, known as *cart abandonment*, occurs when the customer is dissatisfied with the fulfillment options presented.

3.2. Demand

We assume that the arrival of the customer j during a fulfillment cycle τ follows a Poisson process with a time-dependent arrival rate of $\lambda_{j,\tau}(t)$, where t represents the time elapsed since the start of the current fulfillment cycle. Without loss of generality, we normalize the duration of each fulfillment cycle to the interval $[0, 1]$, with $t = 0$ marking the beginning of cycle τ and $t = 1$ indicating its

closure (just before the fulfillment decision is made). This arrival process is independent between locations and does not need to be homogeneous in τ and t . We denote $\tilde{D}_{j,\tau}(t) \geq 0$, $\tilde{D}_{j,\tau}(t) \in \mathbb{Z}$, the cumulative arrivals of customer j by time t during cycle τ . Furthermore, we let $\tilde{D}_{j,\tau} := \tilde{D}_{j,\tau}(1)$ represent the total number of customers from location j arriving during the cycle τ . Consequently, $\tilde{D}_{j,\tau}$ follows a Poisson distribution with parameter $\Lambda_{j,\tau} := \int_0^1 \lambda_{j,\tau}(t)dt$. That is,

$$\mathbb{P}\{\tilde{D}_{j,\tau} = n\} = \frac{(\Lambda_{j,\tau})^n}{n!} e^{-\Lambda_{j,\tau}}$$

When a customer arrives, the seller shows an option list based on some decision rule $S_{j,\tau}(t)$. This decision rule may vary depending on the customer j , the fulfillment cycle τ , and the elapsed time t within the cycle. We denote $D_{k,j,\tau}(t) \geq 0$, $D_{k,j,\tau}(t) \in \mathbb{Z}$, as the cumulative number of customers from location j who choose fulfillment option k by the elapsed time t in cycle τ , representing the *realized* demand.

It is important to note that while the cumulative arrival process $\tilde{D}_{j,\tau}(t)$ follows a Poisson distribution, the realized demand $D_{k,j,\tau}(t)$ may not. This deviation arises because even though $D_{k,j,\tau}(t)$ is derived from $\tilde{D}_{j,\tau}(t)$, it is influenced by the selected fulfillment option, which in turn is influenced by the decision rule $S_{j,\tau}(t)$. Since the decision rule $S_{j,\tau}(t)$ is dynamic and may depend on the history of realized demand, $D_{k,j,\tau}(t)$ also becomes dependent on past events, losing the memoryless property, which is characteristic of a Poisson process. This intricate interplay between option display and realized demand poses significant computational challenges, which will be further elaborated in §3.5.

3.3. System Dynamics

We define the “pre-choice” state, denoted as $x_{k,j,\tau}(t) \geq 0$, $x_{k,j,\tau} \in \mathbb{Z}$, as the cumulative unfulfilled orders received from location j that have k remaining cycles left in the fulfillment window at the beginning of time t during fulfillment cycle τ . Similarly, the “post-choice” state, denoted as $y_{k,j,\tau}(t) \geq 0$, $y_{k,j,\tau} \in \mathbb{Z}$, describes the cumulative unfulfilled orders at the end of time t . When a new customer j arrives at time t and an option list S is displayed, if she selects the fulfillment option k , then $y_{k,j,\tau}(t) = x_{k,j,\tau}(t) + 1$. Otherwise, $y_{k,j,\tau}(t) = x_{k,j,\tau}(t)$. Additionally, we define $x_{k,j,\tau} := x_{k,j,\tau}(0)$ as the pre-choice state of fulfillment cycle τ and $y_{k,j,\tau} := y_{k,j,\tau}(1)$ as the post-choice state of that cycle. Thus, we have the following relationship

$$y_{k,j,\tau} = x_{k,j,\tau} + D_{k,j,\tau}$$

We point out that all the states $x_{k,j,\tau}$ and $y_{k,j,\tau}$ are subject to two sources of randomness: the arrivals of customers are stochastic, and the fulfillment options they choose, conditioned on some lists offered, are also stochastic.

At the end of the fulfillment cycle τ , the seller makes a fulfillment decision denoted by $u_\tau = (u_{k,j,i,\tau} : u_{k,j,i,\tau} \geq 0, u_{k,j,i,\tau} \in \mathbb{Z}, \forall k \in [K], j \in [N], i \in [M])$. In particular, $u_{k,j,i,\tau}$ corresponds to the number of orders shipped by fulfillment cycle τ from warehouse i to demand location j to satisfy the unfulfilled orders with k cycles left in fulfillment window. The seller needs only ship orders that have been submitted by some customers but have not yet been fulfilled. To ensure that, we impose the following feasibility constraints for all $k \in [K]$, $j \in [N]$, and $\tau \in [T]$:

$$\sum_{i \in [M]} u_{k,j,i,\tau} \leq y_{k,j,\tau}$$

Next, we define the state update equations. After the fulfillment decision u_τ is made for cycle τ , a portion of the orders are fulfilled during the cycle, while the remaining orders are deliberately postponed and carried over to the next cycle. For each postponed order, the remaining fulfillment window decreases by one, as the order is one cycle closer to its committed fulfillment deadline. Accordingly, for all $k < K$, the state variable is updated as follows:

$$x_{k,j,\tau+1} = y_{k+1,j,\tau} - \sum_{i \in [M]} u_{k+1,j,i,\tau}$$

Additionally, $x_{K,j,\tau+1} = 0$ since no remaining orders can have K cycles after being carried over. If any orders have only one fulfillment cycle remaining but are not fulfilled in the current cycle, they are removed from the system and incur penalty costs.

3.4. Revenue and Cost

Our focus on optimizing the fulfillment system involves revenue and cost terms associated with the fulfillment process. We denote $r_{k,j} \geq 0$ as the revenue the seller collects when a customer from demand location j selects fulfillment option k and completes the purchase. In our model, $r_{k,j}$ is treated as an exogenous input, and the seller's real-time decision is which options from this pre-set menu to display to each arriving customer. In practice, the platform contracts with logistics carriers whose delivery fees and capacity limits are set externally through carrier contracts. What the platform can control is the mapping from delivery speed to customer-facing price (the surcharge schedule). However, this schedule is a strategic decision set on a slow timescale, and is typically revised in tandem with carrier contract renegotiations rather than adjusted in real time. That said, our modelling framework can endogenize the surcharge schedule with minimal additional structure, see Online Appendix EC.2 for the detail. Finally, the seller does not generate revenue if customers abandon their carts. Therefore, the expected revenue generated during the fulfillment cycle τ can be calculated as follows:

$$\mathbb{E} \left\{ \sum_{k \in [K]} \sum_{j \in [N]} r_{k,j} D_{k,j,\tau} \right\}$$

Next, we consider the fulfillment cost. The seller relies on multiple logistic carriers to execute the fulfillment operations. Each carrier commits some capacity (usually in terms of weight or volume) and charges a flat rate for each unit of the logistic resource (see the left panel in Figure 3). Initially, the seller prioritizes using a more cost-effective carrier. However, if the logistics capacity from the cheaper carrier is insufficient to meet the demand, the seller turns to a carrier with a higher unit cost. This results in a piecewise linear function representing the lower envelope of the total cost curve faced by the retailer, with each breakpoint reflecting cumulative capacity (the right panel in Figure 3).

To model this, we assume that the fulfillment cost function is convex and increasing in the total number of products shipped from each warehouse $i \in [M]$. To summarize, we define $C_i : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ to denote the fulfillment cost function and make the following assumption:

ASSUMPTION 1. $C_i(z)$ is convex and increasing in z and its derivative $C'_i(z) < \bar{C}$ for all $z \in \mathbb{R}_+$, where $0 < \bar{C} < \infty$.

The fulfillment cost for warehouse i , given its fulfillment vector $u_{i,\tau}$, can be obtained as

$$C_i \left(\sum_{k \in [K]} \sum_{j \in [N]} w_j u_{k,j,i,\tau} \right) = C_i (w^T u_{i,\tau})$$

where w_j is the weight conversion factor for the order associated with demand location j . $w = (w_{k,j} : \forall k \in [K], j \in [N])$, $w_{k,j} = w_j$, is the vector of weight factors.

Finally, if the seller cannot fulfill the order from location j within its fulfillment period k using the committed logistic capacity, it incurs a unit penalty b_j . This penalty can be considered the shipping rate of an expedited service that can fulfill an order from j with a single fulfillment window. We assume $C'_i(0) < \max_j b_j$ to avoid triviality, which implies there exists a committed carrier whose shipping rate is lower than the expedited shipping service. Using the state and action

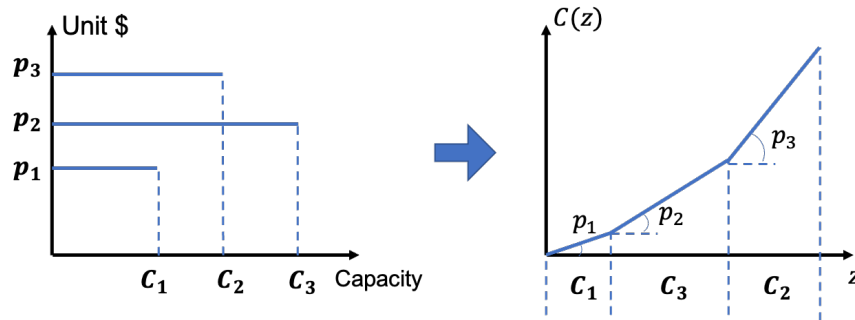


Figure 3 Illustration of fulfillment cost function with three logistic providers. The left side shows the capacity and the unit price for shipping of each carrier, and the right side shows the total fulfillment cost as a function of used capacity.

defined above, we can calculate the total number of outstanding orders from location j in cycle τ fulfilled with the expedited service by $y_{1,j,\tau} - \sum_{i \in [M]} u_{1,j,i,\tau}$. Multiplying this by location-specific unit penalty cost b_j and summing over all locations $j \in [N]$, we obtain the total penalty cost as follows:

$$\sum_{j \in [N]} b_j \left(y_{1,j,\tau} - \sum_{i=1}^M u_{1,j,i,\tau} \right)$$

We now provide several remarks regarding the modeling of fulfillment costs. First, the differentiability assumption is made primarily for analytical convenience and can be relaxed. In practice, if only a finite number of carriers are available, the fulfillment cost function becomes piecewise-linear and convex. In such cases, our analysis remains valid by replacing gradients with subgradients. Second, the assumption $\bar{C} < \infty$ is similarly introduced for technical reasons. It can be relaxed without loss of generality, since the marginal fulfillment cost can always be capped by the maximum penalty cost $\max_j b_j$. Third, we implicitly assume that the fulfillment cost is independent of the demand location. This assumption is reasonable for small- and medium-sized online retailers serving regional markets. For larger retailers, however, the model can be extended by incorporating demand location-dependent fulfillment costs using multiple shipping zones, as logistics carriers typically charge zone-based fixed marginal rates (Shipbob 2022). We operationalize this by defining the shipping cost function with two indices, $C_{i,z}$, where z is a zone index associated with warehouse i . A detailed discussion of this extension is provided in EC.3.

3.5. Joint Optimization

The seller's objective is to maximize its expected profit by deciding which fulfillment options to offer at any time and how to fulfill orders at the end of each fulfillment cycle. We can formulate this problem as a joint stochastic optimization called **OPT**.

$$\begin{aligned} V^*(x) = \max_{S,u} \quad & \mathbb{E} \left\{ \sum_{\tau \in [T]} \sum_{j \in [N]} \sum_{k \in [K]} r_{k,j} D_{k,j,\tau} \right\} \\ & - \mathbb{E} \left\{ \sum_{\tau \in [T]} \left[\sum_{i \in [M]} C_i(w^\top u_{i,\tau}) + \sum_{j \in [N]} b_j \left(y_{1,j,\tau} - \sum_{i \in [M]} u_{1,j,i,\tau} \right) \right] + \sum_{k \in [K]} \sum_{j \in [N]} b_j x_{k,j,T+1} \right\} \\ \text{s.t.} \quad & x_{k,j,1} = x_{k,j}, \quad \forall k, j \tag{1a} \\ & y_{k,j,\tau} = x_{k,j,\tau} + D_{k,j,\tau}, \quad \forall k, j, \tau \tag{1b} \\ & x_{k,j,\tau+1} = y_{k+1,j,\tau} - \sum_i u_{k+1,j,i,\tau}, \quad \forall k < K, j, \tau \tag{1c} \\ & x_{K,j,\tau+1} = 0, \quad \forall j, \tau \tag{1d} \\ & \sum_i u_{k,j,i,\tau} \leq y_{k,j,\tau}, \quad \forall k, j, \tau \tag{1e} \end{aligned}$$

$$u_{k,j,i,\tau} \geq 0, u_{k,j,i,\tau} \in \mathbb{Z} \quad \forall k, j, i, \tau \quad (1f)$$

$$S_{j,\tau}(t) \in \mathcal{N}, \quad \forall j, \tau, t \in [0, 1] \quad (1g)$$

In **OPT**, the first part of the objective function is the expected revenue. The second part is the expected costs, which consist of three components: the first is the fulfillment cost, the second is the penalty cost, and the third is the terminal cost, meaning that all unfulfilled orders at the end of the horizon are handled with expedited shipping services. For constraints, (1a) specifies the initial condition, (1b) to (1d) describe the dynamics of the system, (1e) and (1f) require fulfillment decisions to be feasible, and (1g) ensures that the option list must be selected from the set of possible fulfillment options. We point out that the random variable $D_{k,j,\tau}$ depends on $S_{j,\tau}(t)$. All constraints must hold almost surely.

OPT can theoretically be solved using dynamic programming (DP). However, this approach suffers from the curse of dimensionality, rendering it computationally intractable in practice. To illustrate this, let $V_\tau(x, t)$ denote the maximum total expected profit from time t onward in fulfillment cycle τ , given the pre-choice state x . Define $e_{k,j} \in \{0, 1\}^{KN}$ as a unit vector with a one at the (k, j) -th position and zeros elsewhere. Thus, $e_{k,j}$ represents an order of the type (k, j) . The marginal impact of receiving this additional order on the expected profit is given by

$$\Delta V_\tau^{k,j}(x, t) := V_\tau(x, t) - V_\tau(x + e_{k,j}, t).$$

As shown in EC.4, the optimal option list presented to a customer from location j at time t in cycle τ must solve the following maximization problem:

$$S_{j,\tau}^*(t) \in \arg \max_S \sum_k (r_{k,j} - \Delta V_\tau^{k,j}(x, t)) \pi_k^j(S). \quad (2)$$

Since the value function $V_\tau(x, t)$ does not increase with an additional order, we have $\Delta V_\tau^{k,j}(x, t) \geq 0$, which can be interpreted as the marginal fulfillment cost for an order (k, j) . Equation (2) characterizes an assortment optimization problem with multiple resources: for the same customer j , each fulfillment option k incurs a fulfillment cost $\Delta V_\tau^{k,j}(x, t)$, which generally differs across options. Solving this problem is computationally difficult due to the high dimensionality of the state space (x, t) , which makes it difficult to compute the optimal value functions efficiently.

In addition, the fulfillment decisions at the end of each fulfillment cycle are also intractable. To see this, let G_τ denote the profit function of fulfillment cycle τ , which is defined as follows:

$$G_\tau(x_\tau, u_\tau, S_\tau^*) = \sum_{k,j} r_{k,j} D_{k,j,\tau}^* - \left(\sum_i C_i(w^\top u_{i,\tau}) + \sum_j b_j \left(x_{1,j,\tau} + D_{1,j,\tau}^* - \sum_i u_{1,j,i,\tau} \right) \right)$$

where $S_\tau^* = \{S_{j,\tau}^*(t) \in \mathcal{N} : j \in [N], t \in [0, 1]\}$ denotes the set of optimal option lists presented to customers, and $D_{k,j,\tau}^*$ is the realized demand resulting from S_τ^* and the corresponding customer choices. The DP formulation for fulfillment decisions is given by:

$$V_\tau(x_\tau, 0) = \max_{u_\tau} \mathbb{E} \{G_\tau(x_\tau, u_\tau, S_\tau^*) + V_{\tau+1}(x_{\tau+1}, 0)\}$$

with the boundary condition $V_{T+1}(x, 0) = -\sum_{k,j} b_j x_{k,j}$. Even when the optimal option list strategy S_τ^* is known, solving the above DP remains computationally intractable due to the high dimensional state space x . As a result, our focus shifts to developing computationally efficient heuristic policies and establishing their provable effectiveness.

We close this section with the following structural property, which states that under any option display strategy, the total fulfillment weight dispatched from each warehouse in a given fulfillment cycle—under an optimal fulfillment policy—is bounded above. We refer to this upper bound as the (implicit) fulfillment capacity of the warehouse. The intuition behind this property is that the seller, when optimizing its operations, would prefer to delay an order and potentially incur penalty costs for unfulfilled orders rather than fulfill them now through carriers that impose prohibitively high marginal costs. The formal proof of this result is provided in EC.1.1.

LEMMA 1 (Implicit Fulfillment Capacity). *For any option list strategy $S_{j,\tau}(t)$, if a warehouse i is ever used to fulfill any orders from location j during cycle τ , then the optimal fulfillment from that warehouse must satisfy $C'_i(w^\top u_{i,\tau}^*) w_j \leq b_j$.*

4. Choice-based Deterministic Approximation

To establish the foundation for a computationally tractable heuristic, we define the deterministic (fluid) relaxation as follows: Customers arrive uniformly at the rate of $\lambda_{j,\tau}(t)$ from location j at period t in cycle τ . Integrating this throughout the cycle τ , we obtain a total arrivals from location j as $\Lambda_{j,\tau} = \int_0^1 \lambda_{j,\tau}(t) dt$. Each customer may purchase a fractional quantity of an order. Moreover, if the seller shows an option list S , a deterministic proportion of $\pi_k^j(S)$ of the customer's order is fulfilled using option k , while a fraction $\pi_0^j(S)$ need not be fulfilled due to cart abandonment. Consequently, the seller collects only a partial reward $\sum_{k=1}^K r_{k,j} \pi_k^j(S)$. In addition, we consider a strategy such that during cycle τ , we offer a list S to $\gamma_{j,\tau}(S)$ proportion of customers from location j . Under this strategy, the realized demand indexed by k, j, τ can be calculated as $\sum_S \Lambda_{j,\tau} \gamma_{j,\tau}(S) \pi_k^j(S)$. Since we consider a fluid model, we now allow the fulfillment decisions $u_{k,j,i,\tau}$ to be continuous instead of integral. The deterministic programming can then be formulated as follows, which we refer to as **DET**.

$$\begin{aligned}
V^D(x) &= \max_{\gamma, \bar{u}} \sum_{\tau} \sum_j \sum_k \sum_S r_{k,j} \Lambda_{j,\tau} \gamma_{j,\tau}(S) \pi_k^j(S) \\
&\quad - \sum_{\tau} \left(\sum_i C_i(w^\top \bar{u}_{i,\tau}) + \sum_j b_j \left(\bar{y}_{1,j,\tau} - \sum_i \bar{u}_{1,j,i,\tau} \right) \right) - \sum_{k,j} b_j \bar{x}_{k,j,T+1} \\
\text{s.t. } \quad &\bar{x}_{k,j,1} = x_{k,j}, \quad \forall k, j \tag{3a} \\
&\bar{y}_{k,j,\tau} = \bar{x}_{k,j,\tau} + \Lambda_{j,\tau} \sum_S \gamma_{j,\tau}(S) \pi_k^j(S), \quad \forall k, j, \tau \tag{3b} \\
&\bar{x}_{k,j,\tau+1} = \bar{y}_{k+1,j,\tau} - \sum_i \bar{u}_{k+1,j,i,\tau}, \quad \forall k < K, j, \tau \tag{3c} \\
&\bar{x}_{K,j,\tau+1} = 0, \quad \forall j, \tau \tag{3d} \\
&\sum_i \bar{u}_{k,j,i,\tau} \leq \bar{y}_{k,j,\tau}, \quad \forall k, j, \tau \tag{3e} \\
&\sum_S \gamma_{j,\tau}(S) = 1, \quad \forall j, \tau \tag{3f} \\
&\bar{u}_{k,j,i,\tau} \geq 0 \quad \forall k, j, i, \tau \tag{3g} \\
&\gamma_{j,\tau}(S) \geq 0, \quad \forall j, \tau, S \in \mathcal{N} \tag{3h}
\end{aligned}$$

First, we establish that the optimal function value $V^D(x)$ for **DET** serves as an upper bound for the optimal expected profit $V^*(x)$ in **OPT**.

PROPOSITION 1. $V^*(x) \leq V^D(x)$

The proof is available in EC.1.2. Note that **DET** constitutes a convex optimization problem with linear constraints, which implies that it may be solved using commercial solvers. However, it poses two challenges. Firstly, since we define a decision variable for each option list, the number of decision variables grows *exponentially* with the number of fulfillment options. A common approach to address this issue is the column generation (CG) technique (Gallego et al. 2004, Bront et al. 2009). Nonetheless, this technique requires iteratively solving a sequence of subproblems that involve searching across an exponential number of option lists, which may still be time-consuming. Secondly, unlike **OPT**, **DET** eliminates all sources of randomness and assumes fulfillment actions are continuous. As discussed in §5, the solution obtained from **DET** may not be feasible and does not effectively adapt to demand realizations and dynamic changes in fulfillment capacity, leading to suboptimal performance in practice.

We delegate addressing the second issue to the next section. In this section, our focus is to tackle the first challenge by characterizing specific structural properties of an optimal solution to **DET**.

DEFINITION 1. For any list of fulfillment options $S \in \mathcal{N}$, define the expected revenue for customer j as $R_j(S) = \sum_k r_{k,j} \pi_k^j(S)$ and the probability of purchase $Q_j(S) = \sum_k \pi_k^j(S)$. S is *efficient* for

customer j if there does not exist a set of convex weights $\alpha(S')$ satisfying $\sum_{S' \in \mathcal{N}} \alpha(S') = 1$ and $\alpha(S') \geq 0$ such that

$$\begin{aligned} R_j(S) &< \sum_{S' \in \mathcal{N}} \alpha(S') R_j(S') \\ Q_j(S) &\geq \sum_{S' \in \mathcal{N}} \alpha(S') Q_j(S') \end{aligned}$$

Otherwise, S is *inefficient* for customer j .

PROPOSITION 2. *If $\gamma_{j,\tau}^*(S) > 0$ for some j and τ is an optimal solution to **DET**, then S is efficient for customer j in cycle τ .*

We defer the proof to EC.1.3, which consists of three steps: First, we show that if $\gamma_{j,\tau}^*(S) > 0$, then S solves the following problem

$$\max_{S'} R_j(S') - \sum_k \theta_{k,j} \pi_k^j(S') \quad (4)$$

for some $\theta_j = (\theta_{1,j}, \dots, \theta_{K,j})$. Second, we show that $\theta_{k,j}$ are identical and non-negative for all k . Finally, by the necessary and sufficient condition shown in the technical Lemma EC.2, S must be efficient. To show all $\theta_{k,j}$ are identical for all k , we need to consider several fulfillment scenarios: (1) If orders with different fulfillment options are handled by the same warehouse, the corresponding $\theta_{k,j}$ (which can be interpreted as marginal cost of fulfilling a customer j using option k) is identical since they use the same fulfillment resource. (2) If orders with different fulfillment options are fulfilled by different warehouses, then any cost disparity between them would imply that the seller could reduce total cost by reallocating fulfillment from the higher-cost warehouse to the lower-cost one. Therefore, in equilibrium under an optimal policy, the marginal costs must again be equal to avoid such inefficiency. (3) For orders that are not fulfilled, the associated marginal cost must be at least $\theta_{k,j}$; otherwise, it would be more cost-effective to fulfill those orders rather than leave them unserved. Collectively, these scenarios demonstrate that under an optimal fulfillment strategy for **DET**, all $\theta_{k,j}$ must be equal for a given customer j , regardless of the specific fulfillment option k .

Proposition 2 has two important implications. First, it enables an efficient two-step solution procedure for **DET**. In the first step, efficient option lists are identified for each customer j ; in the second step, a convex optimization problem is solved over these lists. When the number of efficient lists is small, this decomposition significantly reduces computational effort. Second, the proposition reduces the number of efficient option lists and simplifies complexity of identifying them all. If we interpret $\theta_{k,j}$ as the cost of fulfilling order j with option k , the fact that $\theta_{k,j}$ is constant across all k implies uniform fulfillment cost. Even if the resources for different options come from different warehouses or future cycles, they can be virtually pooled as a single resource since their costs are identical.

This simplification is important. Note that each option list can be denoted as a point $v_j(S) = (\pi_1^j(S), \dots, \pi_K^j(S), R_j(S)) \in \mathbb{R}^{K+1}$, where the first K dimensions capture the purchase probability for each option k , and the last dimension reflects the expected revenue. According to (4), which can be reformulated as

$$\max_{S'} c_j^\top v_j(S'), \quad \text{where } c_j = (-\theta_{1,j}, \dots, -\theta_{K,j}, 1)^\top$$

any option list S with positive sampling probability $\gamma_{j,\tau}^*(S)$ maximizes the above linear objective in the direction c_j . When the costs $\theta_{k,j}$ differ across options k , the search must be performed in a $(K+1)$ -dimensional space. This becomes computationally intensive as the number of options K increases. Worse yet, there are no efficient algorithms available to handle this general case.

In contrast, when the costs $\theta_{k,j}$ are identical across k (and denoted by θ_j), (4) reduces to

$$\max_{S'} R_j(S') - \theta_j Q_j(S') \tag{5}$$

which is the necessary and sufficient condition (Lemma EC.2) for an option list to be efficient under Definition 1. Thus, each option list can be represented as a point $(Q_j(S), R_j(S))$, which is independent of K , in the two-dimensional space. We essentially collapse the first K dimensions into a single one, which is only possible because $\theta_{k,j}$ is identical across k . In this much simpler setting, we can efficiently identify all efficient option lists using known algorithms from the literature, which we introduce in the next section. Thus, Proposition 2 enables a major reduction in computational burden by transforming a high-dimensional problem into a tractable one.

It is worth noting that an optimal option list to the original stochastic problem may not necessarily be efficient. This idea is formalized as the following result and we provide a counter example in EC.1.4. This observation motivates us to modify an option list selected in **DET** within the proposed heuristic in §5.

LEMMA 2. *An optimal option list to **OPT** for customer j is not always efficient.*

4.1. Construction of Efficient Sets under a Markov Chain Choice Model

To apply Proposition 2, we must identify all efficient option lists for each customer j , denoted by $\mathcal{E}_j$, and ideally, the number of such lists $|\mathcal{E}_j|$ is much smaller than the total number of possible option lists, 2^K . In this section, we focus on identifying efficient option lists under Definition 1. We also analyze the algorithm's complexity and characterize the number of efficient lists.

For a general discrete choice model, we adopt the well-known *largest marginal revenue (LMR)* algorithm proposed by Talluri and Van Ryzin (2004) to construct all efficient option lists; see Algorithm 5 and the corresponding simulations in Online Appendix EC.7. The simulation results demonstrate that, in practice, the decision space of **DET** can be substantially reduced via efficient

option lists, making the resulting optimization problem computationally tractable for problem sizes of interest, even without additional structural assumptions on the choice model.

Despite that, the worst-case complexity of this method is $\mathcal{O}(|\mathcal{E}_j| \cdot 2^K)$, as identifying each efficient list may require enumerating all 2^K possible option lists. Meanwhile, the number of efficient lists, $|\mathcal{E}_j|$, depends on the structure of the underlying choice model and the specific fulfillment surcharges, and thus cannot be generally bounded. To obtain stronger complexity guarantees and meaningful bounds on $|\mathcal{E}_j|$, we need additional structural assumptions on the choice model.

To this end, we adopt the *Markov chain (MC) choice model* introduced by Zhang and Cooper (2005). In this model, a customer initially considers a specific fulfillment option. If that option is unavailable, the customer may switch to another option or abandon the purchase, subject to transition probabilities. This process repeats until the customer either selects an available option or leaves. The MC model is well-suited for our setting for three reasons. First, it *strictly generalizes* the MNL model by capturing asymmetric substitution patterns and product-specific transition behavior that MNL cannot represent. Second, Blanchet et al. (2016) shows that any random utility maximization-based discrete choice model can be approximated to arbitrary precision by an MC model. Third, Berbeglia et al. (2022) shows that MC achieves comparable or superior predictive accuracy relative to other choice models in retail and e-commerce settings, which is highly relevant to our fulfillment context.

While we defer the detailed discussion to Online Appendix EC.5, as it requires additional notations that are less relevant to the focus of this paper, the key observations are as follows. With an MC choice model, we can derive a dual formulation of (5), parameterized by θ_j . We further show that the dual problem has a unique solution for any $\theta_j \geq 0$, which helps us to construct a special option list (we call it dual-selected option list) $S^{\theta_j} \in \mathcal{E}_j$. The family of the dual-selected option list $\{S^{\theta_j} : \theta_j \geq 0\}$ possesses a monotonic nesting property (Lemma EC.8) and thus has at most $K + 1$ elements that can be enumerated in polynomial time with Algorithm 3, which we term the *Parametric Dual Sweep* (PDS) algorithm (see Online Appendix EC.5).

Furthermore, for any $\theta_j \geq 0$, the optimal dual solution partitions all options into three sets $\mathcal{A}(\theta_j), \mathcal{C}(\theta_j), \mathcal{T}(\theta_j)$: if $k \in \mathcal{A}(\theta_j)$, that option is strictly included in an efficient option list; if $k \in \mathcal{C}(\theta_j)$, that option is strictly excluded; if $k \in \mathcal{T}(\theta_j)$, that option is indifferent between the two actions. We further show that $\mathcal{E}_j$ is equivalent to $\{\mathcal{A}(\theta_j) \cup U : U \subseteq \mathcal{T}(\theta_j)\}$. Under mild Assumption EC.2 in Online Appendix EC.5, we can further show that $\mathcal{E}_j = \{S^{\theta_j} : \theta_j \geq 0\}$, which leads to our main results regarding constructing efficient option lists in Proposition 3. Moreover, we can relax Assumption EC.2 but still guarantee $\mathcal{E}_j$ is polynomially bounded. We refer the readers to Remark EC.1 in Online Appendix EC.5 for details.

PROPOSITION 3. *Under an MC choice model and Assumption EC.2, for each consumer j , there exist a set of fulfillment option lists*

$$S_{j,0} \supsetneq S_{j,1} \supsetneq \cdots \supsetneq S_{j,L} = \emptyset$$

such that

$$\mathcal{E}_j = \{S_{j,0}, S_{j,1}, \dots, S_{j,L}\}$$

and $|\mathcal{E}_j| = L + 1 \leq K + 1$. Moreover, Algorithm 3 enumerates all elements in $\mathcal{E}_j$ in polynomial time.

We also consider alternative choice models. In particular, since a Multinomial Logit (MNL) choice model is a special case of MC, the above results hold. Under a Nested Logit (NL) choice model, we show that the number of efficient option lists is also polynomially bounded under mild assumptions, and we propose another algorithm to identify all of them in polynomial time. We provide a detailed discussion in Online Appendix EC.6. Under a Mixture of MNL (MMNL) choice model, we empirically demonstrate that the number of efficient lists remains well-behaved for problem sizes of practical interest, see Online Appendix EC.7.

Based on the preceding discussion, we make the following assumption for the remainder of the paper. Under this assumption, solving **DET** in two steps—as suggested by Proposition 2—remains computationally efficient.

ASSUMPTION 2. *Given a choice model $\{\pi_k^j(\cdot) : k \in \mathcal{S}\}$ and a list of distinct rewards $\{r_{k,j} : k \in [K]\}$, there is an oracle to construct the set of efficient option lists $\mathcal{E}_j$ for each customer j .*

5. Proposed Algorithm and Performance Analysis

Utilizing the optimal solution $\gamma_{j,\tau}^*(S)$ and $u_{k,j,i,\tau}^*$ obtained from **DET**, we present a heuristic policy for **OPT** with a provable performance guarantee. Specifically, upon the arrival of customer j during cycle τ , we choose an option list S with probability $\gamma_{j,\tau}^*(S)$ and tailor it by removing options that are economically nonviable. We then address the feasibility issue of fulfillment allocation $u_{k,j,i,\tau}^*$ and aim to preserve the structural property described in Lemma 1 by the end of the fulfillment cycle.

5.1. Improving Fulfillment Option List

While one could select a fulfillment option list S with probability $\gamma_{j,\tau}^*(S)$ and present it to customer j during cycle τ , this strategy may perform poorly under **OPT**. The main reason is that in **DET**, an option list S is selected so that the expected marginal fulfillment costs $\theta_{k,j}$ are identical across all $k \in S$. However, under demand uncertainty in **OPT**, the realized marginal cost $\Delta V_\tau^{k,j}(x, t)$ can vary significantly across options, which in turn makes some options in S unprofitable. We refer to such options as *economically nonviable*, a concept that will be defined formally in the following

section. To improve performance, we eliminate such options from the selected list S . To better illustrate the concept of economically nonviable fulfillment options under uncertainty, consider the following toy example.

Example. Suppose the seller operates a single warehouse and contracts with two logistics carriers to fulfill customer orders. The first carrier offers a limited capacity of 10 units per fulfillment cycle at a unit shipping cost of \$10. The second carrier imposes a higher rate of \$20 per unit but is willing to fulfill an unlimited number of orders. This setup can be represented by the following piecewise linear fulfillment cost function:

$$C(z) = \begin{cases} 10z, & z \leq 10 \\ 20z - 100, & z > 10 \end{cases}$$

where z denotes the total fulfillment units from the warehouse in a given cycle. Additionally, if an order is not fulfilled within the promised time window, the seller incurs a penalty cost of \$25 per unfulfilled unit. Consider an efficient option list that includes two options: a faster delivery option with $k = 1$ and $r_1 = 12$, and a slower delivery option with $k = 2$ and $r_2 = 11$. Suppose the seller has already committed to fulfill 10 units in the current cycle. A new customer j arrives, and the seller must decide which options to offer in the list S . If the seller offers both options and the customer selects $k = 1$, the seller collects 12 in revenue but incurs a penalty of 20 due to capacity exhaustion from the first carrier, resulting in a negative marginal profit of $12 - 20 = -8$. In contrast, if only the slow option $k = 2$ is offered and selected, the order can be deferred to the next cycle, where the cheaper capacity may be available, yielding a positive marginal profit of $11 - 10 = 1$.

Despite both $k = 1$ and $k = 2$ being part of the efficient option list in the deterministic setting—where marginal fulfillment costs should be equal—the realized costs can differ substantially. Consequently, options like $k = 1$, which generates negative marginal profit in some system states, are economically nonviable and should be excluded. This motivates the need to estimate marginal fulfillment cost dynamically for each option in real-time.

Formally, when a customer j arrives, the pre-choice state is $x_\tau(t) = (x_{k,j,\tau}(t) : k \in [K], j \in [N])$, $t \in [0, 1]$. The focal customer is presented with a list S and selects a fulfillment option $k' \in S$ with a probability $\pi_{k'}(S)$. Consequently, the post-choice state becomes $y_\tau(t) = x_\tau(t) + e_{k',j}$. We approximate the marginal fulfillment cost by $\mathcal{F}(x_\tau(t) + e_{k',j}) - \mathcal{F}(x_\tau(t))$, where $\mathcal{F}(x)$ is defined as

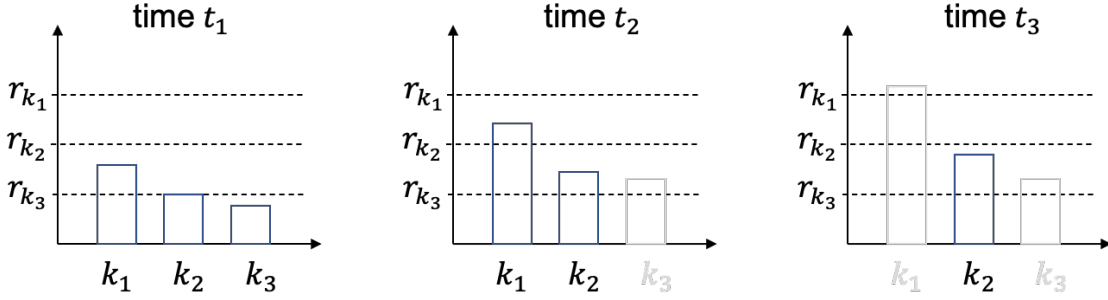


Figure 4 Illustration of dynamically eliminating economically nonviable fulfillment options for some customer j .

follows:

$$\begin{aligned}
 \mathcal{F}(x) = \min_{\tilde{u}} \quad & \sum_{\tau'=\tau}^{\tau+K-1} \left\{ \sum_i C_i(w^\top \tilde{u}_{i,\tau'}) + \sum_j b_j \left(\tilde{y}_{1,j,\tau'} - \sum_i \tilde{u}_{1,j,i,\tau'} \right) \right\} \\
 \text{s.t.} \quad & \tilde{y}_{k,j,\tau} = x_{k,j}, \quad \forall k, j \\
 & \tilde{y}_{k,j,\tau'+1} = \tilde{y}_{k+1,j,\tau'} - \sum_i \tilde{u}_{k+1,j,i,\tau'}, \quad \forall k < K, j, \tau' < \tau + K - 1 \\
 & \sum_i \tilde{u}_{k,j,i,\tau'} \leq \tilde{y}_{k,j,\tau'}, \quad \forall k, j, \tau' \\
 & \tilde{u}_{k,j,i,\tau'} \geq 0, \quad \forall k, j, i, \tau'
 \end{aligned} \tag{6}$$

Clearly, $\mathcal{F}(x)$ is a convex optimization that describes an optimal fulfillment allocation for x orders in K cycles and can be solved efficiently by a commercial solver. An option k' is deemed economically nonviable if its associated revenue is smaller than the approximated marginal fulfillment cost. With that, we can formally define an economically viable fulfillment option.

DEFINITION 2. We say the fulfillment option $k' \in S$ is *economically nonviable* for customer from location j who arrives at time t in cycle τ if $r_{k',j} < \mathcal{F}(x_\tau(t) + e_{k',j}) - \mathcal{F}(x_\tau(t))$; Otherwise, we say k' is *economically viable*.

For all $k' \in S$, we determine its viability using Definition 2. If k' is nonviable, we remove it from the list and only show the economically viable options to a customer j . If there is no viable option, we reject this customer. The elimination process is also illustrated in Figure 4, where we assume to show an efficient list $S = \{(k_1, r_{k_1}), (k_2, r_{k_2}), (k_3, r_{k_3})\}$ to a customer j during fulfillment cycle τ . The figure depicts the marginal fulfillment cost for each option as a rectangle. At elapsed time t_1 , since options are viable by Definition 2, we show the entire list S . After the customer from location j chooses to order, the state increases by 1. Therefore, when another customer j arrives at time t_2 , the marginal fulfillment cost increases such that option k_3 is nonviable. We remove it from S and only show the customer options k_1 and k_2 . Later, at time t_3 , a new customer j arrives. The calculated marginal fulfillment cost of k_1 at time t_3 exceeds its revenue, so k_1 becomes nonviable and removed. Only k_2 is offered to the customer.

Lastly, we discuss the number of times $\mathcal{F}(x)$ needs to be solved. In the worst case, $\mathcal{F}(x)$ must be solved $2K$ times for each customer, as estimating the marginal fulfillment cost requires two evaluations per option. However, in practice, significantly fewer executions are typically needed (see §6.1.3 for numerical evidence). First, we only need to evaluate the option within a selected list S , which may only include a subset of K options. Second, the estimated marginal fulfillment cost only needs to be updated if the system state has changed. Third, we can skip marginal fulfillment cost estimation for some options under the following conditions. We fix a fulfillment cycle τ and demand location j . (1) If an option $k' \in S$ is nonviable at time t , then it must remain nonviable to the end of this fulfillment cycle. This is simply because the unfulfilled orders are monotonically increasing with the fulfillment cycle and the fulfillment cost is convex and increasing. (2) If an option $k' \in S$ is nonviable at time t , any fulfillment option $k \in S$ such that $r_{k,j} < r_{k',j}$ and $k < k'$ must also be nonviable. This is because the fulfillment option k cannot have a lower marginal fulfillment cost than option k' since otherwise, we can always fulfill an order with option k' earlier. (3) If an option $k' \in S$ is viable at time t , then any fulfillment option $k \in S$ such that $r_{k,j} \geq r_{k',j}$ and $k > k'$ must also be viable.

5.2. Ensuring Feasibility of Allocation Decisions

The fulfillment allocations $\bar{u}_{k,j,i,\tau}^*$ from **DET** may not be feasible for **OPT** for two reasons. First, they may violate constraint (1e) if the actual realized post-choice state becomes sufficiently small for some k, j , and τ such that $\sum_i \bar{u}_{k,j,i,\tau}^* > y_{k,j,\tau}$. Second, $\bar{u}_{k,j,i,\tau}^*$ may violate constraint (1f) because it is not necessarily integral.

To address these issues, we adopt a randomized rounding scheme with a threshold check. Specifically, we first scale $\bar{u}_{k,j,i,\tau}^*$ by a factor of $\frac{y_{k,j,\tau}}{\bar{y}_{k,j,\tau}}$ to account for the stochastic realization of actual demand. We denote this scaled action by $\hat{u}_{k,j,i,\tau} := \frac{y_{k,j,\tau}}{\bar{y}_{k,j,\tau}} \bar{u}_{k,j,i,\tau}^*$. We then attempt to determine the actual fulfillment allocation using the following randomized rounding function

$$u_{k,j,i,\tau} = h(\hat{u}_{k,j,i,\tau}) = \begin{cases} \lceil \hat{u}_{k,j,i,\tau} \rceil & \text{with probability } \hat{u}_{k,j,i,\tau} - \lfloor \hat{u}_{k,j,i,\tau} \rfloor \\ \lfloor \hat{u}_{k,j,i,\tau} \rfloor & \text{otherwise} \end{cases} \quad (7)$$

In line with the structural property described in Lemma 1, we adopt a threshold fulfillment strategy, in which unfulfilled orders $y_{k,j,\tau}$ are prioritized according to a predefined sequence over indices k and j , and are fulfilled sequentially within each cycle τ . Each warehouse i fulfills $u_{k,j,i,\tau}$ orders as long as it remains within its implicit capacity. Once the capacity is reached (by checking if $C'_i(w^\top u_{i,\tau}) w_j \geq b_j$), the warehouse becomes unavailable for the remainder of cycle τ . To ensure feasibility with respect to constraint (1e) after the randomized rounding, if there exists any pair (k, j) such that $\sum_i u_{k,j,i,\tau} > y_{k,j,\tau}$, we recursively identify a warehouse $i' \in \arg \max_i C'_i(w^\top u_{i,\tau})$ and reduce $u_{k,j,i',\tau}$ by one unit until the constraint is satisfied.

Algorithm 1: Economic Option, Rounded and Threshold Fulfillment Policy (ERT)**Input:** Customer arrival rate vector $\Lambda_{j,\tau}$, revenue parameters $r_{j,k}$, the fulfillment costfunction $C_i(\cdot)$, the penalty vector b_j , $\forall j \in [N]$, and the choice model $\pi_k^j(S)$.**Output:** $S^{\text{ERT}} = \{S_{j,\tau}(t)\}$ and $u^{\text{ERT}} = \{u_{k,j,i,t}\}$ 1 Construct the set of all efficient option lists $\mathcal{E}_j$ for all $j \in [N]$;2 Solve **DET** to obtain $\gamma_{j,\tau}^*(S)$ for all $S \in \mathcal{E}_j$, $\bar{u}_{k,j,i,\tau}^*$, and $\bar{y}_{k,j,\tau}$;3 **for** each cycle $\tau \in [T]$ **do**4 **for** a particular time t within each cycle τ **do**5 **if** a customer arrives from location j **then**6 Choose one option list $S \in \mathcal{E}_j$ with probability $\gamma_{j,\tau}^*(S)$;7 Set $S_{j,\tau}(t) \leftarrow S$;8 **for** each option $k' \in S$ **do**9 Remove that option if it is *economically nonviable* by Definition 2, i.e.,10 $S_{j,\tau}(t) \leftarrow S_{j,\tau}(t) \setminus \{k'\}$;11 **end**12 Offer a personalized delivery assortment $S_{j,\tau}(t)$ to the customer;13 **end**14 **end**15 At the end of each cycle τ , aggregate all orders and generate fulfillment decisions as follows;16 Set $U_{i,\tau} = 0$ for all i , and $\mathcal{I}_\tau = [M]$;17 **for** each order received from location $j \in [N]$ in descending order by b_j **do**18 **for** each delivery option $k = 1, 2, \dots, K$ **do**19 **for** each warehouse $i \in \mathcal{I}_\tau$ **do**20 Calculate $\hat{u}_{k,j,i,\tau} = \frac{\bar{u}_{k,j,i,\tau}^*}{\bar{y}_{k,j,\tau}} y_{k,j,\tau}$;

21 Determine fulfillment quantity by

$$u_{k,j,i,\tau} = \begin{cases} h(\hat{u}_{k,j,i,\tau}), & \text{if } \mathcal{C}'_i(U_{i,\tau} + w_j \hat{u}_{k,j,i,\tau}) w_j \leq b_j, \\ h(z) \text{ that solves } \mathcal{C}'_i(U_{i,\tau} + w_j z) w_j = b_j, & \text{if } \mathcal{C}'_i(U_{i,\tau}) w_j < b_j \leq \mathcal{C}'_i(U_{i,\tau} + w_j \hat{u}_{k,j,i,\tau}) w_j, \\ 0, & \text{otherwise.} \end{cases}$$

22 where $h(\cdot)$ is defined in Eq. (7);23 Update $U_{i,\tau} \leftarrow U_{i,\tau} + w_j u_{k,j,i,\tau}$;24 Remove warehouse i from $\mathcal{I}_\tau$ if $\mathcal{C}'_i(U_{i,\tau}) w_j \geq b_j$;25 **end**26 **while** $\sum_i u_{k,j,i,\tau} > y_{k,j,\tau}$ **do**27 Let $i' \in \arg \max_i \mathcal{C}'_i(w^\top u_{i,\tau})$ and set $u_{k,j,i',\tau} \leftarrow u_{k,j,i',\tau} - 1$;28 **end**29 **end**30 **end**

5.3. Performance Analysis and Implications

Combining the option list part and fulfillment part, we present the complete heuristic policy in Algorithm 1, and we name it the Economic option, Rounded and Threshold Fulfillment Policy (**ERT**). Note that by design **ERT** is a dynamic policy and depends on the pre-assortment state $x_{k,j,\tau}(t)$. Steps 1 and 2 in Algorithm 1 represent the offline phase in which the set of efficient assortments are constructed and **DET** is optimized. For each fulfillment cycle τ , steps 4 to 12 dynamically determine the personalized fulfillment options by eliminating economically nonviable options from the efficient assortment, and steps 14 to 27 provide the rounded and threshold fulfillment decisions.

Next, we upper bound the performance of our algorithm **ERT** in the following theorem:

THEOREM 1. *Let $\Lambda = \max_{j,\tau} \Lambda_{j,\tau}$, $\bar{w} = \max_j w_j$. Then, the optimality gap of **ERT** is bounded by*

$$V^*(x) - V^{\text{ERT}}(x) \leq \bar{w}\bar{C}\sqrt{MT}\sqrt{K\Lambda N}$$

Delegating the detailed proof to EC.1.5, here, we outline the main steps of our proof: First, we define a hypothetical policy (denoted by **HYP**) that utilizes the same option lists constructed by step 1 of **ERT** algorithm. However, **HYP** does three things differently. First, it does not eliminate economically nonviable assortments dynamically. Second, it uses continuous fulfillment decision $\frac{\bar{u}_{k,j,i,\tau}^* y_{k,j,\tau}}{\bar{y}_{k,j,\tau}}$ rather than the rounded version. Third, it disregards the implicit fulfillment capacity in Lemma 1 and may incur higher costs for fulfilling excessive orders. The benefit of employing **HYP** is that it follows the solution to **DET** in expectation and $u_{k,j,i,\tau}^{\text{HYP}}$ fulfills a deterministic proportion of arrived customers from cycles $\tau - (k - 1)$ to τ . Utilizing this property, we can show $V^D(x) - V^{\text{HYP}}(x) \leq \bar{w}\bar{C}\sqrt{MT}\sqrt{K\Lambda N}$ where $V^{\text{HYP}}(x)$ is the expected total profit of **HYP**. Then, we show that **ERT** cannot yield worse performance than **HYP**, i.e., $V^{\text{HYP}}(x) \leq V^{\text{ERT}}(x)$, using the sample path arguments. Together with Proposition 1, we can prove this theorem.

Implications: The performance bound characterized in Theorem 1 implies that the **ERT** policy is asymptotically optimal as either the number of demand location N or the maximum cycle arrival rate Λ becomes large. This is because, on the one hand, the optimality gap of **ERT** is growing sub-linearly with respect to N and Λ . On the other hand, if either N or Λ increases and the implicit fulfillment capacity also increases accordingly, the optimal profit $V^*(x)$ also increases linearly in N and Λ . As a result, the relative optimality gap converges to 0 as either N or Λ increases. This implication is particularly important in the context of online retailing, where the number of demand locations N often reaches the tens of thousands, while the number of warehouses M remains in the low single digits. Therefore, the fact that our algorithm's performance converges to optimality as the number of demand locations N increases shows its scalability for real-world applications.

Next, we conduct a numerical study to evaluate how the performance of our algorithm varies with the planning horizon T , the number of warehouses M , and the length of the fulfillment window

K . Delegating the details to Section EC.9, our numerical experiments show that the optimality gap increases linearly with T (see Figure EC.2), consistent with our theoretical bound in Theorem 1 and confirming that the bound is tight with respect to T . As the number of warehouses M increases, our numerical results show that the empirical optimality gap remains approximately constant (see Figure EC.3), which is tighter than the $\sqrt{M}$ theoretical bound in Theorem 1. We attribute this gap between theory and practice to two algorithmic features of **ERT** that the worst-case analysis does not fully exploit: (i) the threshold fulfillment rule (“T” in **ERT**) caps each warehouse’s contribution once marginal cost exceeds penalty b_j , limiting the realized variance below the partition bound; and (ii) the dynamic elimination of nonviable options (“E” in **ERT**) removes high-cost states before they occur, so the variance of **ERT** is strictly smaller than that of **HYP**, the policy through which the worst-case bound is derived. Finally, Theorem 1 establishes that the optimality gap grows sublinearly with the length of the fulfillment window K . To assess the implication of this result, we conduct a numerical study by varying K . Our results indicate diminishing returns as K increases—namely, the expected profit grows sub-linearly in K . Consequently, the performance of our algorithm remains stable as the fulfillment window increases. Overall, our proposed **ERT** policy has provable performance guarantees and demonstrates scalability with respect to both demand- and supply-side parameters.

6. Computational Experiments

In this section, we conduct a series of computational experiments to evaluate the performance of the **ERT** policy and generate additional insights. In §6.1, we analyze the computational complexity of the proposed algorithm using synthetic data. In §6.2, we calibrate an MNL choice model using real data from our industrial partner. By comparing the performance of **ERT** policy against several benchmark policies, we then examine the benefits of joint optimization. A key limitation of the MNL model is its assumption of homogeneous customer preferences across fulfillment options, which may not reflect real-world heterogeneity. To address this, in §6.3, we calibrate an MMNL model and repeat the same analysis. For additional numerical studies, we refer readers to EC.9. All simulations are conducted on a personal laptop equipped with an 8-core 3.2 GHz CPU and 16 GB of RAM. We use Gurobi 11.0.3 as the convex optimization solver.

6.1. Computation

6.1.1. Constructing efficient fulfillment option lists. We begin by evaluating the computational cost of constructing efficient option lists under an MC choice model using Algorithm 3. For each value of K ranging from 3 to 30, we randomly generate the choice model parameters and repeat the process 100 times. We report the average wall-clock time in milliseconds (ms), as well as the average ($|\mathcal{E}|_{\text{avg}}$) and worst-case ($|\mathcal{E}|_{\text{max}}$) number of efficient option lists in Table 1. It is

evident that all efficient option lists can be efficiently identified. $|\mathcal{E}|_{\text{avg}}$ significantly smaller than 2^K , highlighting that the decision space is significantly reduced for **DET**. Moreover, since all instances satisfy Assumption EC.2, the empirical worst-case number of efficient lists $|\mathcal{E}|_{\text{max}}$ is bounded by $K + 1$, supporting our theoretical result in Proposition 3.

Table 1 Constructing efficient option lists under an MC Choice Model.

K	3	6	9	12	15	18	21	24	27	30
wall-clock time (ms)	1.8	1.8	2.4	3.0	3.7	4.2	6.2	7.1	8.4	9.8
$ \mathcal{E} _{\text{avg}}$	2.9	4.2	4.8	5.5	6.1	6.7	7.3	7.3	7.9	8.1
$ \mathcal{E} _{\text{max}}$	4	6	8	8	10	10	10	10	14	12

6.1.2. Solving DET. After constructing efficient option lists, **DET** is solved over efficient option lists only. The number of decision variables comprises $\sum_j |\mathcal{E}_j| \cdot T$ list-selection variables $\gamma_{j,\tau}(S)$ and $K \cdot N \cdot M \cdot T$ fulfillment variables $u_{k,j,i,\tau}$. Under the MC choice model, this leads to number of decision variables being in the order of $\mathcal{O}(KNMT)$ —linear in every problem parameter. The constraint count is of the same order. For retail-relevant scales, this yields an LP that is well within the reach of any modern commercial solver. To further evaluate the advantage provided by Proposition 2, we stress test the wall-clock time for solving **DET** under various scenarios. Specifically, we consider three representative fulfillment network sizes: small ($M = 1$, $N = 50$), medium ($M = 3$, $N = 200$), and large ($M = 5$, $N = 500$), and cross them with the number of fulfillment options ($K \in \{3, 5, 10\}$) and the planning horizon ($T \in \{30, 60, 90\}$), producing 27 scenarios in total. Customer choice is generated under an MC model with random parameters. We report wall-clock time for efficient-list construction and for solving **DET** in Table 2. We observe that efficient-list construction is negligible relative to solving **DET**. Moreover, **DET** remains tractable across the full range of scenarios. Indeed, even for the largest instance considered ($M = 5$, $N = 500$, $K = 10$, $T = 90$, corresponding to a quarterly plan for a 5-warehouse network serving 500 customer locations), **DET** solves to optimality in under 45 minutes on commodity hardware.

6.1.3. Removing economically nonviable options. We also investigate the average per-period computation in the online phase, which dynamically removes nonviable options for all customers arriving within the period. As shown in Table 2, the per-period online computation stays at $\mathcal{O}(0.1)$ seconds throughout, well within the latency budget of any checkout environment.

6.2. Real Data 1: Homogeneous customer preference

In this section, we evaluate the performance of **ERT** and generate additional insights. Specifically, we examine the value of jointly optimizing assortments and fulfillment decisions by comparing **ERT** with benchmark policies across scenarios.

Table 2 Wall Clock Time for Implementing Algorithm 1

M	N	K	T	construction (sec)	DET solver (sec)	Online Decision (sec)
1	50	3	30	0.0002	2.8	0.0102
1	50	3	60	0.0002	8.3	0.0058
1	50	3	90	0.0003	10.4	0.0078
1	50	5	30	0.0003	2.0	0.0056
1	50	5	60	0.0003	9.1	0.0064
1	50	5	90	0.0003	31.6	0.0116
1	50	10	30	0.0005	4.5	0.0073
1	50	10	60	0.0004	14.0	0.0087
1	50	10	90	0.0005	13.6	0.0069
3	200	3	30	0.0002	17.7	0.0275
3	200	3	60	0.0002	55.9	0.0245
3	200	3	90	0.0002	116.3	0.0257
3	200	5	30	0.0003	27.2	0.0287
3	200	5	60	0.0003	91.6	0.0288
3	200	5	90	0.0003	198.7	0.0303
3	200	10	30	0.0006	93.0	0.0408
3	200	10	60	0.0004	104.5	0.0383
3	200	10	90	0.0004	314.7	0.0452
5	500	3	30	0.0003	85.1	0.0692
5	500	3	60	0.0002	320.5	0.0841
5	500	3	90	0.0002	1157.2	0.0654
5	500	5	30	0.0002	87.4	0.0836
5	500	5	60	0.0003	310.5	0.0814
5	500	5	90	0.0003	1119.0	0.0831
5	500	10	30	0.0004	320.0	0.1327
5	500	10	60	0.0004	877.5	0.1324
5	500	10	90	0.0005	2624.8	0.1339

Since computing the optimal policy is computationally intractable, based on Proposition 1, we use V^{DET} as an upper bound for V^{OPT} . We calculate percentage profit loss of a policy as its performance measure and compute this metric as follows: $(V^{\text{DET}} - V^{\text{policy}})/V^{\text{DET}} \times 100\%$, where V^{policy} is the expected profit obtained from some policy, and V^{DET} is the expected profit from **DET**.

6.2.1. Calibration Our industrial partner is a growing online grocery retailer in North America that began its business within east coast and is gradually expanding nationwide. Customers visit its website to shop for groceries. At the checkout, they have to choose either home-delivery or self-pickup and specify a date. Due to the retailer’s business scale and the operational limit, orders

are fulfilled once every week. Thus, customers select the week they wish to receive their orders. A shipping surcharge is displayed based on the selected fulfillment option, allowing customers to explore the cost differences among various options and choose their preferred ones. The transaction is complete once the fulfillment option is finalized and the payment is processed. The retailer fulfills orders from a single warehouse within the customer-selected fulfillment window, using either a local trucking company or a national carrier; both incur flat-rate costs.

The dataset comprises 6,511 transactions (including abandoned transactions) from 2,961 customers from January 1, 2023, to December 31, 2023. Our analysis focuses on 2,833 transactions from 21 cities, as the local trucking company and self-pickup service are available only to customers in these cities. Due to practical constraints, customers from other locations can only choose home delivery with the national carrier. The dataset contains detailed information on each order, including the number of products, total basket value, the selected fulfillment option, customer location, and whether the transaction was completed.

MNL choice model. The dataset allows us to calibrate an MNL choice model for customer behavior. To achieve that, we construct three fulfillment options based on the length of the fulfillment window: fulfill by this week with a mean surcharge of \$6.46 (Option 1), fulfill by next week with a mean surcharge of \$4.37 (Option 2), and fulfill after two weeks with a mean surcharge of \$1.69 (Option 3). We then estimate the attraction of each option v_k based on the proportion of customers who selected the corresponding time frame. Normalizing the attraction of abandonment to one ($v_0 = 1.00$), we find $v_1 = 3.47$, $v_2 = 10.48$, $v_3 = 0.12$. These results indicate that Option 2, which balances waiting time and fulfillment surcharge, is the most attractive to customers.

Baseline setup. We assume a homogeneous demand generation process: customers from different locations arrive following the same stationary Poisson process, purchase the same products, and choose a fulfillment option following the same choice model. The billable weight is set to one due to the lack of specific information.

We set $M = 1$ and $N = 21$, as the retailer fulfills orders from 21 cities using a single warehouse. Customers have three options regarding the fulfillment window, i.e., $K = 3$. We consider a planning horizon of one year, which includes 52 fulfillment cycles ($T = 52$). Each week, customers arrive at the system following a stationary Poisson process with an intensity of 19.5 ($\lambda_{j,\tau} = 19.5$ for all j and τ). Each order is valued \$140.00 ($p_j = 140.00$ for all j). Based on the discussion with our industrial partner, we set the percentage applied to each basket size as 5% ($\kappa = 0.05$). Together with the fulfillment surcharge, the revenue generated by each option is $r_{1,j} = 13.46$, $r_{2,j} = 11.37$, and $r_{3,j} = 8.69$, respectively.

Logistic carriers. The retailer employs two trucking companies to fulfill the orders. The local company has a fixed capacity and can fulfill up to 220 orders per week for \$8.00 per order. The

retailer turns to the national carrier for orders exceeding this number, which can provide sufficient capacity for \$16.00 per order. Hence, we can consider the national carrier to provide expedited fulfillment service (i.e., $b_j = 16$ for all j). Clearly, using the national carrier results in a profit loss due to the higher cost than the revenue. Regarding the fulfillment cost function, we use $C(z) = \max\{8z, \alpha(z - 220) + 1760\}$ with $\alpha > 16$. According to Lemma 1, we only fulfill up to 220 orders with the local company, which matches the business constraints. Moreover, we call this limit the *fulfillment capacity* and the *relative capacity* as the ratio of fulfillment capacity to the expected number of arriving customers ($\sum_j \lambda_{j,\tau} = 21 \times 19.5 = 409.5$). In the baseline case, the capacity is $\Xi = 220$, and the relative capacity is 53.7%.

6.2.2. Benchmark Policies. E-commerce practitioners often delegate the design of fulfillment options at checkout and the execution of fulfillment decisions to separate teams. As a result, each team may operate based on its own objectives without fully considering the system-wide impact. To emulate such operational practices, we introduce three benchmark policies, each characterized by whether or not the seller optimizes the option lists and/or the fulfillment decisions. Specifically, the Static-Myopic policy (**SM**) optimizes neither component. The Optimized List policy (**OL**) optimizes only the option lists, while the Optimized Fulfillment policy (**OF**) optimizes only the fulfillment decisions. In addition to these three main benchmarks, we introduce two supplementary policies to underscore the importance of removing nonviable fulfillment options and imposing threshold-based fulfillment decisions in the proposed **ERT** policy. Implementation details for all benchmark policies can be found in EC.8.1 and the additional numerical results can be seen in EC.9.4. In the main text, we focus on reporting the value of jointly optimizing both the fulfillment option lists and the assignment decisions across various scenarios.

6.2.3. Varying Customers Preferences. Our estimated choice model represents a market where most customers prefer a balance between fulfillment cost and waiting time. We simulate different customer preferences by swapping the attraction values among the choices. For instance, swapping the values v_1 and v_2 imitates a scenario where customers prefer faster fulfillment (Case 2). We also increase v_0 to 3.0 to simulate the scenarios when more customers abandon carts if they are dissatisfied with offered fulfillment options.

To summarize, we create 6 cases: Case 1 represents the actual market, Case 2 represents customers who prefer faster fulfillment, Case 3 represents customers who prefer cheaper fulfillment, and Cases 4 to 6 replicate Cases 1 to 3 with a higher abandonment probability. We report the percentage profit loss in Table 3. Note that a smaller profit loss indicates a better policy performance. Unsurprisingly, we observe that **ERT** outperforms all benchmark policies in all scenarios.

Case	(v_0, v_1, v_2, v_3)	ERT	OL	OF	SM
1	(1.00, 3.47, 10.48, 0.12)	2.4%	4.0%	24.0%	20.8%
2	(1.00, 10.48, 3.47, 0.12)	2.8%	4.0%	35.6%	32.4%
3	(1.00, 0.12, 3.47, 10.48)	1.3%	15.8%	65.5%	69.6%
4	(3.00, 3.47, 10.48, 0.12)	2.6%	4.0%	3.8%	3.9%
5	(3.00, 10.48, 3.47, 0.12)	2.7%	4.1%	24.1%	20.7%
6	(3.00, 0.12, 3.47, 10.48)	1.3%	19.2%	3.6%	25.7%

Table 3 Comparison of ERT with respect to various benchmark policies under different customer preferences

Upon closer examination, it is evident that demand management (via optimizing assortment decisions) is more crucial than capacity management (via optimizing fulfillment decisions) when fewer customers abandon their carts. This can be seen by comparing **ERT** to fixed lists policies in Cases 1 to 3. Moreover, when customers prefer fast fulfillment in Cases 1 and 2, many will choose Option 1. This necessitates fulfilling most orders by the end of this week, leaving little room for optimized fulfillment decisions to improve performance. As a result, all the additional benefits stem from dynamic assortment management, reflected in the smaller percentage profit losses of **ERT** compared to **OL**. In contrast, when customers prefer cheaper fulfillment in Case 3, more orders will have longer fulfillment windows. This scenario requires careful consideration of fulfillment capacity and active postponement of some fulfillments to later cycles. Thus, joint optimization leads to significant improvements.

Analyzing Cases 4 to 6, where the abandonment probability is higher, provides further insights. Customer dissatisfaction with fulfillment options leads to increased cart abandonment in these cases. This emphasizes the importance of offering attractive fulfillment options but simplifies the option list decisions. Specifically, in Cases 4 and 6, where customers less prefer fast fulfillment, fixed list policies achieve comparable, sometimes even better, performance than **ERT**. In contrast, **ERT** still performs better than the benchmarks, highlighting its consistency and robustness.

6.2.4. Varying Number of Locations. This section aims to investigate the benefits of adopting **ERT** when the retailer expands the business to a larger area. We vary the value of N from 5 to 50. In the meantime, we increase the local fulfillment quantity so that the relative fulfillment capacity remains unchanged. The capacity is set to $\frac{220}{21}N$. We report the results in Table 4, columns 2 to 5.

Consistent with previous results, **ERT** performs the best in all cases. Moreover, the profit loss decreases as N increases. This supports our theoretical performance guarantee in Theorem 1, implying that the retailer will benefit from expanding the business to a larger area, provided it can increase the fulfillment capacity accordingly. In contrast, the profit losses for **OF** and **SM** do not decrease as N increases.

	When customers prefer faster delivery				When customers prefer cheaper delivery			
N	ERT	OL	OF	SM	ERT	OL	OF	SM
5	6.1%	8.6%	27.9%	21.6%	3.8%	39.7%	69.6%	69.7%
10	4.0%	6.2%	25.8%	21.9%	2.4%	26.8%	67.7%	69.8%
20	2.8%	4.2%	24.2%	21.1%	1.8%	18.9%	65.9%	69.7%
30	2.3%	3.4%	23.6%	20.8%	1.4%	13.7%	64.9%	69.6%
40	1.9%	2.8%	23.1%	20.7%	0.9%	12.2%	64.5%	69.7%
50	1.8%	2.6%	22.9%	21.0%	1.1%	11.1%	64.4%	69.6%

Table 4 Comparison of ERT with various benchmark policies under different number of customer locations

Regarding **OL**, the profit loss also decreases similarly to **ERT**. To further understand whether the asymptotic optimality arises from joint optimization or assortment optimization, we repeat the test in a counterfactual market where customers are more price-sensitive and willing to wait (i.e., Case 3 in §6.2.3). The results, shown in Table 4, columns 6 to 9, reveal that better fulfillment decisions are crucial in this scenario. Therefore, we conclude that asymptotic optimality is a unique benefit of joint optimization.

6.2.5. Varying Fulfillment Capacity. This section examines the benefit of **ERT** when the seller negotiates different capacities with the local trucking company. We adjust the maximum fulfillment quantity from the local trucking company so that the relative capacity varies from 30% to 80%. We report the simulation results in Table 5.

Relative capacity	ERT	OL	OF	SM
80%	1.0%	9.4%	1.6%	1.5%
70%	2.5%	3.8%	8.5%	5.5%
60%	2.5%	3.9%	17.1%	14.2%
50%	3.0%	4.4%	29.2%	26.2%
40%	3.3%	5.0%	47.5%	44.5%
30%	4.0%	5.7%	77.8%	75.4%

Table 5 Performance comparison of ERT with benchmark policies under different fulfillment capacity

As observed, **ERT** outperforms the benchmarks in all the cases. Interestingly, when capacity is relatively insufficient (less than or equal to 70%), meaning the availability of cheaper logistics is limited, the retailer must control the quantity of realized demand with optimized assortment decisions. If too many orders are realized (as with **OF** and **SM**), the expected profit will be lower due to excessive usage of the more expensive carrier. However, the results change drastically when the capacity is relatively sufficient (80%). **OL** tends to accept more orders with a lower surcharge than the other three policies, which backfires. It has to use the national carrier more to fulfill those orders, resulting in a profit loss. This interesting finding emphasizes the necessity to align demand management with capacity management.

6.3. Real Data 2: Value of Personalization

In this section, we show the value of personalized option lists via consumer segmentation. Specifically, we segment consumers into three types based on basket value, as basket value affects the shipping surcharge and, in turn, consumer preferences. The basket values are determined using the average order values within specific percentiles: from the minimum to the 50th percentile for low-value consumers, from the 50th to the 80th percentile for medium-value consumers, and from the 80th percentile to the maximum for high-value consumers. The basket values for low-, medium-, and high-valued consumers are 64.06, 132.31, and 249.70 dollars, respectively.

We then calibrate an MNL choice model for each consumer segment. According to §6.2, most orders are fulfilled within two weeks. Thus, we identify fulfillment options with two windows: a fast option (fulfilled this week) and a slow option (fulfilled next week). We calibrate a basic attraction choice model for each consumer segment, similar to the previous section. Regarding the fulfillment surcharge, we aggregate the dataset to the segment-fulfillment-option level and compute the average surcharge. The attractions, surcharges, and, hence, the revenue for each fulfillment option and consumer segment are shown in Table 6. The rest of the calibration follows the baseline setup in § 6.2.

	Attraction		Surcharge (\$)		Revenue (\$)	
	Fast	Slow	Fast	Slow	Fast	Slow
Low	1.77	4.93	11.49	8.69	14.69	11.89
Medium	5.46	24.32	5.62	3.58	12.24	10.20
High	3.36	21.27	8.16	5.87	20.65	18.36

Table 6 Calibrated fulfillment surcharges and attractions for each option for each consumer segment.

To show the value of personalization, we consider two benchmarks. The first one is called the Common-menu policy, which discard personalization while preserving dynamic assortment. This policy is analogous to the demand management approach in Lei et al. (2018), who apply the same pricing menu to all consumers. Specifically, Common-menu is identical to **ERT** except that it requires the platform to show the same option lists (with the same probabilities) to all consumers in each period. It is implemented by adding a constraint $\gamma_{j,\tau}(S) = \gamma_\tau(S)$ for all j and τ in **DET**. The second benchmark is to show all options (Display-all), which imposes no demand management and is analogous to the exogenous-demand setting of Zhou et al. (2025). Meanwhile, we vary the relative capacity from 30% to 100% to investigate the impact of logistic capacity on the value of personalization.

The results in Table 7 demonstrate that personalization yields significant profit improvements, particularly under tight capacity. The mechanism is selective restriction: when fulfillment capacity

Relative capacity	ERT	Common-menu	Display-all
100%	0.0%	1.9%	8.7%
70%	1.3%	3.0%	41.1%
50%	1.5%	16.7%	67.4%
30%	2.0%	28.4%	107.8%

Table 7 Value of personalization under segmented customer preferences and various capacity constraints.

is scarce, **ERT** steers low-value consumers toward the slow option (preserving fast-delivery capacity for high-value consumers who generate more revenue per order), while the common-menu policy cannot differentiate and must apply the same restriction uniformly. As capacity tightens, this selective steering becomes more valuable, and the gap between **ERT** and common-menu widens. The display-all policy, which imposes no demand management, incurs substantially larger losses.

7. Conclusion

In collaboration with one of the online grocery platforms in North America, this paper investigates the fulfillment optimization problem with personalized delivery choices. A primary challenge for the online retailer is to decide which delivery options to offer, given customer heterogeneity for differentiated delivery options, and how to fulfill these orders with multiple logistic providers. To the best of our knowledge, our paper is among the first in the literature to address this problem.

To address this challenge, we developed an algorithm called **ERT** that runs in two phases. In the first (offline) phase, considering the customer heterogeneity for differentiated delivery options, **ERT** generates a set of efficient option lists that maximizes the expected revenue while achieving the best possible utilization of fulfillment resources. It then calculates the optimal certainty equivalent fulfillment decisions using the those lists. In the second (online) phase, **ERT** iterates between the personalization and fulfillment stages for the planning cycle. At the delivery personalization stage, **ERT** starts with an efficient option list and dynamically removes the economically nonviable options based on the customer’s location and the current state of unfulfilled orders. At the fulfillment stage, the algorithm aggregates all the orders and assigns them to the appropriate warehouses according to the certainty equivalent plan as much as possible. **ERT** is proved to have a worst-case optimality gap that grows sub-linearly in the number of demand locations and demand arrival rate. This implies that **ERT** is asymptotically optimal as the customer network expands, which corresponds to a typical case faced by many online retailers, including our industrial partner. We performed two series of simulations and demonstrated the value of joint optimization and the value of option personalization.

Our numerical experiments reveal several managerial insights. First, while optimizing personalized fulfillment options and assignments are both crucial, their significance varies across different scenarios. Specifically, when customers prioritize faster fulfillment, demand management is more

critical, whereas capacity management plays a minor role. Conversely, capacity management gains importance when customers prefer more cost-effective fulfillment options. Second, online retailers who serve larger areas and/or have more customers from each location benefit more from adopting the **ERT** policy as it is asymptotically optimal. Third, demand management takes precedence when fulfillment capacity is limited, but the focus shifts toward capacity management as capacity increases.

Finally, we would like to conclude by exploring potential avenues for future research. Firstly, our paper assumes that efficient option lists are constructed based on a known choice model. However, in reality, the parameters of a choice model need to be probed and learned jointly, along with the optimal lists and fulfillment decisions. This can be conducted within an online learning framework. Secondly, we assume flat-rate fulfillment costs for all logistic providers. Future research could extend this framework by considering a different fulfillment cost structure. For example, the fulfillment cost consists of fixed and variable parts, which incentivizes operations to ship a large number of orders to pickup locations to lower the average per-order cost. Indeed, our industrial partner is actively forming new pickup partnerships with local convenience stores, believing it is crucial to improve service further and reduce operational costs. We firmly believe that our paper spurs further research on the above extensions, offering valuable insights to the managers of online retailers in improving their customer satisfaction through delivery personalization while minimizing the cost of fulfillment.

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Electronic Companion

EC.1. Proofs

EC.1.1. Lemma 1

Proof Since we want to show this lemma hold with an arbitrary display strategy $S_{j,\tau}(t)$, without loss of generality, we can consider realized demand $D_{k,j,\tau}$ as exogenous with an arbitrary distribution. Therefore, **OPT** is equivalent to a cost minimization problem, which is convex optimization with linear constraints. In the meantime, we can simplify the DP value function to $V_\tau(y_\tau)$. It is easy to see that $\frac{\partial V_\tau(y_\tau)}{\partial y_{k,j,\tau}} \leq b_j$. This is because penalize all orders are feasible.

Define $\delta_{1,j,\tau} := b_j$ and $\delta_{k,j,\tau} := \frac{\partial V_{\tau+1}(y_{\tau+1})}{\partial y_{k-1,j,\tau+1}}$ for $k > 1$. Consequently, $\delta_{k,j,\tau} \leq b_j$. Define u_τ^* the optimal fulfillment decisions in cycle τ . We fix an arbitrary fulfillment cycle τ . There must exist dual variables $\lambda^* = (\lambda_{k,j}^* : k \in [K], j \in [N])$, together with u_τ^* , solving the following KKT conditions.

$$\begin{aligned} C'_i(w^\top u_{i,\tau})w_j - \delta_{k,j,\tau} + \lambda_{k,j} &\geq 0, \quad \forall k, j, i \\ y_{k,j,\tau} - \sum_{i=1}^M u_{k,j,i,\tau} &\geq 0, \quad \forall k, j \\ \lambda_{k,j} \left(y_{k,j,\tau} - \sum_{i=1}^M u_{k,j,i,\tau} \right) &= 0, \quad \forall k, j \\ u_{k,j,i,\tau} (C'_i(w^\top u_{i,\tau})w_j - \delta_{k,j,\tau} + \lambda_{k,j}) &= 0, \quad \forall k, j, i \\ \lambda_{k,j} &\geq 0, \quad \forall k, j \\ u_{k,j,i,\tau} &\geq 0, \quad \forall k, j, i \end{aligned}$$

For those orders indexed by j and k such that $\delta_{k,j,\tau} \geq \bar{C}w_j$, we have $C'_i(w^\top u_{i,\tau}^*)w_j < \bar{C}w_j \leq \delta_{k,j,\tau} \leq b_j$. Meanwhile, we must fulfill all those orders immediately since $C'_i(w^\top u_{i,\tau}^*) - \delta_{k,j,\tau} < 0$ implies $\lambda_{k,j,\tau}^* > 0$. Therefore, we must have $y_{k,j,\tau} = \sum_i u_{k,j,i,\tau}^*$. For the rest orders such that $\delta_{k,j,\tau} \leq \bar{C}w_j$, if they are ever fulfilled by warehouse i , meaning $u_{k,j,i,\tau}^* > 0$, we must have $C'_i(w^\top u_{i,\tau}^*)w_j - \delta_{k,j,\tau} + \lambda_{k,j,\tau}^* = 0$. This implies $C'_i(w^\top u_{i,\tau}^*)w_j \leq \delta_{k,j,\tau} \leq b_j$. $\square$

EC.1.2. Proposition 1

Before proving Proposition 1, we present a technical lemma and its proof.

LEMMA EC.1. *Let $f(x)$ be concave in x , and c is a vector of constant. Then, the following parametric optimization is concave in β*

$$P(\beta) := \max_x f(x) + c^\top \beta \quad s.t. \quad Ax \leq \beta$$

Proof Consider two parameters β_1 and β_2 , and x_1^* and x_2^* are maximizers for $P(\beta_1)$ and $P(\beta_2)$, respectively. Let $\lambda \in [0, 1]$. Note that

$$A(\lambda x_1^*) + A((1 - \lambda)x_2^*) \leq \lambda \beta_1 + (1 - \lambda)\beta_2$$

which implies $x = \lambda x_1^* + (1 - \lambda)x_2^*$ is feasible for $P(\lambda \beta_1 + (1 - \lambda)\beta_2)$. Therefore

$$\begin{aligned} P(\lambda \beta_1 + (1 - \lambda)\beta_2) &\geq f(\lambda x_1^* + (1 - \lambda)x_2^*) + c^\top (\lambda \beta_1 + (1 - \lambda)\beta_2) \\ &\geq \lambda f(x_1^*) + (1 - \lambda)f(x_2^*) + c^\top (\lambda \beta_1 + (1 - \lambda)\beta_2) \\ &= \lambda(f(x_1^*) + c^\top \beta_1) + (1 - \lambda)(f(x_2^*) + c^\top \beta_2) \\ &= \lambda P(\beta_1) + (1 - \lambda)P(\beta_2) \end{aligned}$$

The second inequality holds since f is a concave function. $\square$

Proof of Proposition 1 Consider an arbitrary feasible joint strategy $\rho = (S^\rho, u^\rho)$ to **OPT**, where S^ρ is the assortment decisions and u^ρ is fulfillment assignments, and denote $D = (D_{k,j,\tau} \in \mathbb{Z} : k \in [K], j \in [N], \tau \in [T])$ a sample path of realized demand generated under S^ρ . We consider the following deterministic programming conditioning on D ,

$$\begin{aligned} V(x \mid D) &= \max_{\bar{u}} \sum_{\tau} \sum_j \sum_k r_{k,j} D_{k,j,\tau} \\ &\quad - \sum_{\tau} \left(\sum_i C_i(w^\top \bar{u}_{i,\tau}) + \sum_j b_j \left(\bar{y}_{1,j,\tau} - \sum_i \bar{u}_{1,j,i,\tau} \right) \right) - \sum_{k,j} b_j \bar{x}_{k,j,T+1} \\ \text{s.t. } \quad &\bar{x}_{k,j,1} = x_{k,j}, \quad \forall k, j \\ &\bar{y}_{k,j,\tau} = \bar{x}_{k,j,\tau} + D_{k,j,\tau}, \quad \forall k, j, \tau \\ &\bar{x}_{k,j,\tau+1} = \bar{y}_{k+1,j,\tau} - \sum_i \bar{u}_{k+1,j,i,\tau}, \quad \forall k < K, j, \tau \\ &\bar{x}_{K,j,\tau+1} = 0, \quad \forall j, \tau \\ &\sum_i \bar{u}_{k,j,i,\tau} \leq \bar{y}_{k,j,\tau}, \quad \forall k, j, \tau \\ &\bar{u}_{k,j,i,\tau} \geq 0 \quad \forall k, j, i, \tau \end{aligned}$$

Note that $V(x \mid D)$ is a pure fulfillment optimization problem. We wish to show

$$V^*(x) \leq \mathbb{E}[V(x \mid D)] \leq V(x \mid \mathbb{E}[D]) \leq V^D(x)$$

The first inequality. Since ρ is feasible to **OPT**, u^ρ must be a feasible fulfillment assignment when the realized demand is D . Hence, u^ρ is feasible to $V(x \mid D)$. Taking expectations yields that the expected profit under joint strategy ρ is upper bounded by $\mathbb{E}[V(x \mid D)]$. Since ρ is arbitrary, the relation must also hold for the optimal joint policy ρ^* .

The second inequality. we claim that $V(x \mid D)$ is concave in D . To show that, first note that the objective function in $V(x \mid D)$ is concave in x and that the constraints are linear. The concavity follows the technical Lemma EC.1. Then, by Jensen's inequality, we have $\mathbb{E}[V(x \mid D)] \leq V(x \mid \mathbb{E}[D])$.

The last inequality. It is sufficient to show that we can construct a feasible assortment policy $\gamma_{j,\tau}(S)$ for $V^D(x)$ such that $\sum_S \Lambda_{j,\tau} \gamma_{j,\tau}(S) \pi_k^j(S) = \mathbb{E}[D_{k,j,\tau}]$ for all k, j, τ . Let $\bar{u}^*$ an optimal fulfillment assignment to $V(x \mid \mathbb{E}[D])$. Then $(\gamma_{j,\tau}(S), \bar{u}^*)$ is a feasible joint strategy for $V^D(x)$, implying $V^D(x)$ serves as an upper bound. The constructed assortment policy $\gamma_{j,\tau}(S)$ should solve the following problem for all j and τ :

$$\begin{aligned} \sum_S \gamma_{j,\tau}(S) \Lambda_{j,\tau} \pi_k^j(S) &= \mathbb{E}[D_{k,j,\tau}], \quad \forall k \\ \sum_S \gamma_{j,\tau}(S) &= 1 \\ \gamma_{j,\tau}(S) &\geq 0, \quad \forall S \end{aligned}$$

The above system of equations can be compressed as

$$A\gamma = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \Lambda_{j,\tau} \pi_1^j(S_1) & \Lambda_{j,\tau} \pi_1^j(S_2) & \cdots & \Lambda_{j,\tau} \pi_1^j(S_{|\mathcal{N}|}) \\ \vdots & \vdots & \vdots & \vdots \\ \Lambda_{j,\tau} \pi_K^j(S_1) & \Lambda_{j,\tau} \pi_K^j(S_2) & \cdots & \Lambda_{j,\tau} \pi_K^j(S_{|\mathcal{N}|}) \end{bmatrix} \begin{bmatrix} \gamma_{j,\tau}(S_1) \\ \gamma_{j,\tau}(S_2) \\ \vdots \\ \gamma_{j,\tau}(S_{|\mathcal{N}|}) \end{bmatrix} = \begin{bmatrix} 1 \\ \mathbb{E}[D_{1,j,\tau}] \\ \vdots \\ \mathbb{E}[D_{K,j,\tau}] \end{bmatrix}$$

It is easy to show that the matrix A has a full-row rank, which guarantees a solution of γ . For example, let $\mathcal{N} = (S_1, S_2, \dots, S_{|\mathcal{N}|})$. Moreover, $S_1 = \emptyset, S_2 = \{1\}, S_3 = \{2\}, \dots, S_{K+1} = \{K\}$. We can construct such a solution as follows: $\gamma_{j,\tau}(S_1) = 1 - \sum_{k=1}^K \frac{\mathbb{E}[D_{k,j,\tau}]}{\Lambda_{j,\tau} \pi_k^j(S_{k+1})}$, $\gamma_{j,\tau}(S_{k+1}) = \frac{\mathbb{E}[D_{k,j,\tau}]}{\Lambda_{j,\tau} \pi_k^j(S_{k+1})}$ for $k \in [K]$, and $\gamma_{j,\tau}(S) = 0$ otherwise. We still need to verify that $\gamma_{j,\tau}(S_1) \geq 0$. Indeed, it holds by the definition of $D_{k,j,\tau}$. $\square$

EC.1.3. Proposition 2

We present the necessary and sufficient condition for an option list to be efficient from Talluri and Van Ryzin (2004) as a technical lemma before proving Proposition 2.

LEMMA EC.2. *For any customers j , a fulfillment option list S is efficient if and only if there exists some real number $\theta_j \geq 0$ such that S is the optimal solution to*

$$\max_{S'} R_j(S') - \theta_j Q_j(S')$$

Proof of Proposition 2 The proof consists of two steps: First, we show that if $\gamma_{j,\tau}^*(S) > 0$, then S solves the following problem:

$$\max_{S'} R_j(S') + \sum_k \theta_{k,j} \pi_k^j(S')$$

for some $\theta_j = (\theta_{1,j}, \dots, \theta_{K,j})$. Next, we show that $\theta_{k,j}$ are identical and *non-positive* for all k . Then, by Lemma EC.2, S must be efficient.

Note that **DET** is feasible and bounded, implying it has a finite optimum with some $\gamma_{j,\tau}^*$ and $\bar{u}_{k,j,i,\tau}^*$. Moreover, strong duality holds since **DET** is a convex optimization with linear constraints.

Let $\theta = (\theta_{k,j,\tau} : k \in [K], j \in [N], \tau \in [T])$, $\eta = (\eta_{k,j,\tau} : k \in [K-1], j \in [N], \tau \in [T])$, $\xi = (\xi_{k,j,\tau} \geq 0 : k \in [K], j \in [N], \tau \in [T])$, $\sigma = (\sigma_{j,\tau} : j \in [N], \tau \in [T])$, $\phi = (\phi_{k,j,i,\tau} \geq 0 : k \in [K], j \in [N], i \in [M], \tau \in [T])$, and $\mu = (\mu_{j,\tau,S} \geq 0 : j \in [N], \tau \in [T], S \in \mathcal{N})$ be the dual variables for constraints (3b), (3c), (3e), (3f), (3g), and (3h), respectively. We indicate the optimal dual variables with superscript $*$.

The Lagrangian function can be formulated as

$$\begin{aligned} & \mathcal{L}(\gamma, \bar{u}, \bar{x}, \bar{y}, \theta, \eta, \xi, \sigma, \phi, \mu) \\ &= \sum_{\tau,j} \sigma_{j,\tau} + \sum_{\tau,j,S} \left(\Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_k \theta_{k,j,\tau} \pi_k^j(S) - \sigma_{j,\tau} + \mu_{j,\tau,S} \right) \gamma_{j,\tau}(S) \\ & \quad - \sum_{\tau,i} \left(C_i(w^\top \bar{u}_{i,\tau}) - \sum_j b_j \bar{u}_{1,j,i,\tau} + \sum_{k < K,j} \eta_{k,j,\tau} \bar{u}_{k+1,j,i,\tau} + \sum_{k,j} \xi_{k,j,\tau} \bar{u}_{k,j,i,\tau} - \sum_{k,j} \phi_{k,j,i,\tau} \bar{u}_{k,j,i,\tau} \right) \\ & \quad - \sum_{\tau,j} \left(b_j \bar{y}_{1,j,\tau} + \sum_k \theta_{k,j,\tau} \bar{y}_{k,j,\tau} - \sum_{k < K} \eta_{k,j,\tau} \bar{y}_{k+1,j,\tau} - \sum_k \xi_{k,j,\tau} \bar{y}_{k,j,\tau} \right) \\ & \quad + \sum_{k,j,\tau} \theta_{k,j,\tau} \bar{x}_{k,j,\tau} - \sum_{k < K,j,\tau} \eta_{k,j,\tau} \bar{x}_{k,j,\tau+1} - \sum_{k,j} b_j \bar{x}_{k,j,T+1} \end{aligned}$$

Treating $\bar{x}$ and $\bar{y}$ as primal decision variables, the dual problem can be formulated as follows.

$$\begin{aligned} \mathcal{D} &= \min_{\theta, \eta, \xi, \sigma, \phi, \mu} \max_{\gamma, \bar{u}, \bar{x}, \bar{y}} \mathcal{L}(\gamma, \bar{u}, \bar{x}, \bar{y}, \theta, \eta, \xi, \sigma, \phi, \mu) \\ &= \min_{\theta, \eta, \xi, \sigma, \phi, \mu} \left\{ \sum_{\tau,j} \sigma_{j,\tau} + \underbrace{\max_{\gamma} \sum_{\tau,j,S} \left(\Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_k \theta_{k,j,\tau} \pi_k^j(S) - \sigma_{j,\tau} + \mu_{j,\tau,S} \right) \gamma_{j,\tau}(S)}_{\text{I}} \right. \\ & \quad + \underbrace{\max_{\bar{u}} - \sum_{\tau,i} \left(C_i(w^\top \bar{u}_{i,\tau}) - \sum_j b_j \bar{u}_{1,j,i,\tau} + \sum_{k < K,j} \eta_{k,j,\tau} \bar{u}_{k+1,j,i,\tau} + \sum_{k,j} \xi_{k,j,\tau} \bar{u}_{k,j,i,\tau} - \sum_{k,j} \phi_{k,j,i,\tau} \bar{u}_{k,j,i,\tau} \right)}_{\text{II}} \\ & \quad \left. + \underbrace{\max_{\bar{y}} - \sum_{\tau,j} \left(b_j \bar{y}_{1,j,\tau} + \sum_k \theta_{k,j,\tau} \bar{y}_{k,j,\tau} - \sum_{k < K} \eta_{k,j,\tau} \bar{y}_{k+1,j,\tau} - \sum_k \xi_{k,j,\tau} \bar{y}_{k,j,\tau} \right)}_{\text{III}} \right\} \end{aligned}$$

$$\left. \underbrace{+ \max_{\bar{x}} \sum_{k,j,\tau} \theta_{k,j,\tau} \bar{x}_{k,j,\tau} - \sum_{k < K,j,\tau} \eta_{k,j,\tau} \bar{x}_{k,j,\tau+1} - \sum_{k,j} b_j \bar{x}_{k,j,T+1}}_{\textcircled{\text{IV}}} \right\}$$

We take the dual variables at their optimal values for the subsequent analysis.

Part One: For $\textcircled{\text{I}}$, for all τ , j , and S' , according to first-order optimality condition, we must have

$$\Lambda_{j,\tau} R_j(S') + \Lambda_{j,\tau} \sum_k \theta_{k,j,\tau}^* \pi_k^j(S') - \sigma_{j,\tau}^* + \mu_{j,\tau,S}^* = 0$$

implying

$$\Lambda_{j,\tau} R_j(S') + \Lambda_{j,\tau} \sum_k \theta_{k,j,\tau}^* \pi_k^j(S') - \sigma_{j,\tau}^* = -\mu_{j,\tau,S}^* \leq 0 \quad (\text{EC.1})$$

By the complementary slackness condition $\mu_{j,\tau,S}^* \gamma_{j,\tau}^*(S) = 0$, if $\gamma_{j,\tau}^*(S) > 0$, we must have $\mu_{j,\tau,S}^* = 0$, implying

$$\Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_k \theta_{k,j,\tau}^* \pi_k^j(S) - \sigma_{j,\tau}^* = 0$$

Subtracting the above equation from (EC.1) yields

$$R_j(S) + \sum_k \theta_{k,j,\tau}^* \pi_k^j(S) \geq R_j(S') + \sum_k \theta_{k,j,\tau}^* \pi_k^j(S')$$

which indicates that if $\gamma_{j,\tau}^*(S) > 0$ for some j and τ , then S is an maximizer for $\max_{S'} R_j(S') + \sum_k \theta_{k,j,\tau}^* \pi_k^j(S')$. This completes the first step.

Part two: we show $\theta_{k,j,\tau}^*$ are identical for all k and that $\theta_{k,j,\tau}^* \leq 0$. To do so, we first identify the relationships among dual variables. We start with $\textcircled{\text{III}}$.

$$\textcircled{\text{III}} = \max_{\bar{y}} \sum_{\tau,j} (-b_j - \theta_{1,j,\tau}^* + \xi_{1,j,\tau}^*) \bar{y}_{1,j,\tau} + \sum_{\tau,j,k > 1} (-\theta_{k,j,\tau}^* + \eta_{k-1,j,\tau}^* + \xi_{k,j,\tau}^*) \bar{y}_{k,j,\tau}$$

The first-order optimality condition implies

$$-b_j - \theta_{1,j,\tau}^* + \xi_{1,j,\tau}^* = 0, \quad \forall j, \tau \quad (\text{EC.2})$$

$$-\theta_{k,j,\tau}^* + \eta_{k-1,j,\tau}^* + \xi_{k,j,\tau}^* = 0, \quad \forall j, \tau, k > 1 \quad (\text{EC.3})$$

Next, we examine $\textcircled{\text{IV}}$. Since $\bar{x}_1$ is given and $\bar{x}_{K,j,\tau+1} = 0$ for all τ , we have the following

$$\textcircled{\text{IV}} = \sum_{k,j} \theta_{k,j,1} \bar{x}_{k,j,1} + \max_{\bar{x}_2, \dots, \bar{x}_{T+1}} \left\{ \sum_{\tau > 1} \sum_{k < K,j} (\theta_{k,j,\tau}^* - \eta_{k,j,\tau-1}^*) \bar{x}_{k,j,\tau} + \sum_{k < K,j} (-b_j - \eta_{k,j,T}^*) \bar{x}_{k,j,T+1} \right\}$$

which yields the following relations:

$$\eta_{k,j,T}^* = -b_j \quad \forall k < K, j \quad (\text{EC.4})$$

$$\theta_{k,j,\tau}^* = \eta_{k,j,\tau-1}^* \quad \forall k < K, j, \tau > 1 \quad (\text{EC.5})$$

(EC.3) and (EC.5) implies

$$\theta_{k,j,\tau}^* = \theta_{k-1,j,\tau+1}^* + \xi_{k,j,\tau}^* \quad (\text{EC.6})$$

(EC.2), (EC.3), and (EC.4) imply

$$\theta_{k,j,T}^* = -b_j + \xi_{k,j,T}^* \geq -b_j$$

for all k and j . The inequality holds since $\xi_{k,j,\tau}^* \geq 0$. By backward induction with (EC.6) and the relation in (EC.2), we can further show that, for all k, j , and τ

$$\theta_{k,j,\tau}^* \geq -b_j \quad (\text{EC.7})$$

Lastly, we apply the first-order optimality condition to $\textcircled{\text{II}}$. For all τ and i ,

$$\begin{aligned} C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - b_j + \xi_{1,j,\tau}^* - \phi_{1,j,i,\tau}^* &= 0 \\ C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + \eta_{k-1,j,\tau}^* + \xi_{k,j,\tau}^* - \phi_{k,j,i,\tau}^* &= 0, \quad \forall k > 1 \end{aligned}$$

Using (EC.2) and (EC.3), above are equivalent to, for all k, j, i, τ ,

$$C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + \theta_{k,j,\tau}^* - \phi_{k,j,i,\tau}^* = 0 \quad (\text{EC.8})$$

Using (EC.7) and $\phi_{k,j,i,\tau}^* \geq 0$, we conclude that $\theta_{k,j,\tau}^*$ satisfies the follow constraint.

$$\theta_{k,j,\tau}^* \geq -\min\{b_j, \min_i C'_i(w^\top \bar{u}_{i,\tau}^*) w_j\} \quad (\text{EC.9})$$

In addition, the complementary slackness condition $\bar{u}_{k,j,i,\tau}^* \phi_{k,j,i,\tau}^* = 0$ implies

$$\bar{u}_{k,j,i,\tau}^* (C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + \theta_{k,j,\tau}^*) = 0 \quad (\text{EC.10})$$

We can interpret $-\theta_{k,j,\tau}^*$ as the marginal value for fulfilling an order indexed by k, j , and τ . When $C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - (-\theta_{k,j,\tau}^*) > 0$, it implies that the weighted marginal fulfillment cost of warehouse i is higher than the marginal benefit of fulfillment. As such, it is not optimal to fulfill such orders from warehouse i , i.e., $\bar{u}_{k,j,i,\tau}^* = 0$. On the other hand, when $C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - (-\theta_{k,j,\tau}^*) = 0$, we can fulfill those orders from warehouse i , i.e., $\bar{u}_{k,j,i,\tau}^* \geq 0$.

Next, we will show that $\theta_{k,j,\tau}^*$ are identical and *negative* for all k . We arbitrarily fix the index j and τ . We need to consider two cases.

Case 1, When some orders from j are fulfilled. We define an index set $\mathcal{K}_{j,i,\tau} = \{k \in [K] : \bar{u}_{k,j,i,\tau}^* > 0\}$. Case 1 implies $\cup_i \mathcal{K}_{j,i,\tau} \neq \emptyset$. We consider the following three sub-cases.

Case 1.1: Orders with different fulfillment window are fulfilled by the *same* warehouse. That is, if there exists an index i such that for all $k, k' \in \mathcal{K}_{j,i,\tau}$, $k \neq k'$,

$$\theta_{k,j,\tau}^* = -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j = \theta_{k',j,\tau}^*$$

The relationship holds according to (EC.10). The above relationship indicates that if orders with different windows are fulfilled from the same warehouse, they must have the same marginal value, which equals the negative weighted marginal fulfillment cost of that fulfilled warehouse.

Case 1.2: Orders with different fulfillment window are fulfilled by *different* warehouses. That is, if there exists i, i', k , and k' such that $i \neq i'$, $k \in \mathcal{K}_{j,i,\tau}$, $k' \in \mathcal{K}_{j,i',\tau}$, $k \neq k'$,

$$\theta_{k,j,\tau}^* \geq -C'_{i'}(w^\top \bar{u}_{i',\tau}^*)w_j = \theta_{k',j,\tau}^*$$

$$\theta_{k',j,\tau}^* \geq -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j = \theta_{k,j,\tau}^*$$

The inequality follows (EC.8), and the equality follows (EC.10). This implies

$$\theta_{k,j,\tau}^* = -C'_{i'}(w^\top \bar{u}_{i',\tau}^*)w_j = -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j = \theta_{k',j,\tau}^*$$

Cases 1 and 2 imply that if orders with fulfillment window k are ever fulfilled, they have the same marginal value. In other words, $\theta_{k,j,\tau}^*$ takes the same value for all $k \in \cup_i \mathcal{K}_{j,i,\tau}$. Moreover, any warehouse that fulfills the order from location j during cycle τ has the same weighted marginal fulfillment cost. To simplify the notation for the following discussion, we denote those warehouses by an index set $\mathcal{I}_{j,\tau} = \{i \in [M] : \sum_{k'} \bar{u}_{k',j,i,\tau}^* > 0\}$. We denote this weighted marginal fulfillment cost as C' . That is, $C' = C'_i(w^\top \bar{u}_{i,\tau}^*)w_j$ for all $i \in \mathcal{I}_{j,\tau}$ and $\theta_{k,j,\tau}^* = -C'$ for all $k \in \cup_i \mathcal{K}_{j,i,\tau}$.

Case 1.3: Orders with fulfillment window k are not fulfilled (i.e., exists some k such that $k \notin \cup_i \mathcal{K}_{j,i,\tau}$). We assume there exists at least one $k \notin \cup_i \mathcal{K}_{j,i,\tau}$ such that $\theta_{k,j,\tau}^* > -C'$ since otherwise all $\theta_{k',j,\tau}^*$ are identical for all k' . Our goal is to construct another set of dual variables $\hat{\theta}, \hat{\eta}, \hat{\xi}, \hat{\sigma}$, and $\hat{\phi}$, and show that $(\gamma^*, \bar{u}^*, \bar{x}^*, \bar{y}^*, \hat{\theta}, \hat{\eta}, \hat{\xi}, \hat{\sigma}, \hat{\phi}, \mu^*)$ satisfy Karush–Kuhn–Tucker (KKT) conditions.

Denote $\bar{\mathcal{K}}_{j,\tau} = \{k \notin \cup_i \mathcal{K}_{j,i,\tau} : \theta_{k,j,\tau}^* > -C'\}$. Define $\delta_k := \theta_{k,j,\tau}^* + C'$ for all $k \in \bar{\mathcal{K}}_{j,\tau}$. Let $\hat{\theta}$ equals to θ^* everywhere except $\hat{\theta}_{k,j,\tau} = -C'$ for all $k \in \bar{\mathcal{K}}_{j,\tau}$; $\hat{\eta}$ equals η^* everywhere except $\hat{\eta}_{k-1,j,\tau} = \eta_{k-1,j,\tau}^* - \delta_k$ for all $k \in \bar{\mathcal{K}}_{j,\tau}$ and $k \neq 1$, $\hat{\eta}_{k,j,\tau-1} = \eta_{k,j,\tau-1}^* - \delta$ for all $k \in \bar{\mathcal{K}}_{j,\tau}$ and $\tau > 1$; $\hat{\xi}$ equals to ξ^* everywhere except $\hat{\xi}_{1,j,\tau} = b_j - C'$ if $\{1\} \subset \bar{\mathcal{K}}_{j,\tau}$; $\hat{\sigma}$ equals σ^* everywhere except $\hat{\sigma}_{j,\tau} = \sigma_{j,\tau}^* - \Lambda_{j,\tau} \sum_{k \in \bar{\mathcal{K}}_{j,\tau}} \pi_k^j(S) \delta_k$; $\hat{\phi}$ equals to ϕ^* everywhere except $\hat{\phi}_{k,j,i,\tau} = \phi_{k,j,i,\tau}^* - \delta_k$ for all i and $k \in \bar{\mathcal{K}}_{j,\tau}$.

First, we verify the dual feasibility conditions. It is sufficient only to show that $\hat{\phi}_{k,j,i,\tau} \geq 0$ for all i and $k \in \bar{\mathcal{K}}_{j,\tau}$ and $\hat{\xi}_{1,j,\tau} \geq 0$ if $\{1\} \subset \bar{\mathcal{K}}_{j,\tau}$. The latter must hold according to Lemma 1. For the formal, according to (EC.8), for all $k \in \bar{\mathcal{K}}_{j,\tau}$,

$$\begin{aligned}\hat{\phi}_{k,j,i,\tau} &= \phi_{k,j,i,\tau}^* - \delta_k \\ &= C'_i(w^\top \bar{u}_{i,\tau}^*)w_j + \theta_{k,j,\tau}^* - \theta_{k,j,\tau}^* - C' \\ &= \begin{cases} 0 & \text{if } i \in \mathcal{I}_{j,\tau} \\ C'_i(w^\top \bar{u}_{i,\tau}^*)w_j - C' & \text{otherwise} \end{cases}\end{aligned}$$

When $i \in \mathcal{I}_{j,\tau}$, the last equality holds according to the definition of C' . For $i \notin \mathcal{I}_{j,\tau}$, we can show $C'_i(w^\top \bar{u}_{i,\tau}^*)w_j \geq C'$ by contradiction. Suppose that the relationship does not hold. For any $i' \in \mathcal{I}_{j,\tau}$, there must exist some k' such that $\bar{u}_{k',j,i',\tau}^* > 0$. Then, we can increase the fulfillment $\bar{u}_{k',j,i,\tau}^*$ and decrease $\bar{u}_{k',j,i',\tau}^*$ by a small amount. This means that we re-assign some fulfillment with a fulfillment window of k' from warehouse i' to i , leading to a reduction in the fulfillment cost since, according to the assumption, $C'_i(w^\top \bar{u}_{i,\tau}^*)w_j < C' = C'_{i'}(w^\top \bar{u}_{i',\tau}^*)w_j$. This contradicts the fact that $\bar{u}^*$ is optimal. Therefore, we conclude $\hat{\phi}_{k,j,i,\tau} \geq 0$ for all i and $k \in \bar{\mathcal{K}}_{j,\tau}$.

Next, we verify the stationarity conditions $\nabla \mathcal{L} = 0$ with respect to the primal variables. Specifically, with respect to $\gamma_{j,\tau}(S)$,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \gamma_{j,\tau}(S)} &= \Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_{k'} \hat{\theta}_{k',j,\tau} \pi_{k'}^j(S) - \hat{\sigma}_{j,\tau} + \hat{\mu}_{j,\tau,S} \\ &= \Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_{k' \notin \bar{\mathcal{K}}_{j,\tau}} \theta_{k',j,\tau}^* \pi_{k'}^j(S) + \Lambda_{j,\tau} \sum_{k \in \bar{\mathcal{K}}_{j,\tau}} \pi_k^j(S) (\theta_{k,j,\tau}^* - \delta_k) \\ &\quad - \sigma_{j,\tau}^* + \Lambda_{j,\tau} \sum_{k \in \bar{\mathcal{K}}_{j,\tau}} \pi_k^j(S) \delta_k + \mu_{j,\tau,S}^* \\ &= \Lambda_{j,\tau} R_j(S) + \Lambda_{j,\tau} \sum_{k'} \theta_{k',j,\tau}^* \pi_{k'}^j(S) - \sigma_{j,\tau}^* + \mu_{j,\tau,S}^* \\ &= 0\end{aligned}$$

With respect to $\bar{u}_{k,j,i,\tau}$ such that $k \in \bar{\mathcal{K}}_{j,\tau}$ and $k > 1$,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \bar{u}_{k,j,i,\tau}} &= -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j - \hat{\eta}_{k-1,j,\tau} - \xi_{k,j,\tau}^* + \hat{\phi}_{k,j,i,\tau} \\ &= -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j - \eta_{k-1,j,\tau}^* + \delta - \xi_{k,j,\tau}^* + \phi_{k,j,i,\tau}^* - \delta \\ &= -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j - \eta_{k-1,j,\tau}^* - \xi_{k,j,\tau}^* + \phi_{k,j,i,\tau}^* \\ &= 0\end{aligned}$$

when $k \in \bar{\mathcal{K}}_{j,\tau}$ and $k = 1$,

$$\frac{\partial \mathcal{L}}{\partial \bar{u}_{1,j,i,\tau}} = -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j + b_j - \hat{\xi}_{1,j,\tau} + \hat{\phi}_{1,j,i,\tau}$$

$$\begin{aligned}
&= -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j + b_j - b_j + C' + \phi_{1,j,i,\tau}^* - \theta_{1,j,\tau}^* - C' \\
&= -C'_i(w^\top \bar{u}_{i,\tau}^*)w_j + \phi_{1,j,i,\tau}^* - \theta_{1,j,\tau}^* \\
&= 0
\end{aligned}$$

where the last equality follows (EC.8). For $\bar{y}_{k,j,\tau}$ and $\bar{x}_{k,j,\tau}$, $k \in \bar{\mathcal{K}}_{j,\tau}$,

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \bar{y}_{k,j,\tau}} &= \hat{\theta}_{k,j,\tau} - \hat{\eta}_{k-1,j,\tau} - \xi_{k,j,\tau}^* \\
&= -C' - (\eta_{k-1,j,\tau}^* - \theta_{k,j,\tau}^* - C') - \xi_{k,j,\tau}^* \\
&= \theta_{k,j,\tau}^* - \eta_{k-1,j,\tau}^* - \xi_{k,j,\tau}^* \\
&= 0 \\
\frac{\partial \mathcal{L}}{\partial \bar{x}_{k,j,\tau}} &= \hat{\theta}_{k,j,\tau} - \hat{\eta}_{k,j,\tau-1} \\
&= -C' - (\eta_{k,j,\tau-1}^* - \theta_{k,j,\tau}^* - C') \\
&= \theta_{k,j,\tau}^* - \eta_{k,j,\tau-1}^* \\
&= 0
\end{aligned}$$

Note that $\frac{\partial \mathcal{L}}{\partial \bar{x}_{k,j,\tau}} = 0$ is required to hold only for $\tau > 1$ since $\bar{x}_{k,j,1} = x_{k,j}$ is not a decision variable.

Lastly, regarding the complementary slackness conditions, since $\hat{\theta}$, $\hat{\eta}$, and $\hat{\sigma}$ correspond to equality constraints, we only have to verify $\hat{\phi}_{k,j,i,\tau} \bar{u}_{k,j,i,\tau}^* = 0$ for all i and $k \in \bar{\mathcal{K}}_{j,\tau}$ and $\hat{\xi}_{1,j,\tau}(\sum_i \bar{u}_{1,j,i,\tau}^* - \bar{y}_{1,j,\tau}^*) = 0$ if $\{1\} \subset \bar{\mathcal{K}}_{j,\tau}$. The former holds since $\bar{u}_{k,j,i,\tau}^* = 0$ according to the assumption. The latter holds since the only scenario when $\{1\} \subset \bar{\mathcal{K}}_{j,\tau}$ is when $\bar{y}_{1,j,\tau}^* = 0$. To see that, suppose $\bar{y}_{1,j,\tau}^* > 0$, which indicates, according to the assumption, it is optimal to fulfill some orders with $k' > 1$ but not with $k = 1$. However, we can always find some i such that $\bar{u}_{k',j,i,\tau}^* > 0$. Then, we increase $\bar{u}_{1,j,i,\tau}^*$ by a small amount and reduce $\bar{u}_{k',j,i,\tau}^*$ by the same amount. The resulting fulfillment decision achieves a lower fulfillment cost, which contradicts that $\bar{u}^*$ is optimal.

Therefore, $(\gamma^*, \bar{u}^*, \bar{x}^*, \bar{y}^*)$ and $(\hat{\theta}, \hat{\eta}, \xi^*, \hat{\sigma}, \hat{\phi}, \mu^*)$ jointly satisfy KKT conditions. To show that the constructed dual variables are optimal, we still need Slater's condition to hold, which is clearly the case for our problem. For instance, we can find a relative interior and feasible primal point by letting $\gamma_{j,\tau}(S) = 1/|\mathcal{N}|$ for all $S \in \mathcal{N}$ and equally assigning $(1 - \varepsilon)$ proportion of unfulfilled orders to each warehouse, for any $\varepsilon \in (0, 1)$. Thus, we conclude that $\hat{\theta}$ is also an optimal dual and $\hat{\theta}_{k,j,\tau} = -C'$ for all k .

Case 2: no order from j is fulfilled. Once again, our goal is to construct dual variables $(\hat{\theta}, \hat{\eta}, \hat{\xi}, \hat{\sigma}, \hat{\phi}, \mu^*)$, and show that, together with primal optimal variables $(\gamma^*, \bar{u}^*, \bar{x}^*, \bar{y}^*)$, they satisfy KKT conditions.

Denote $C' := \min\{b_j, \min_i C'_i(w^\top \bar{u}_{i,\tau}^*)w_j\}$. We set $\hat{\theta}_{k,j,\tau} = -C'$ for all k ; $\hat{\eta}$ equals to η^* everywhere except $\hat{\eta}_{k-1,j,\tau} = \eta_{k-1,j,\tau}^* - \theta_{k,j,\tau}^* - C'$ for all $k > 1$ and $\hat{\eta}_{k,j,\tau-1} = \eta_{k,j,\tau-1}^* - \theta_{k,j,\tau}^* - C'$ for $\tau > 1$; $\hat{\xi}$

equals to ξ^* everywhere except $\hat{\xi}_{1,j,\tau} = b_j - C'$; $\hat{\sigma}$ equals to σ^* everywhere except $\hat{\sigma}_{j,\tau} = \sigma_{j,\tau}^* - \Lambda_{j,\tau} \sum_{k'} (\theta_{k,j,\tau}^* + C') \pi_{k'}^j(S)$; $\hat{\phi}$ equals to ϕ^* everywhere except $\hat{\phi}_{k,j,i,\tau} = \phi_{k,j,i,\tau}^* - \theta_{k,j,\tau}^* - C'$ for all i .

For feasibility conditions, we need to show $\hat{\xi}_{1,j,\tau} \geq 0$ and $\hat{\phi}_{k,j,i,\tau} \geq 0$ for all i . The first holds since $\hat{\xi}_{1,j,\tau} = \max\{0, b_j - \min_i C'_i(w^\top \bar{u}_{i,\tau}^*) w_j\} \geq 0$. The second holds since, according to (EC.8),

$$\begin{aligned} \hat{\phi}_{k,j,i,\tau} &= \phi_{k,j,i,\tau}^* - \theta_{k,j,\tau}^* - C' \\ &= C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + \theta_{k,j,\tau}^* - \theta_{k,j,\tau}^* - C' \\ &= C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - \min\{b_j, \min_{i'} C'_{i'}(w^\top \bar{u}_{i',\tau}^*) w_j\} \\ &= \begin{cases} C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - b_j & \text{if } b_j \leq \min_{i'} C'_{i'}(w^\top \bar{u}_{i',\tau}^*) w_j \\ C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - \min_{i'} C'_{i'}(w^\top \bar{u}_{i',\tau}^*) w_j & \text{otherwise} \end{cases} \\ &\geq 0 \end{aligned}$$

For the stationarity conditions $\nabla \mathcal{L} = 0$, similar to the previous discussion in Case 1.3, we can show our constructed dual variables satisfy $\frac{\partial \mathcal{L}}{\partial \gamma_{j,\tau}(S)} = 0$, $\frac{\partial \mathcal{L}}{\partial \bar{u}_{k,j,i,\tau}} = 0$ when $k > 1$, $\frac{\partial \mathcal{L}}{\partial \bar{y}_{k,j,\tau}} = 0$ when $k > 1$, and $\frac{\partial \mathcal{L}}{\partial \bar{x}_{k,j,\tau}} = 0$ for $\tau > 1$. For $k = 1$,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \bar{u}_{1,j,i,\tau}} &= -C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + b_j - \hat{\xi}_{1,j,\tau} + \hat{\phi}_{1,j,i,\tau} \\ &= -C'_i(w^\top \bar{u}_{i,\tau}^*) w_j + b_j - (b_j - C') + (\phi_{1,j,i,\tau}^* - \theta_{1,j,\tau}^* - C') \\ &= -C'_i(w^\top \bar{u}_{i,\tau}^*) w_j - \theta_{1,j,\tau}^* + \phi_{1,j,i,\tau}^* \\ &= 0 \end{aligned}$$

where the last equality follows (EC.8), and

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \bar{y}_{1,j,\tau}} &= -b_j - \hat{\theta}_{1,j,\tau} + \hat{\xi}_{1,j,\tau} \\ &= -b_j + C' + b - C' \\ &= 0 \end{aligned}$$

For complementary slackness conditions, it is sufficient only to verify $\hat{\xi}_{1,j,\tau} (\sum_i \bar{u}_{1,j,i,\tau}^* - \bar{y}_{1,j,\tau}^*) = 0$. If $\bar{y}_{1,j,\tau}^* = 0$, the condition holds since $\sum_i \bar{u}_{1,j,i,\tau}^* - \bar{y}_{1,j,\tau}^* = 0$. Otherwise, if $\bar{y}_{1,j,\tau}^* > 0$, we need to show $\hat{\xi}_{1,j,\tau} = 0$, which is equivalent to show $b_j \leq C'_i(w^\top \bar{u}_{i,\tau}^*) w_j$ for all i . This is indeed the case: If there exists a warehouse i that does not satisfy that requirement, we are better off by fulfilling some orders with $k = 1$ from that warehouse to location j , i.e., $\bar{u}_{1,j,i,\tau}^* > 0$, which contradicts that it is optimal to not fulfill all orders from j .

Therefore, we conclude that $(\gamma^*, \bar{u}^*, \bar{x}^*, \bar{y}^*)$ and $(\hat{\theta}, \hat{\eta}, \hat{\xi}, \hat{\sigma})$ satisfy KKT conditions. As discussed previously, Slater's condition also holds. Thus, $\hat{\theta}_{k,j,\tau} = -\min\{b_j, \min_i C'_i(w^\top \bar{u}_{i,\tau}^*) w_j\} < 0$ for all k is also an optimal dual.

Since the index j and τ are arbitrarily picked, the above completes the second part. $\square$

EC.1.4. Lemma 2

We prove this lemma by contradiction. Assume $S_{j,\tau}^*(t)$ is efficient for all j , t , and τ . $S_{j,\tau}^*(t)$ solves the following maximization,

$$\max_S \sum_k (r_{k,j} - \Delta V_\tau^{k,j}(x, t)) \pi_k^j(S)$$

which is equivalent to

$$\max_S \sum_k R(S) - \sum_k \Delta V_\tau^{k,j}(x, t) \pi_k^j(S)$$

With the necessary and sufficient condition (Lemma EC.2), we must have $\Delta V_\tau^{k,j}(x, t) = \theta$ for all k . Note that θ may take various values according to t , τ , and x . However, we can construct an instance as follows and show the above does not hold.

Fix a cycle τ . Suppose $t = 1$, and x_τ satisfies $x_{1,j,\tau} > 0$ for all j , $x_{k,j,\tau} = 0$ for all k, j , and all $x_{1,j,\tau}$ will be fulfilled completely but there is no additional capacity to take another order with $k = 1$. In that case, we must have $\Delta V_\tau^{1,j}(x, t) = b_j$ since any additional orders with $k = 1$ must be penalized. In contrast, we can take an order with $k > 1$ and postpone its fulfillment to a later fulfillment cycle and fulfill it by a carrier, resulting in $\Delta V_\tau^{k,j}(x, t) < b_j$.

EC.1.5. Theorem 1

Before providing the proof for Theorem 1, we first give the formal definition of the **HYP** policy in Algorithm 2 and two technical lemmas and their proof. The proof of the theorem directly comes from the lemmas.

Algorithm 2: Hypothetical Policy (HYP)

Input: Customer arrival rate vector $\Lambda_{j,\tau}$, $\forall j \in [N]$, $\tau \in [T]$, Fulfillment cost vector

C_i , $\forall i \in [M]$, penalty vector b_j , $\forall j \in [N]$

Output: Fulfillment option display decision $S^{\text{HYP}} = \{S_{j,\tau}^{\text{HYP}}\}$ and fulfillment decision

$$u^{\text{HYP}} = \{u_{k,j,i,t}^{\text{HYP}}\}$$

- 1 Construct the set of all *efficient assortments* $\mathcal{E}_j$ for all $j \in [N]$.
 - 2 Solve **DET** to obtain $\gamma_{j,\tau}^*(S)$ for all $S \in \mathcal{E}_j$ and $\bar{u}_{k,j,i,\tau}^*$.
 - 3 **for** $\tau \in [T]$ **do**
 - 4 At any time $t \in [0, 1]$, if there is a customer from location j , offer an assortment $S \in \mathcal{E}$ with probability $\gamma_{j,\tau}^*(S)$.
 - 5 At the end of the cycle, fulfill orders according to a scaling rule $u_{k,j,i,t}^{\text{HYP}} = \frac{y_{k,j,\tau}}{y_{k,j,\tau}^*} \cdot \bar{u}_{k,j,i,\tau}^*$.
 - 6 **end**
-

LEMMA EC.3. Let $\Lambda = \max_{j,\tau} \Lambda_{j,\tau}$, $\bar{w} = \max_j w_j$. The optimality gap of HYP is bounded by

$$V^D(x) - V^{\text{HYP}}(x) \leq \bar{w} \bar{C} \sqrt{MT} \sqrt{\Lambda K N}$$

Proof Let $(\tilde{D}_{j,\tau} : k \in [K], \tau \in [T])$ denote a sample path of customer arrivals, $(D_{k,j,\tau}^{\text{HYP}} : k \in [K], j \in [N], \tau \in [T])$ denote the realized demand under HYP and the sample path of arrivals. Note that $D_{k,j,\tau}^{\text{HYP}}$ is a counting process with the parameter $\Lambda_{j,\tau} \sum_S \gamma_{j,\tau}^*(S) \pi_k^j(S)$ over a unit time span, hence,

$$\mathbb{E}[D_{k,j,\tau}^{\text{HYP}}] = \Lambda_{j,\tau} \sum_S \gamma_{j,\tau}^*(S) \pi_k^j(S)$$

The expected revenue collected during some fulfillment cycle τ can be calculated by

$$\mathbb{E} \left[\sum_j \sum_k r_{k,j} D_{k,j,\tau}^{\text{HYP}} \right] = \sum_{k,j,S} r_{k,j} \Lambda_{j,\tau} \gamma_{j,\tau}^*(S) \pi_k^j(S)$$

which equals the revenue in **DET** with an optimal solution. Hence, the difference between the expected profit under **HYP** and **DET** can be bounded as follows:

$$\begin{aligned} V^D(x) - V^{\text{HYP}}(x) \\ = \mathbb{E} \left\{ \sum_{\tau=1}^T (g(y_\tau, u_\tau^{\text{HYP}}) - g(\bar{y}_\tau^*, \bar{u}_\tau^*)) + \sum_{k,j} b_j(x_{k,j,T+1} - \bar{x}_{k,j,T+1}^*) \right\} \end{aligned}$$

where g is the immediate cost function defined as

$$g(y, u) = \sum_i C_i \left(\sum_{k,j} w_j u_{k,j,i} \right) + \sum_j b_j \left(y_{1,j} - \sum_i u_{1,j,i} \right)$$

x_τ and y_τ are the pre- and post-choice state under **HYP**, respectively. $\bar{x}_\tau^*$ and $\bar{y}_\tau^*$ are the pre- and post-choice state along the “optimal fluid path”.

We first show that the system’s evolution under **HYP** follows the “optimal fluid path” in expectation. Specifically, we want to show that

$$\mathbb{E}[x_\tau] = \bar{x}_\tau^*, \quad \mathbb{E}[y_\tau] = \bar{y}_\tau^*, \quad \mathbb{E}[u_\tau^{\text{HYP}}] = \bar{u}_\tau^*$$

for all τ . We can show this by induction. When $\tau = 1$

$$\begin{aligned} \mathbb{E}[y_{k,j,1}] &= \mathbb{E}[x_{k,j,1} + D_{k,j,1}^{\text{HYP}}] \\ &= x_{k,j,1} + \mathbb{E}[D_{k,j,1}^{\text{HYP}}] \\ &= x_{k,j,1} + \Lambda_{j,1} \sum_S \gamma_{j,1}^*(S) \pi_k^j(S) \\ &= \bar{y}_{k,j,1}^* \end{aligned}$$

and $\mathbb{E}[u_{k,j,i,1}^{\text{HYP}}] = \bar{u}_{k,j,i,1}^*$ follows the definition in Algorithm 2. Since $x_{k,j,2}$ is linear in both $y_{k,j,1}$ and $u_{k,j,1}^{\text{HYP}}$, then we must have $\mathbb{E}[x_{k,j,2}] = \bar{x}_{k,j,2}^*$.

Suppose for any $\tau > 1$ we have $\mathbb{E}[x_\tau] = \bar{x}_\tau^*$, $\mathbb{E}[y_{\tau-1}] = \bar{y}_{\tau-1}^*$, and $\mathbb{E}[u_{\tau-1}^{\text{HYP}}] = \bar{u}_{\tau-1}^*$ by the inductive assumption. We can easily show the following holds:

$$\mathbb{E}[y_\tau] = \bar{y}_\tau^*, \quad \mathbb{E}[u_\tau^{\text{HYP}}] = \bar{u}_\tau^*, \quad \mathbb{E}[x_{\tau+1}] = \bar{x}_{\tau+1}^*$$

which completes the induction. Therefore,

$$\begin{aligned} V^D(x) - V^{\text{HYP}}(x) &\leq \mathbb{E} \left\{ \sum_{\tau=1}^T (g(y_\tau, u_\tau^{\text{HYP}}) - g(\bar{y}_\tau^*, \bar{u}_\tau^*)) + \sum_{k,j} b_j(x_{k,j,T+1} - \bar{x}_{k,j,T+1}^*) \right\} \\ &= \mathbb{E} \left\{ \sum_{\tau=1}^T \sum_{i=1}^M (C_i(w^\top u_{i,\tau}^{\text{HYP}}) - C_i(w^\top \bar{u}_{i,\tau}^*)) \right\} \\ &\leq \sum_{\tau=1}^T \sum_{i=1}^M \bar{C} \mathbb{E} \{ |w^\top u_{i,\tau}^{\text{HYP}} - w^\top \bar{u}_{i,\tau}^*| \} \\ &\leq \sum_{\tau=1}^T \sum_{i=1}^M \bar{C} \sqrt{\text{Var} \left(\sum_{k,j} w_j u_{k,j,i,\tau}^{\text{HYP}} \right)} \end{aligned}$$

The second inequality holds due to Mean Value Theorem and Assumption 1. The last inequality is held by Jensen's inequality and the definition of variance.

To further upper bound the variance, we must carefully examine the relation of the fulfillment from the warehouse i to different demand indices k and j . Note that customers' arrival process and the option list display decision are independent in j . Hence, the post-assortment state $y_{k,j,\tau}$ is independent in j . Since $u_{k,j,i,\tau}^{\text{HYP}} = \frac{\bar{u}_{k,j,i,\tau}^*}{\bar{y}_{k,j,\tau}^*} y_{k,j,\tau}$, where $\frac{\bar{u}_{k,j,i,\tau}^*}{\bar{y}_{k,j,\tau}^*}$ is deterministic, the randomness of $u_{k,j,i,\tau}^{\text{HYP}}$ completely comes from $y_{k,j,\tau}$. Therefore, $u_{k,j,i,\tau}^{\text{HYP}}$ is independent in j , implying

$$\text{Var} \left(\sum_{k,j} w_j u_{k,j,i,\tau}^{\text{HYP}} \right) = \sum_j w_j^2 \text{Var} \left(\sum_k u_{k,j,i,\tau}^{\text{HYP}} \right)$$

Moreover, using the technical Lemma EC.4 we have

$$\text{Var} \left(\sum_k u_{k,j,i,\tau}^{\text{HYP}} \right) = \sum_k c_{k,j,i}^2 \text{Var}(\tilde{D}_{j,\tau+1-k}) \leq \sum_k c_{k,j,i}^2 \Lambda_{j,\tau+1-k}$$

Denote $A_i := \text{Var} \left(\sum_{k,j} w_j u_{k,j,i,\tau}^{\text{HYP}} \right)$ for each warehouse i . To aggregate across warehouses, we apply the Cauchy–Schwarz inequality:

$$\sum_{i=1}^M \sqrt{A_i} \leq \sqrt{M \sum_{i=1}^M A_i}$$

It remains to bound $\sum_i A_i$. Substituting the variance bound and exchanging the order of summation,

$$\sum_{i=1}^M A_i \leq \sum_j w_j^2 \sum_k \Lambda_{j,\tau+1-k} \underbrace{\sum_i c_{k,j,i}^2}_{\leq 1}$$

where the last step uses $c_{k,j,i} \in [0, 1]$ and $\sum_i c_{k,j,i} \leq 1$ from Lemma EC.4, so that $\sum_i c_{k,j,i}^2 \leq \sum_i c_{k,j,i} \leq 1$. Letting $\Lambda := \max_{j,\tau} \Lambda_{j,\tau}$, we obtain

$$\sum_{i=1}^M A_i \leq \sum_j w_j^2 K \Lambda \leq \bar{w}^2 K N \Lambda$$

Combining,

$$V^D(x) - V^{\text{HYP}}(x) \leq \sum_{\tau=1}^T \bar{C} \sqrt{M \cdot \bar{w}^2 K N \Lambda} = \bar{w} \bar{C} \sqrt{M} T \sqrt{K N \Lambda}$$

□

LEMMA EC.4. *There exists a sequence of constants $c_{1,j,i}, \dots, c_{K,j,i}$ with $c_{k,j,i} \in [0, 1]$ and $\sum_i c_{k,j,i} \leq 1$ for all k such that*

$$\sum_k u_{k,j,i,\tau}^{\text{HYP}} = \sum_k c_{k,j,i} \tilde{D}_{j,\tau+1-k}$$

Proof Define $\rho_{k,j,i,\tau} := \frac{\bar{u}_{k,j,i,\tau}^*}{\bar{y}_{k,j,\tau}^*}$, and we must have $\rho_{k,j,i,\tau} \in [0, 1]$ and $\sum_i \rho_{k,j,i,\tau} \in [0, 1]$ since $\sum_i \bar{u}_{k,j,i,\tau}^* \leq \bar{y}_{k,j,\tau}^*$. With that, we have $u_{k,j,i,\tau}^{\text{HYP}} = \rho_{k,j,i,\tau} y_{k,j,\tau}$.

Using the system dynamics, we have the following

$$\begin{aligned} y_{k,j,\tau} &= D_{k,j,\tau}^{\text{HYP}} + \left(D_{k+1,j,\tau-1}^{\text{HYP}} - \sum_i u_{k+1,j,i,\tau-1}^{\text{HYP}} \right) + \dots + \left(D_{K,j,\tau-(K-k)}^{\text{HYP}} - \sum_i u_{K,j,i,\tau-(K-k)}^{\text{HYP}} \right) \\ &= D_{k,j,\tau}^{\text{HYP}} + \left(D_{k+1,j,\tau-1}^{\text{HYP}} - \sum_i \rho_{k+1,j,i,\tau-1} y_{k+1,j,\tau-1} \right) + \dots \\ &\quad + \left(D_{K,j,\tau-(K-k)}^{\text{HYP}} - \sum_i \rho_{K,j,i,\tau-(K-k)} y_{K,j,\tau-(K-k)} \right) \\ &= D_{k,j,\tau}^{\text{HYP}} + (D_{k+1,j,\tau-1}^{\text{HYP}} - \rho_{k+1,j,\tau-1} y_{k+1,j,\tau-1}) + \dots + (D_{K,j,\tau-(K-k)}^{\text{HYP}} - \rho_{K,j,\tau-(K-k)} y_{K,j,\tau-(K-k)}) \end{aligned}$$

where $\rho_{k,j,\tau} := \sum_i \rho_{k,j,i,\tau}$. The first term is realized demand at the current cycle τ , the second is realized but unfulfilled orders at cycle $\tau - 1$, and the last term is unfulfilled orders from $K - k$ cycles ago. The above presents an iterative relationship between $y_{k,j,\tau}$, which can be further reduced to

$$y_{k,j,\tau} = D_{k,j,\tau}^{\text{HYP}} + \bar{\rho}_{k+1,j,\tau-1} D_{k+1,j,\tau-1}^{\text{HYP}} + \dots + \prod_{k'=1}^{K-k} \bar{\rho}_{k+k',j,\tau-k'} D_{k+(K-k),j,\tau-(K-k)}^{\text{HYP}} \quad (\text{EC.11})$$

where $\bar{\rho}_{k,j,\tau} := 1 - \rho_{k,j,\tau}$, $\bar{\rho}_{k,j,\tau} \leq 1$. Under the **HYP** policy, the realized demand is determined by

$$D_{k,j,\tau}^{\text{HYP}} = \sum_S \gamma_{j,\tau}^*(S) \pi_k^j(S) \tilde{D}_{j,\tau} = \eta_{k,j,\tau} \tilde{D}_{j,\tau} \quad (\text{EC.12})$$

where $\eta_{k,j,\tau} := \sum_S \gamma_{j,\tau}^*(S) \pi_k^j(S)$, $\eta_{k,j,\tau} \leq 1$, is a constant that can be interpreted as the proportion of customers who select option k . Finally,

$$\begin{aligned} \sum_k u_{k,j,i,\tau}^{\text{HYP}} &= \sum_k \rho_{k,j,i,\tau} y_{k,j,\tau} \\ &= \rho_{K,j,i,\tau} \eta_{K,j,\tau} \tilde{D}_{j,\tau} \\ &\quad + \rho_{K-1,j,i,\tau} \eta_{K-1,j,\tau} \tilde{D}_{j,\tau} + \rho_{K-1,j,i,\tau} \bar{\rho}_{K,j,\tau-1} \eta_{K,j,\tau-1} \tilde{D}_{j,\tau-1} \\ &\quad + \dots \\ &\quad + \rho_{1,j,i,\tau} \eta_{1,j,\tau} \tilde{D}_{j,\tau} + \rho_{1,j,i,\tau} \bar{\rho}_{2,j,\tau-1} \eta_{2,j,\tau-1} \tilde{D}_{j,\tau-1} + \dots + \rho_{1,j,i,\tau} \prod_{k'=1}^{K-1} \bar{\rho}_{1+k',j,\tau-k'} \eta_{K,j,\tau-(K-1)} \tilde{D}_{j,\tau-(K-1)} \\ &= \sum_{k=1}^K \rho_{k,j,i,\tau} \eta_{k,j,\tau} \tilde{D}_{j,\tau} + \sum_{k=1}^{K-1} \rho_{k,j,i,\tau} \bar{\rho}_{k+1,j,\tau-1} \eta_{k+1,j,\tau-1} \tilde{D}_{j,\tau-1} + \dots \\ &\quad + \sum_{k=1}^1 \rho_{k,j,i,\tau} \prod_{k'=1}^{K-k} \bar{\rho}_{k+k',j,\tau-k'} \eta_{k+(K-1),j,\tau-(K-1)} \tilde{D}_{j,\tau-(K-1)} \end{aligned}$$

We define $c_{k,j,i}$ as follows

$$c_{k,j,i} = \sum_{k'=1}^{K+1-k} \rho_{k',j,i,\tau} \prod_{k''=1}^{k-1} \bar{\rho}_{k'+k'',j,\tau-k''} \eta_{k'+(k-1),j,\tau-(k-1)}$$

Since $\rho_{k',j,i,\tau} \leq 1$ and each $\bar{\rho}_{k'+k'',j,\tau-k''} \leq 1$, each term in the sum is bounded by $\eta_{k'+(k-1),j,\tau-(k-1)}$. It is clear that the sum over η ranges over $m = k, \dots, K$ if we substituting $m = k' + (k-1)$. Since $\sum_{m=1}^K \eta_{m,j,\tau'} \leq 1$ for any τ' (at most all arriving customers make a purchase), we have $c_{k,j,i} \leq 1$. Moreover, summing over warehouses replaces $\rho_{k',j,i,\tau}$ with $\rho_{k',j,\tau} := \sum_i \rho_{k',j,i,\tau} \leq 1$, so $\sum_i c_{k,j,i} \leq 1$ by the same argument. Therefore $c_{k,j,i} \leq 1$ and $\sum_i c_{k,j,i} \leq 1$ for all $k \in [K]$. $\square$

LEMMA EC.5. $V^{\text{HYP}}(x) \leq V^{\text{ERT}}(x)$

Proof Consider a sequence of customer arrivals, denoted as $\tilde{D} = (\tilde{D}_{j,\tau}(t) : j \in [N], \tau \in [T], t \in [0, 1])$. This process can also be represented by the specific times of arrival. For each fulfillment cycle τ , let the total number of customers be $L_\tau := \sum_j \tilde{D}_{j,\tau}$. These arrivals are indexed by $l \in [1, L_\tau]$ with corresponding arrival times $t_{l,\tau}$. Each order is identified by its arrival time.

Applying both **ERT** and **HYP** to $\tilde{D}$, the total profit from each policy can be obtained by the cumulative profit from each customer, termed as marginal profit, which can be calculated as the marginal revenue subtracting the marginal fulfillment cost. That is,

$$V^\rho(x \mid \tilde{D}) = \sum_{\tau=1}^T \sum_{l=1}^{L_\tau} r_{l,\tau}^\rho - c_{l,\tau}^\rho$$

where $\rho \in \{\mathbf{ERT}, \mathbf{HYP}\}$, $r_{l,\tau}^\rho$ and $c_{l,\tau}^\rho$ are the marginal revenue and cost for the orders arriving at time $t_{l,\tau}$ during cycle τ under policy ρ . The expected profit $V^{\mathbf{ERT}}(x)$ and $V^{\mathbf{HYP}}(x)$ can be obtained by taking the expectation of the profit with respect to the sample path.

$$V^\rho(x) = \mathbb{E}_{\tilde{D}} \left[V^\rho(x \mid \tilde{D}) \right]$$

Therefore, to prove this lemma, it is sufficient to show that

$$r_{l,\tau}^{\mathbf{ERT}} - c_{l,\tau}^{\mathbf{ERT}} \geq r_{l,\tau}^{\mathbf{HYP}} - c_{l,\tau}^{\mathbf{HYP}}$$

for all l and τ .

To control the randomness due to customer choice, we assume that if the same option list is offered under both **ERT** and **HYP** to the customer at time $t_{l,\tau}$, she will choose the same fulfillment option under both policies.

If both policies offer identical option list for all customer $t_{l,\tau}$ across all τ , the lemma holds trivially. Therefore, we assume, for the first time, **ERT** and **HYP** offer different lists during cycle τ_0 to the customer at t_{l_0,τ_0} from location j_0 . Denote the offered list by $S^{\mathbf{HYP}}$ and $S^{\mathbf{ERT}}$, respectively. The lists are different only when some option in $S^{\mathbf{HYP}}$ becomes economically nonviable and removed, thus $S^{\mathbf{ERT}} \subsetneq S^{\mathbf{HYP}}$.

Let us focus on this customer at t_{l_0,τ_0} . If she chooses to leave without the purchase under both policies, $r_{l_0,\tau_0}^\rho = c_{l_0,\tau_0}^\rho = 0$. Otherwise, if she chooses the same fulfillment option $k \in S^{\mathbf{ERT}}$ under both policies, then the marginal revenue will be identical, i.e., $r_{l_0,\tau_0}^\rho = r_{k,j}$. However, the marginal fulfillment cost associated with this order under **ERT** will not be larger than that under **HYP**. To show that, we first notice that this order will not be fulfilled in a later cycle under **HYP** than under **ERT** according to Algorithm 1. If this order is fulfilled in the same cycle under both policies, then the marginal fulfillment cost can be considered equal under both policies. However, if this order is fulfilled in a later cycle under **ERT** than under **HYP**, then, according to Algorithm 1, $c_{l_0,\tau_0}^{\mathbf{HYP}} > b_{j_0} \geq c_{l_0,\tau_0}^{\mathbf{ERT}}$.

Conversely, if the customer chooses $k' \in S^{\mathbf{HYP}} \setminus S^{\mathbf{ERT}}$ under **HYP** but $k \in S^{\mathbf{ERT}} \cup \{\emptyset\}$ under **ERT**. Since k' is eliminated in **ERT**, it implies k' is economically nonviable while k is viable. According to Definition 2, the marginal profit with k' is strictly negative, whereas, with k , it is non-negative. This rationale applies to any customers from location j_0 after t_{l_0,τ_0} during cycle τ_0 , as k' remains nonviable. Therefore, we can conclude that **ERT** will not have worse performance than **HYP** by the end of cycle τ .

For the subsequent cycle $\tau_1 := \tau_0 + 1$, while the pre-assortment state $x_{\tau_1}^\rho$ may differ under **ERT** and **HYP**, this does not impact the assortment strategy $S^{\mathbf{HYP}}$ for cycle τ_1 . Nor does it affect

S^{ERT} until the time instance t_{l_1, τ_1} , when some fulfillment option $k' \in S^{\text{HYP}}$ becomes economically nonviable. Before this time, both policies offer the same assortment. After t_{l_1, τ_1} , k' is removed from S^{ERT} . Using the same logic as before, we can demonstrate that the profit collected under **ERT** during the fulfillment cycle τ_1 will not be less than that under **HYP**. $\square$

EC.2. Endogenize Surcharge Schedules

This section shows how to extend our framework to endogenize surcharge schedules with minimal additional structure. Suppose the platform offers K delivery speeds (characterized by the latest delivery time) but has not yet committed to a surcharge for each speed. Instead, the platform considers a set of candidate surcharges $\mathcal{L}_k$ for each speed k .

We construct the expanded fulfillment option set $\mathcal{S} = \{(k, r) : k \in [K], r \in \mathcal{L}_k\}$ with $|\mathcal{S}| = \sum_k |\mathcal{L}_k|$. We assume the candidate price sets satisfy a natural non-overlapping condition.

ASSUMPTION EC.1 (Non-overlapping Surcharge). *For any $k, k' \in [K]$, if $k > k'$, then $r < r'$ for all $r \in \mathcal{L}_k, r' \in \mathcal{L}_{k'}$.*

This assumption reflects the standard practice that faster delivery commands a higher surcharge. It is highly possible that an arbitrary fulfillment option list $S \subset \mathcal{S}$ contains two or more options with the same speed but different surcharges. The price difference may reflect other add-on services we do not explicitly model. For instance, the more expensive fulfillment option includes a tracking service, whereas the cheaper option does not. Thus, we can still use a discrete model (e.g., an MC model) to describe consumers' choice on some option $(k, r) \in S$.

Once a consumer submits an order with a specific fulfillment option (k, r) , the revenue $r_{k,j}$ is obtained in parts. The first part is the fulfillment reserve, a fixed percentage of the order's value. In general, consumers place orders that vary in size and weight, reflecting the diversity of purchase needs. We thus can consider a demand location j as a combination of a specific geographic region and a specific order type, where the order type is determined by the order's value and weight (this can be operationalized by discretizing order sizes and weights into representative groups). Thus, the first part of revenue is κp_j , where $\kappa \in [0, 1)$ specifies a fixed percentage and p_j is the order value from location j . The second part is simply the shipping surcharge r . This two-part strategy is common in the industry, where the first term is derived from the displayed price, and the second term corresponds to a separate surcharges. Therefore, $r_{k,j} = \kappa p_j + r$.

Our formulation remains untouched and the theoretical results carry over to this setting. Specifically, efficient option lists lie on the $(R_j(S), Q_j(S))$ Pareto frontier and solve the parametric problem $\max_S R_j(S) - \theta_j Q_j(S)$ for some $\theta_j \geq 0$. If we use an MC model to describe consumer behavior, the monotonic nesting property (Lemma 3) holds over $\mathcal{S}$, since it is a property of the MC choice

structure and does not depend on how options are constructed. Proposition 3 then applies with $|\mathcal{S}|$ replacing K : the number of efficient option lists is bounded by $|\mathcal{S}| + 1$, and the parametric dual sweep (Algorithm 3) enumerates all of them in polynomial time in $|\mathcal{S}|$. Thus, the efficient option lists can be identified in polynomial time. Once they are identified, we apply Proposition 2 to solve **DET**, and the ERT heuristic proceeds as before.

EC.3. Multiple Shipping Zones

This section extends the formulation to scenarios with multiple shipping zones and non-zero lead time requirements. Specifically, we introduced warehouse-specific *multiple shipping zones* to account for location heterogeneity and lead-time requirements: demands from different shipping zones incur different marginal fulfillment costs and require different lead times. This leads to warehouse-shipping zone-specific fulfillment cost functions $C_{i,z}$ and to an additional lead-time constraint in the definition of the feasible action sets.

Logistics carriers often determine fulfillment costs based on the origin-destination (OD) pair and the type of service. For the same OD pair, a standard fulfillment service takes a longer time (lead time) to send the product, while a dedicated fulfillment service takes a shorter lead time. However, for different OD pairs, the same type of fulfillment service takes different times. For a specific warehouse, carriers partition geographical areas into zones, so sending a unit weight to different locations within the same zone with the same service incurs the same cost. For instance, UPS segments shipments within the United States into eight distinct zones, labeled Zone 1 to Zone 8, and applies different but fixed marginal rates based on zone codes rather than miles from the origin to the destination (Shipbob 2022). Moreover, for the same origin, different destinations in the same shipping zone incur the same lead time. This incentivizes the fulfillment cost function to be warehouse- and shipping-zone-specific.

Formally, we denote Z_i the number of zones for warehouse i , accounting for both geographic area and fulfillment service. An order from location j with a fulfillment option k belongs to zone $z \in [Z_i]$ if the *lead time* (time to physically ship an order from warehouse i to location j) requirement is smaller than k periods. Denote $S_{i,z}$ the set of (k, j) indices in zone z for warehouse i . If $(k, j) \in S_{i,z}$, then $(k', j) \in S_{i,z}$ for all $k' > k$. This indicates that an order from j with a longer fulfillment window can be fulfilled with the same service. Note that the sets $S_{i,z}$ can be constructed using the fulfillment network and the available fulfillment services.

Actions. For each warehouse i , the sets $S_{i,z}$ for all z are not mutually exclusive since an order (k, j) can be fulfilled with multiple services. Thus, we need an additional index in the fulfillment decision to capture this differentiation. Specifically, we denote a fulfillment decision is $u_t = (u_{k,j,z,i,t} \in \mathbb{Z} : k \in [K], j \in [N], i \in [M], z \in [Z_i])$, where each element $u_{k,j,z,i,t}$ corresponds to the

number of the orders shipped in period t from warehouse i to demand location j to satisfy the unfulfilled orders with k periods left in fulfillment window with a service specified by zone z . It is worth pointing out the following

$$u_{k,j,z,i,t} = 0, \quad \forall i \in [M], z \in [Z_i], (k,j) \notin S_{i,z} \quad (\text{EC.13})$$

which we call the lead time constraint, as it prevents fulfillment that does not meet the lead time requirements.

Fulfillment cost. We define a convex fulfillment cost function $\mathcal{C}_{i,z}$ for warehouse i and its corresponding shipping zone z to capture heterogeneity in fulfillment costs across shipping zones. The convexity reflects the seller using multiple logistic carriers. Although different carriers may define their shipping zones differently regarding the geographic areas, we can always align them and define the proper fulfillment cost functions. Thus, the total fulfillment cost for the period t can be obtained by

$$\sum_{i \in [M]} \sum_{z \in [Z_i]} C_{i,z} \left(\sum_{k,j \in S_{i,z}} w_j u_{k,j,z,i,t} \right) = \sum_{i,z} C_{i,z} (w^\top u_{i,z,t})$$

We highlight that introducing multiple shipping zones and non-zero lead time in transportation does not affect customer choice but only the sellers fulfillment decisions. With this, we can update **OPT** as follows.

$$V^*(x) = \max_{S,u} \mathbb{E} \left\{ \sum_{\tau,j,k} r_{k,j} D_{k,j,\tau} \right\} - \mathbb{E} \left\{ \sum_{\tau} \left[\sum_{i,z} C_{i,z} (w^\top u_{i,z,\tau}) + \sum_j b_j \left(y_{1,j,\tau} - \sum_{i,z} u_{1,j,z,i,\tau} \right) \right] + \sum_{k,j} b_j x_{k,j,T+1} \right\}$$

$$\begin{aligned} \text{s.t.} \quad & x_{k,j,1} = x_{k,j}, \quad \forall k,j \\ & y_{k,j,\tau} = x_{k,j,\tau} + D_{k,j,\tau}, \quad \forall k,j,\tau \\ & x_{k,j,\tau+1} = y_{k+1,j,\tau} - \sum_{i,z} u_{k+1,j,z,i,\tau}, \quad \forall k < K, j, \tau \\ & x_{K,j,\tau+1} = 0, \quad \forall j, \tau \\ & \sum_{i,z} u_{k,j,z,i,\tau} \leq y_{k,j,\tau}, \quad \forall k,j,\tau \\ & u_{k,j,z,i,\tau} = 0, \quad \forall i,z, (k,j) \notin S_{i,z} \\ & u_{k,j,i,\tau} \geq 0, u_{k,j,i,\tau} \in \mathbb{Z} \quad \forall k,j,i,\tau \\ & S_{j,\tau}(t) \in \mathcal{N}, \quad \forall j, \tau, t \in [0,1] \end{aligned}$$

It is evident that the multi-zone extension modifies only the fulfillment side of the problem. The demand side, including customer arrivals, discrete choice probabilities $\pi_k^j(S)$, and option display decisions $S_{j,\tau}(t)$, is entirely unaffected. Consequently, all theoretical results related to the demand side remain the same, and the **ERT** heuristic carries over with straightforward modifications.

Efficient option lists (Propositions 2 and 3, Lemma EC.8). The deterministic relaxation **DET** extends by replacing $\sum_i C_i(w^\top u_{i,\tau})$ with $\sum_{i,z} C_{i,z}(w^\top u_{i,z,\tau})$ and appending the linear lead time constraint (EC.13). Since each $C_{i,z}$ remains convex and the lead time constraint is linear, the Lagrangian structure underlying Proposition 2 is preserved: efficient option lists are still characterized by the parametric assortment optimization $\max_S R_j(S) - \theta_j Q_j(S)$ for some $\theta_j \geq 0$. Because Proposition 3 and Lemma EC.8 depend solely on the demand-side quantities $(R_j(S), Q_j(S))$ and are independent of the fulfillment cost structure, they apply verbatim.

ERT heuristic. The three steps of **ERT** (i.e., economic option screening, randomized rounding, and threshold fulfillment) extend in the natural way. Economic screening uses the same $\mathcal{F}(x)$ function with C_i replaced by $C_{i,z}$ and the lead time constraint imposed; convexity is preserved, so $\mathcal{F}(x)$ remains tractable. The rounding step assigns individual orders to (warehouse, zone) pairs, not to warehouses alone, and restricts them to pairs (i, z) with $(k, j) \in S_{i,z}$. The threshold fulfillment step treats each pair (i, z) as an independent fulfillment channel with its own implicit capacity $C'_{i,z}(w^\top u_{i,z,\tau})w_j \leq b_j$; that is, Lemma 1 applies to each (i, z) pair.

Performance guarantee (Theorem 1). The proof of Theorem 1 aggregates per-warehouse variance contributions via Cauchy–Schwarz: $\sum_{i=1}^M \sqrt{A_i} \leq \sqrt{M \sum_i A_i}$, where $A_i = \text{Var}(\sum_{k,j} w_j u_{k,j,i,\tau}^{\text{HYP}})$, and bounds $\sum_i A_i$ using the partition property $\sum_i c_{k,j,i} \leq 1$ from Lemma EC.4. The same argument applies when each (i, z) pair is treated as an independent channel: Cauchy–Schwarz gives $\sum_{(i,z)} \sqrt{A_{i,z}} \leq \sqrt{\bar{M} \sum_{(i,z)} A_{i,z}}$ with $\bar{M} := \sum_{i \in [M]} Z_i$ and Lemma EC.4 yields $\sum_{(i,z)} c_{k,j,z,i} \leq 1$, implying $\sum_{(i,z)} A_{i,z} \leq \bar{w}^2 K \Lambda N$. Consequently, the only change in the optimality gap is $\sqrt{M} \rightarrow \sqrt{\bar{M}}$:

$$V^*(x) - V^{\text{ERT}}(x) \leq \bar{w} \bar{C} \sqrt{\bar{M}} T \sqrt{K \Lambda N} = \mathcal{O}\left(T \sqrt{\bar{M} K \Lambda N}\right).$$

All asymptotic implications of Theorem 1 remain valid: **ERT** is asymptotically optimal as the number of demand locations N or maximum cycle arrival rate Λ grows. Notably, the scaling is $\sqrt{\bar{M}}$ in the total number of fulfillment channels rather than linear in $\bar{M}$, so adding shipping zones inflates the bound only mildly.

EC.4. Dynamic Program Formulation of OPT

let $V_\tau(x, t)$ denote the maximum total expected profit that can be obtained since time t during fulfillment cycle τ with x unfulfilled orders. Let δt satisfies $\lambda_{j,\tau}(t)\delta t \ll 1$ for all j . We can formulate the original problem as follows:

$$V_\tau(x, t) = \max_{S_j} \sum_j \lambda_{j,\tau}(t) \delta t \left[\sum_k \pi_k^j(S_j) (r_{k,j} + V_\tau(x + e_{k,j}, t + \delta t)) \right]$$

$$\begin{aligned}
& + \left(\sum_j \pi_{\emptyset}^j(S_j) + 1 - \sum_j \lambda_{j,\tau}(t) \delta t \right) V_{\tau}(x, t + \delta t) + o(\delta t) \\
& = \sum_j \lambda_{j,\tau}(t) \delta t \left[\max_{S_j} \sum_k \pi_k^j(S) (r_{k,j} - \Delta V_{\tau}^{k,j}(x, t + \delta t)) \right] + V_{\tau}(x, t + \delta t) + o(\delta t)
\end{aligned}$$

where the boundary condition is

$$V_{\tau}(y_{\tau}, 1) = \max_u - \left(\sum_i C_i(w^{\top} u_i) + \sum_j b_j \left(u_{1,j} - \sum_i u_{1,j,i} \right) \right) + V_{\tau+1}(x_{\tau+1})$$

where $x_{\tau+1}$ is the pre-choice state for the fulfillment cycle $\tau + 1$, and it is obtained by the system dynamics. $V_{\tau+1}(\cdot)$ is the maximized expected profit since cycle $\tau + 1$. In the DP formulation, $e_{k,j} \in \{0, 1\}^{KN}$ is a unit vector where the k, j -th element is one and the rest are zero, and

$$\Delta V_{\tau}^{k,j}(x, t) = V_{\tau}(x, t) - V_{\tau}(x + e_{k,j}, t)$$

represents the difference in expected profit with or without this additional order $e_{k,j}$. It can be interpreted as a measure of the marginal fulfillment cost of that order. Note that with one more order to fulfill, $V_{\tau}(x + e_{k,j}, t) \leq V_{\tau}(x, t)$, and hence $\Delta V(x, t) \geq 0$. Taking limit as $\delta t \downarrow 0$, we know the expected profit $V_{\tau}(x, t)$ needs to satisfy the Hamilton-Jacobi-Bellman (HJB) equation:

$$\frac{\partial V_{\tau}(x, t)}{\partial t} = \sum_j \lambda_{j,\tau}(t) \left[\max_{S_j} \sum_k \pi_k^j(S) (r_{k,j} - \Delta V_{\tau}^{k,j}(x, t)) \right]$$

and that the optimal assortment to show to a customer from j should solve the following problem

$$\max_S \sum_k (r_{k,j} - \Delta V_{\tau}^{k,j}(x, t)) \pi_k^j(S)$$

which is exactly (2).

EC.5. Construct Efficient Option List under an MC Choice Model

We fix a customer j and suppress the subscript j throughout this section. Consider an MC choice model with K fulfillment options. $\lambda_k > 0$ represents the probability that option k is the first choice of a consumer, and $\rho_{k,k'} \geq 0$ is the probability of the next choice is k' given that her current choice k is not available, and $\sum_{k' \in [K]} \rho_{k,k'} < 1$ for all $k \in [K]$. According to Davis et al. (2014),

$$\pi_k(S) = \lambda_k + \lambda_W^{\top} (I - \rho_{W,W})^{-1} \rho_{W,S} e_k$$

where $W = [K] \setminus S$ is the non-offered option list, $\lambda_W = (\lambda_{k'} : k' \in W)$ is the vector of the first choice probability of non-offered options, $\rho_{S_1, S_2} = (\rho_{k,k'} : k \in S_1, k' \in S_2)$ for any $S_1, S_2 \subseteq [K]$ is the transition submatrix. Since $\sum_{k'} \rho_{k,k'} < 1$ for all k , the matrix $I - \rho_{W,W}$ is non-singular.

Note that an option list is efficient if and only if it solves $\max_S R(S) - \theta Q(S)$ for some $\theta \geq 0$. Under an MC model, Feldman and Topaloglu (2017) suggests the dual formulation of the problem, parameterized by $\theta \geq 0$:

$$\mathcal{D}(\theta) : \min_{v \in \mathbb{R}^K} \left\{ \sum_{k \in [K]} \lambda_k v_k : v_k \geq r_k - \theta \ \forall k \in [K], \quad v_k \geq \sum_{k' \in [K]} \rho_{k,k'} v_{k'} \ \forall k \in [K] \right\}. \quad (\text{EC.14})$$

Let v^θ be an optimal solution to the dual problem. We first show the following properties of v^θ . Note that all the proof is dedicated to Section EC.5.1.

LEMMA EC.6. *For any $\theta \geq 0$, v^θ is a unique solution for $\mathcal{D}(\theta)$.*

LEMMA EC.7. *The map $\theta \mapsto v^\theta$ is Lipschitz continuous and piecewise affine on $[0, \infty)$.*

Define the *dual-selected* option list

$$S^\theta := \{k \in [K] : v_k^\theta = r_k - \theta\}$$

According to Theorem 2 of Feldman and Topaloglu (2017), S^θ is optimal (not necessarily the only one) for

$$\max_{S \subseteq [K]} \Phi(S; \theta) := R(S) - \theta \cdot Q(S) \quad (\text{EC.15})$$

Moreover, S^θ possesses the following property.

LEMMA EC.8 (**Monotonic nesting**). *For any $0 \leq \theta_1 < \theta_2$, $S^{\theta_2} \subseteq S^{\theta_1}$.*

COROLLARY EC.1. *The family of dual-selected option lists $\{S^\theta : \theta \geq 0\}$ has at most $K + 1$ elements.*

The corollary only bounds the number of dual-selected option lists. However, for some $\theta \geq 0$, there may be multiple efficient option lists $S^*(\theta)$ other than S^θ . To bound the number of all efficient option lists requires further analysis. Let us fix a value for θ , and denote v^θ the unique optimal solution for $\mathcal{D}(\theta)$. Define

$$c_k^\theta := \sum_{l=1}^K \rho_{k,l} v_l^\theta$$

which shows the expected revenue when option k is not offered. For each option k , if it is offered to the consumer, it generates revenue $r_k - \theta$; if it is not offered, the consumer chooses other options according to the transition probability, and the generated revenue is c_k^θ . The optimal dual value satisfies

$$v_k^\theta = \max\{r_k - \theta, c_k^\theta\}$$

Next, we partition the fulfillment options into three sets:

$$\mathcal{A}(\theta) := \{k : r_k - \theta > c_k^\theta\}, \quad \mathcal{T}(\theta) := \{k : r_k - \theta = c_k^\theta\}, \quad \mathcal{C}(\theta) := \{k : r_k - \theta < c_k^\theta\}$$

Algorithm 3: Parametric Dual Sweep for Dual-selected Option Lists under MC.

Input: Fulfillment options $[K]$ with distinct revenues $\{r_1, \dots, r_K\}$, and MC parameters (λ, ρ) .**Output:** The set of all efficient option lists $\mathcal{E}$ and the set of breakpoints Θ .1 Set an arbitrary small $\varepsilon > 0$.2 Initialize $a \leftarrow 0$, $\mathcal{E} \leftarrow \{\emptyset\}$, and $\Theta \leftarrow \{0\}$.3 **while** $a < \infty$ **do**4 Set $a \leftarrow a + \varepsilon$.5 Solve $\mathcal{D}(a)$, obtain the dual optimal v^a and $S = \{k : v_k^a = r_k - a\}$.6 Calculate the parametric (by θ) gap function

$$g_k(\theta) = (r_k - \theta) - \sum_{k'=1}^K \rho_{k,k'} v_{k'}^\theta$$

where $v_S^\theta = r_S - \theta \cdot \mathbf{1}$, $v_W^\theta = (I - \rho_{W,W})^{-1} \rho_{W,S} (r_S - \theta \cdot \mathbf{1})$, $W = [K] \setminus S$.

7 Let $b = \inf\{\theta > a : g_k(\theta) = 0, \forall k \in S \text{ such that } g_k(\theta) \neq 0\}$.8 Set $\mathcal{E} \leftarrow \mathcal{E} \cup \{S\}$, $\Theta \leftarrow \Theta \cup \{b\}$.9 Set $a \leftarrow b$.10 **return** $\mathcal{E}$ and Θ .

So, if $k \in \mathcal{A}(\theta)$, the dual strictly prefers offer this option; if $k \in \mathcal{C}(\theta)$, the dual strictly prefers not offer this option; if $k \in \mathcal{T}(\theta)$, the dual is indifferent between the two actions. Importantly, we have the following results

PROPOSITION EC.1 (Decomposition). *For every $\theta \geq 0$, $\arg \max_S \Phi(S; \theta) = \{\mathcal{A}(\theta) \cup U : U \subseteq \mathcal{T}(\theta)\}$.*

It becomes clear that the number of efficient option lists depends entirely on the indifference set $\mathcal{T}(\theta)$. The next assumption rules out indifference options over an interval of θ and multiple indifference options at a single point. The first part of the assumption says that the indifference sets are isolated in θ , and the second part says the indifference sets have at most one element. With such an assumption, we can show $\mathcal{E} = \{S^\theta : \theta \geq 0\}$. Consequently, $|\mathcal{E}| \leq K + 1$, and we can enumerate all the elements in polynomial time with Algorithm 3. Such results are formalized in Proposition 3 in the main text, and we re-state here (with consumer index j omitted) as a reference.

ASSUMPTION EC.2 (Simple breakpoints). (1) *For any $\theta_0 \geq 0$ such that $\mathcal{T}(\theta_0) \neq \emptyset$, there exists some $\varepsilon > 0$ such that $\mathcal{T}(\theta) \equiv \emptyset$ for all $\theta \in (\theta_0 - \varepsilon, \theta_0 + \varepsilon) \setminus \{\theta_0\}$; (2) $|\mathcal{T}_j(\theta)| \leq 1$ for all $\theta \geq 0$;*

PROPOSITION EC.2. *Under an MC choice model and Assumption EC.2, there exists a set of fulfillment option lists*

$$S_0 \supsetneq S_1 \supsetneq \cdots \supsetneq S_L = \emptyset$$

such that

$$\mathcal{E} = \{S_0, S_1, \dots, S_L\}$$

and $|\mathcal{E}| = L + 1 \leq K + 1$. Moreover, Algorithm 3 enumerates all elements in $\mathcal{E}$ in polynomial time.

REMARK EC.1. Algorithm 3 works even when Assumption EC.2 does not hold. However, the output is not the set of all efficient option lists but rather the set of dual-selected option lists $\{S^\theta : \theta \geq 0\}$. This is because Corollary EC.1 holds even without that assumption. After obtaining the set of breakpoints Θ , one can verify whether Assumption EC.2 holds by obtaining $\mathcal{T}(\theta)$ for all $\theta \in \Theta$. If it does not hold, the complete efficient option list can be recovered by

$$\mathcal{E} = \{S^\theta : \theta \geq 0\} \cup \bigcup_{\theta \in \Theta} \{\mathcal{A}(\theta) \cup U : U \subseteq \mathcal{T}(\theta)\}$$

Moreover, we can show that $|\mathcal{E}| \leq 1 + \sum_{\theta \in \Theta} (2^{|\mathcal{T}(\theta)|} - 1)$. In particular, even when Assumption EC.2 does not hold, $\mathcal{E}$ remains polynomially bounded whenever $\max_{\theta \in \Theta} |\mathcal{T}(\theta)| = \mathcal{O}(1)$.

REMARK EC.2. It is well known that a Multinomial Logit (MNL) choice model is a special case of an MC choice model (Blanchet et al. 2016). Moreover, with an MNL, Assumption EC.2 reduces to the requirement that each fulfillment option has distinct revenue for each consumer. Thus, the $K + 1$ worst-case bound still applies.

EC.5.1. Proof

EC.5.1.1. Proof of Lemma EC.6 We show this lemma by contradiction. Assume $x, y \in \mathbb{R}^K$ are both feasible for $\mathcal{D}(\theta)$. Define $z := x \wedge y$ (the coordinate-wise minimum), then z is also feasible to $\mathcal{D}(\theta)$. To see this, note that $z_k \geq r_k - \theta$ for all k by definition. For some k , if $z_k = x_k$, then $z_k = x_k \geq \sum_l \rho_{k,l} x_l \geq \sum_l \rho_{k,l} z_l$. The same argument for other k when $z_k = y_k$. Hence, z is feasible.

Now assume $x \neq y$ and both are optimal solutions. Then z would satisfy $z \leq x$, $z \leq y$, and $z_k < x_k$ or $z_k < y_k$ for at least one k . Because $\lambda_k > 0$ for all k , this gives

$$\sum_k \lambda_k z_k < \sum_k \lambda_k x_k = \sum_k \lambda_k y_k$$

which contradicts the fact that x and y are both optimal. Hence, v^θ is unique.

EC.5.1.2. Proof of Lemma EC.7 Let

$$T_\theta(v) := \max\{r - \theta \mathbf{1}, \rho v\}$$

where the maximum is taken coordinate-wise, and define

$$\beta := \max_{k \in [K]} \sum_{l=1}^K \rho_{k,l} < 1.$$

Because the map $x \mapsto \max\{a, x\}$ is non-expansive coordinate-wise and

$$\|\rho x - \rho y\|_\infty \leq \beta \|x - y\|_\infty,$$

we have, for any $x, y \in \mathbb{R}^K$,

$$\|T_\theta(x) - T_\theta(y)\|_\infty \leq \beta \|x - y\|_\infty.$$

Hence T_θ is a contraction under the sup norm. By Banach's fixed-point theorem, it admits a unique fixed point, denoted by u^θ .

We next show that u^θ is exactly the unique optimizer of $\mathcal{D}(\theta)$. A vector v is feasible for $\mathcal{D}(\theta)$ if and only if

$$v \geq r - \theta \mathbf{1} \quad \text{and} \quad v \geq \rho v,$$

which is equivalent to $v \geq T_\theta(v)$. Since T_θ is monotone, any feasible v satisfies

$$v \geq T_\theta(v) \geq T_\theta^2(v) \geq \dots \geq u^\theta,$$

where the last limit follows from the contraction property. Therefore, u^θ is the coordinate-wise minimal feasible vector. Because the objective of $\mathcal{D}(\theta)$ is $\sum_k \lambda_k v_k$ with $\lambda_k > 0$ for all k , the coordinate-wise minimal feasible vector is also the unique optimal solution. Thus $u^\theta = v^\theta$.

To prove continuity, take any $\theta_1, \theta_2 \geq 0$. Using the fixed-point identities $v^{\theta_i} = T_{\theta_i}(v^{\theta_i})$, we obtain

$$\begin{aligned} \|v^{\theta_1} - v^{\theta_2}\|_\infty &= \|T_{\theta_1}(v^{\theta_1}) - T_{\theta_2}(v^{\theta_2})\|_\infty \\ &\leq \|T_{\theta_1}(v^{\theta_1}) - T_{\theta_1}(v^{\theta_2})\|_\infty + \|T_{\theta_1}(v^{\theta_2}) - T_{\theta_2}(v^{\theta_2})\|_\infty \\ &\leq \beta \|v^{\theta_1} - v^{\theta_2}\|_\infty + |\theta_1 - \theta_2|. \end{aligned}$$

Rearranging yields

$$\|v^{\theta_1} - v^{\theta_2}\|_\infty \leq \frac{|\theta_1 - \theta_2|}{1 - \beta},$$

so $\theta \mapsto v^\theta$ is Lipschitz continuous.

We now prove piecewise affinity. Fix any subset $S \subseteq [K]$ and let $W = [K] \setminus S$. Suppose that $S^\theta = S$ for some θ . By definition of S^θ ,

$$v_k^\theta = r_k - \theta \quad \forall k \in S, \quad v_k^\theta = \sum_{l=1}^K \rho_{k,l} v_l^\theta \quad \forall k \in W.$$

Equivalently,

$$v_S^\theta = r_S - \theta \mathbf{1}, \quad (I - \rho_{W,W})v_W^\theta = \rho_{W,S}(r_S - \theta \mathbf{1}).$$

Since every row sum of $\rho_{W,W}$ is strictly less than one, $I - \rho_{W,W}$ is nonsingular. Therefore,

$$v_W^\theta = (I - \rho_{W,W})^{-1} \rho_{W,S}(r_S - \theta \mathbf{1}),$$

which is affine in θ . Hence, in any region where S^θ is fixed and equal to S , the whole vector v^θ is affine in θ .

Conversely, for this fixed subset S , define the affine candidate path $v^S(\theta)$ by

$$v_S^S(\theta) = r_S - \theta \mathbf{1}, \quad v_W^S(\theta) = (I - \rho_{W,W})^{-1} \rho_{W,S}(r_S - \theta \mathbf{1}).$$

Then $S^\theta = S$ holds exactly when the sign conditions

$$\begin{aligned} (r_k - \theta) - \sum_{l=1}^K \rho_{k,l} v_l^S(\theta) &\geq 0 \quad \forall k \in S, \\ (r_k - \theta) - \sum_{l=1}^K \rho_{k,l} v_l^S(\theta) &\leq 0 \quad \forall k \in W \end{aligned}$$

are satisfied. Each inequality is affine in θ , so the set of parameters supporting the pattern S is an interval (possibly empty). Since there are only finitely many subsets $S \subseteq [K]$, the domain $[0, \infty)$ is covered by finitely many such intervals, and v^θ is affine on each of them. Thus $\theta \mapsto v^\theta$ is piecewise affine.

EC.5.1.3. Proof of Lemma EC.8 we show the monotonic nesting property by contradiction. Let $\delta := \theta_2 - \theta_1 > 0$. Define

$$w_k := v_k^{\theta_2} + \delta \quad \forall k$$

Then, w is feasible for $\mathcal{D}(\theta_1)$. To see that, note for each k ,

$$w_k = v_k^{\theta_2} + \delta \geq (r_k - \theta_2) + \delta = r_k - \theta_1$$

Also,

$$w_k - \sum_l \rho_{k,l} w_l = \left(v_k^{\theta_2} - \sum_l \rho_{k,l} v_l^{\theta_2} \right) + \delta \left(1 - \sum_l \rho_{k,l} \right) \geq 0$$

because v^{θ_2} is feasible for $\mathcal{D}(\theta_2)$ and $\sum_l \rho_{k,l} \leq 1$.

Define $v := v^{\theta_1} \wedge w$. By the same coordinate-wise minimum argument, v is feasible for $\mathcal{D}(\theta_1)$. Suppose there exists some option k such that $k \in S^{\theta_2}$ and $k \notin S^{\theta_1}$. Then,

$$v_k^{\theta_2} = r_k - \theta_2 \quad \Rightarrow \quad w_k = v_k^{\theta_2} + \delta = r_k - \theta_1$$

But since $k \notin S^{\theta_1}$,

$$v_k^{\theta_1} > r_k - \theta_1 = w_k$$

Hence,

$$v_k = w_k < v_k^{\theta_1}$$

Since $v \leq v^{\theta_1}$ coordinate-wise and the inequality is strict at k , we get

$$\sum_l \lambda_l v_l < \sum_l \lambda_l v_l^{\theta_1}$$

contradicting the optimality of v^{θ_1} for $\mathcal{D}(\theta_1)$. Therefore, no such k exists, and

$$S^{\theta_2} \subseteq S^{\theta_1}$$

EC.5.1.4. Proof of Corollary EC.1 By Lemma EC.8, the family $\{S^\theta : \theta \geq 0\}$ is nested. Every strict inclusion removes at least one option; thus, there can be at most $K + 1$ distinct sets.

It only remains to observe that $S^\theta = \emptyset$ for all sufficiently large θ . Indeed, if $S \neq \emptyset$, for any $\theta > \max r_k$,

$$\Phi(S; \theta) = R(S) - \theta Q(S) = \sum_{k \in S} (r_k - \theta) \pi_k(S) < 0 = \Phi(\emptyset; \theta)$$

Hence, the empty set is optimal for all sufficiently large θ , and the bound follows.

EC.5.1.5. Proof of Proposition EC.1 For any fulfillment option list S , define

$$u_k^S(\theta) := \begin{cases} r_k - \theta & k \in S \\ \sum_{l=1}^K \rho_{k,l} u_l^S(\theta) & k \notin S \end{cases}$$

We can show that $\Phi(S; \theta) = \sum_k \lambda_k u_k^S(\theta)$. To see that, define

$$\tilde{u}_k := \sum_{h \in S} (r_h - \theta) P_k^S(h)$$

where $P_k^S(h)$ denotes the probability that the Markov chain, starting from state k , eventually ends with the purchase of option $h \in S$. If $k \in S$, then $P_k^S(k) = 1$ and $P_k^S(h) = 0$ for all $h \neq k$, and hence

$$\tilde{u}_k = r_k - \theta$$

If $k \notin S$, we have

$$P_k^S(h) = \sum_l \rho_{k,l} P_l^S(h)$$

Then,

$$\tilde{u}_k = \sum_{h \in S} (r_h - \theta) \sum_l \rho_{k,l} P_l^S(h) = \sum_l \rho_{k,l} \sum_{h \in S} (r_h - \theta) P_l^S(h) = \sum_l \rho_{k,l} \tilde{u}_l$$

Clearly, both $\tilde{u}$ and u^S satisfy the same boundary condition on S and the same linear recursion on the complementary set of S . Therefore,

$$u_k^S(\theta) = \tilde{u}_k = \sum_{h \in S} (r_h - \theta) P_k^S(h)$$

Consequently,

$$\sum_{k=1}^K \lambda_k u_k^S(\theta) = \sum_{k=1}^K \lambda_k \sum_{h \in S} (r_h - \theta) P_k^S(h) = \sum_{h \in S} (r_h - \theta) \sum_{k=1}^K \lambda_k P_k^S(h) = \sum_{h=1}^K (r_h - \theta) \pi_h(S) = \Phi(S; \theta).$$

Denote v^θ the optimal solution for $\mathcal{D}(\theta)$. We can show that $u_k^S \leq v_k^\theta$ for all k using the dual feasibility of v^θ . Hence,

$$\Phi(S; \theta) \leq \sum_k \lambda_k v_k^\theta$$

By Strong duality, we have

$$\max_S \sum_k \lambda_k u_k^S(\theta) = \max_S \Phi(S; \theta) = \sum_k \lambda_k v_k^\theta$$

Since $\lambda_k > 0$, equality in the weighted sum implies coordinate-wise equality. Therefore, S maximizes $\Phi(S; \theta)$ if and only if

$$u^S = v^\theta$$

We first show that the right-hand side is a subset of the left-hand side in two steps. In the first step, we show every efficient S must satisfy $\mathcal{A}(\theta) \subseteq S$ by contradiction. Suppose $S \in \arg \max_S \Phi(S; \theta)$ but exists $k \in \mathcal{A}(\theta)$ and $k \notin S$. Since S is a maximizer, we have

$$v_k^\theta = u_k^S = \sum_l \rho_{k,l} u_l^S = \sum_l \rho_{k,l} v_l^\theta = c_k^\theta$$

But since $k \in \mathcal{A}(\theta)$,

$$v_k^\theta = r_k - \theta > c_k^\theta$$

a contradiction.

In the second step, we show $S \cap \mathcal{C}(\theta) = \emptyset$ by contradiction. Take any $k \in \mathcal{C}(\theta)$. Suppose S is a maximizer and $k \in S$. Then

$$v_k^\theta = u_k^S = r_k - \theta$$

But since $k \in \mathcal{C}(\theta)$,

$$v_k^\theta = c_k^\theta > r_k - \theta$$

a contradiction.

Combining the two steps, every efficient option list must be of the form

$$S = \mathcal{A}(\theta) \cup U, \text{ for some } U \subseteq \mathcal{T}(\theta)$$

We then show that the left-hand side is a subset of the right-hand side. We fix any $U \subseteq \mathcal{T}(\theta)$ and let $S := \mathcal{A}(\theta) \cup U$. we show that $u^S = v^\theta$.

For $k \in S$, then either $k \in \mathcal{A}(\theta)$ or $k \in U$, and in both cases,

$$v_k^\theta = r_k - \theta = u_k^S$$

For any $k \notin S$, then either $k \in \mathcal{T}(\theta) \setminus U$ or $k \in \mathcal{C}(\theta)$. In both cases,

$$v_k^\theta = c_k^\theta = \sum_l \rho_{k,l} v_l^\theta$$

Since $k \notin S$, we have $u_k^S = \sum_l \rho_{k,l} u_l^S$. But we already have equality for offered states, and the same recursive relation for unoffered states, so by the uniqueness of the linear system on the continuation states,

$$v_k^\theta = u_k^S$$

Hence, $u_k^S = v_k^\theta$ for all k , and S is a maximizer.

EC.5.1.6. Proof of Proposition 3 Denote $\mathcal{S}^{\text{dual}} := \{S^\theta : \theta \geq 0\}$. By the above results, we have $\mathcal{S}^{\text{dual}} = \{S_0, S_1, \dots, S_L\}$ such that $S_0 \supsetneq S_1 \supsetneq \dots \supsetneq S_L = \emptyset$ and $L \leq K$. It only remains to show $\mathcal{E} = \mathcal{S}^{\text{dual}}$. Since S^θ is optimal for $\Phi(\cdot; \theta)$ for every $\theta \geq 0$, by Feldman and Topaloglu (2017, Theorem 2), we have $\mathcal{S}^{\text{dual}} \subseteq \mathcal{E}$.

To prove the reverse inclusion, take any $S \in \mathcal{E}$. Then there must exist $\bar{\theta} \geq 0$ such that

$$S \in \arg \max_{S'} \Phi(S'; \bar{\theta}) = \{\mathcal{A}(\bar{\theta}) \cup U : U \subseteq \mathcal{T}(\bar{\theta})\}$$

Under Assumption EC.2, $|\mathcal{T}(\bar{\theta})| \leq 1$.

If $\mathcal{T}(\bar{\theta}) = \emptyset$, then the maximizer is unique and hence

$$S = S^{\bar{\theta}} \in \mathcal{S}^{\text{dual}}$$

Now suppose $\mathcal{T}(\bar{\theta}) = \{k\}$ for some option k . Then the optimal solutions are $\mathcal{A}(\bar{\theta})$ and $\mathcal{A}(\bar{\theta}) \cup \{k\} = S^{\bar{\theta}}$. Therefore, it suffices to show that $\mathcal{A}(\bar{\theta})$ is also dual-selected for some nearby parameter value.

Since $\mathcal{T}(\bar{\theta}) = \{k\}$, we have $g_k(\bar{\theta}) = 0$ and $g_l(\bar{\theta}) \neq 0$ for all $l \neq k$. By Lemma EC.7, there exists $\varepsilon > 0$ such that the sign of $g_l(\theta)$ does not change on $(\bar{\theta} - \varepsilon, \bar{\theta} + \varepsilon)$ for every $l \neq k$. Thus, in a neighborhood of $\bar{\theta}$, only option k can change status.

By Lemma EC.8, $k \in S^{\bar{\theta} - \varepsilon}$. Suppose, by contradiction, that $k \in S^{\bar{\theta} + \varepsilon}$ as well. Since no other option changes status near $\bar{\theta}$, the selected set would be the same on both sides of $\bar{\theta}$.

On that neighborhood, $g_k(\theta)$ would therefore be an affine function (Lemma EC.7), nonnegative on both sides, and equal to zero at $\bar{\theta}$. This is only possible if $g_k(\theta) = 0$ on the whole neighborhood, contradicting the isolation part of Assumption EC.2. Hence $k \notin S^{\bar{\theta} + \varepsilon}$.

Therefore,

$$S^{\bar{\theta}-\varepsilon} = \mathcal{A}(\bar{\theta}) \cup \{k\} \quad \text{and} \quad S^{\bar{\theta}+\varepsilon} = \mathcal{A}(\bar{\theta}).$$

and we conclude that $\mathcal{E} \subseteq \mathcal{S}^{\text{dual}}$. Combining the two inclusions gives

$$\mathcal{E} = \mathcal{S}^{\text{dual}} = \{S_0, S_1, \dots, S_L\}, \quad |\mathcal{E}| = L + 1 \leq K + 1.$$

It remains to justify the enumeration claim. Start at $\theta = 0$ and record the set S^0 . If $\mathcal{T}(0) = \{k\}$, then the same breakpoint argument used above shows that the right-limit selected set is $\mathcal{A}(0) = S^0 \setminus \{k\}$; otherwise, the right-limit selected set is simply S^0 . Call this right-limit set the *active set* and denote it by $\widehat{S}$.

Suppose the current breakpoint is a and the active set on the interval immediately to the right of a is $\widehat{S}$. By the continuity of $g_k(\theta)$, next breakpoint is

$$b = \inf\{\theta > a : g_k(\theta) = 0, \forall k \in \widehat{S} \text{ such that } g_k(\theta) \neq 0\}.$$

Note that any option k such that the gap function $g_k(\theta)$ constantly equals zero is removed since it does not contribute to finding the next breakpoint. By Assumption EC.2, exactly one option leaves the active set at b . Thus, each iteration removes one option, so the sweep performs at most K iterations. During each iteration, the dual problem $\mathcal{D}(\cdot)$ with K decision variables and $2K$ constraints is solved exactly once. Thus, Algorithm 3 can finish in polynomial time.

EC.6. Construct Efficient Option Lists under a Nested Logit Choice Model

We fix a customer j and suppress the subscript j throughout this section. The set of fulfillment options is partitioned into H nests, denoted by $\mathcal{K}_1, \dots, \mathcal{K}_H$, where nest n contains K_n options and $\sum_{n=1}^H K_n = K$. For option $k \in \mathcal{K}_n$, let $v_{n,k} > 0$ be its attraction parameter and let $\gamma_n \in (0, 1]$ be the dissimilarity parameter of nest n . We assume the no-purchase alternative forms its own singleton nest, so once a customer selects a nest, she purchases one of the offered options in that nest. For an option list S , define $S_n := S \cap \mathcal{K}_n$ and

$$V_n(S_n) := \sum_{k \in S_n} v_{n,k}.$$

Under the nested logit model, the probability that option $k \in S_n$ is chosen from the list S is

$$\pi_k(S) = \frac{V_n(S_n)^{\gamma_n-1} v_{n,k}}{v_0 + \sum_{m=1}^H V_m(S_m)^{\gamma_m}},$$

where $v_0 > 0$ is the attraction of the no-purchase alternative. Therefore, the purchase probability and expected revenue can be written as

$$Q(S) = \frac{\sum_{n=1}^H V_n(S_n)^{\gamma_n}}{v_0 + \sum_{m=1}^H V_m(S_m)^{\gamma_m}},$$

$$R(S) = \frac{\sum_{n=1}^H V_n(S_n)^{\gamma_n-1} \sum_{k \in S_n} r_k v_{n,k}}{v_0 + \sum_{m=1}^H V_m(S_m)^{\gamma_m}}.$$

Hence, for each $\theta \geq 0$,

$$\Phi(S; \theta) = R(S) - \theta Q(S) = \frac{\sum_{n=1}^H V_n(S_n)^{\gamma_n-1} \sum_{k \in S_n} (r_k - \theta) v_{n,k}}{v_0 + \sum_{m=1}^H V_m(S_m)^{\gamma_m}}. \quad (\text{EC.16})$$

According to Davis et al. (2014), an optimal solution to the above parametric assortment optimization problem is *revenue-ordered within each nest*. Therefore, assuming the revenues within each nest are distinct, we can re-index the option within each nest so that

$$r_{n,1} > r_{n,2} > \dots > r_{n,K_n}, \quad n = 1, \dots, H.$$

For $i_n \in \{0, \dots, K_n\}$, define the prefix assortment in nest n by

$$S_n(i_n) := \{1, \dots, i_n\},$$

with the convention $S_n(0) = \emptyset$. For an index vector

$$\mathbf{i} = (i_1, \dots, i_H) \in \mathcal{I} := \prod_{n=1}^H \{0, \dots, K_n\},$$

define the candidate option list

$$S(\mathbf{i}) := \bigcup_{n=1}^H S_n(i_n).$$

Hence, every efficient option list must be one of these candidates. For later use, define the prefix sums

$$A_n(i) := \sum_{\ell=1}^i v_{n,\ell}, \quad B_n(i) := \sum_{\ell=1}^i r_{n,\ell} v_{n,\ell}, \quad A_n(0) = B_n(0) = 0. \quad (\text{EC.17})$$

For any $\mathbf{i} \in \mathcal{I}$, let

$$D(\mathbf{i}) := v_0 + \sum_{n=1}^H A_n(i_n)^{\gamma_n}.$$

Then the purchase probability and expected revenue associated with $S(\mathbf{i})$ are

$$Q(S(\mathbf{i})) = \frac{\sum_{n=1}^H A_n(i_n)^{\gamma_n}}{D(\mathbf{i})}, \quad (\text{EC.18})$$

$$R(S(\mathbf{i})) = \frac{\sum_{n=1}^H A_n(i_n)^{\gamma_n-1} B_n(i_n)}{D(\mathbf{i})}, \quad (\text{EC.19})$$

where we interpret the term $A_n(0)^{\gamma_n-1} B_n(0)$ as zero.

PROPOSITION EC.3. *Every efficient option list is equal to $S(\mathbf{i})$ for some $\mathbf{i} \in \mathcal{I}$. Consequently,*

$$|\mathcal{E}| \leq |\mathcal{I}| = \prod_{n=1}^H (K_n + 1) \leq (K + 1)^H.$$

Moreover, all efficient option lists can be identified by Algorithm 4, with time complexity

$$\mathcal{O}\left(K \log K + H \prod_{n=1}^H (K_n + 1) + \prod_{n=1}^H (K_n + 1) \log \left(\prod_{n=1}^H (K_n + 1) \right)\right).$$

In particular, if H is fixed, then the number of efficient option lists and the running time are polynomial in K .

Proof. Recall that an option list is efficient if and only if it maximizes $\Phi(S; \theta) = R(S) - \theta Q(S)$ for some $\theta \geq 0$. Fix such a value of θ and let S be any maximizer. By Davis et al. (2014) Theorem 4, S is revenue-ordered within each nest. Hence there exists $\mathbf{i} \in \mathcal{I}$ such that $S = S(\mathbf{i})$. This proves that every efficient option list is one of the $|\mathcal{I}|$ candidates, and therefore

$$|\mathcal{E}| \leq |\mathcal{I}| = \prod_{n=1}^H (K_n + 1).$$

Now fix a candidate $\mathbf{i} \in \mathcal{I}$. Since the nested-logit choice probabilities do not depend on θ , both $R(S(\mathbf{i}))$ and $Q(S(\mathbf{i}))$ are constants. Therefore,

$$f_{\mathbf{i}}(\theta) = R(S(\mathbf{i})) - \theta Q(S(\mathbf{i}))$$

is affine in θ . A candidate $S(\mathbf{i})$ is efficient if and only if there exists $\theta \geq 0$ such that

$$f_{\mathbf{i}}(\theta) = \max_{\mathbf{i}' \in \mathcal{I}} f_{\mathbf{i}'}(\theta).$$

Thus, the efficient option lists are exactly the candidates whose associated affine functions appear on the upper envelope of the family $\{f_{\mathbf{i}} : \mathbf{i} \in \mathcal{I}\}$ over $[0, \infty)$.

It remains to justify that the running time is polynomial. First, sorting the options within each nest by revenue requires $\mathcal{O}(K \log K)$ time in total. Second, after the prefix sums $A_n(i)$ and $B_n(i)$ are precomputed, evaluating $R(S(\mathbf{i}))$ and $Q(S(\mathbf{i}))$ for a given $\mathbf{i}$ takes $\mathcal{O}(H)$ time. Hence, all candidate lines can be constructed in

$$\mathcal{O}\left(H \prod_{n=1}^H (K_n + 1)\right)$$

time. Finally, the upper envelope of $N := \prod_{n=1}^H (K_n + 1)$ affine functions on the real line can be computed in $\mathcal{O}(N \log N)$ time after sorting by slope. If several candidates induce the same affine function, we keep the full bucket of associated option lists. Combining the three steps yields the stated complexity bound. $\square$

EC.7. Largest Marginal Revenue Algorithm

For choice models outside the MC and NL classes, such as the mixture of multinomial logit (MMNL) choice models, we can apply Algorithm 5 to construct all efficient option lists. Unfortunately, the assortment optimization subproblem under an MMNL is known to be NP-hard in general (Bront et al. 2009, Rusmevichientong et al. 2014), precluding analogous worst-case guarantees.

Nonetheless, we conduct a numerical study using synthetic data to evaluate the number of efficient option lists under these models. We include both MNL and NL choice models as benchmarks. An MNL model is a special case of an MC choice model and thus has a provable linear worst-case bound, while an NL choice model has a polynomial worst-case bound.

We vary the number of fulfillment options K from 3 to 20. For each value of K , we generate 100 random instances. For each instance, we compute the number of efficient option lists and report the maximum observed value across all instances. We consider only a single consumer type and omit the index j . Each instance is generated as follows:

Surcharge. We generate surcharges r_k for $k \in [K]$ using a uniform distribution $\mathcal{U}(0, 1)$ and make sure that all values are distinct.

MNL. An MNL choice model is defined as

$$\pi_k(S) = \frac{v_k}{v_0 + \sum_{k' \in S} v_{k'}}$$

where v_0 is the attraction of outside options, and v_k for $k \in [K]$ is the attraction for option k . To generate each random instance, we sample the attraction values from a log-normal distribution. Specifically, for each option $k \in [K] \cup \{0\}$, we draw a standard random variable Z_k , and compute the corresponding attraction as $v_k = e^{Z_k}$.

NL. When fulfillment options can be grouped such that options within each group are close substitutes, customer behavior can be modeled using an NL choice model. We consider an NL model with two nests, where the full option list S can be partitioned into two disjoint subsets: $S = S_1 \cup S_2$, with $S_1 \cap S_2 = \emptyset$. The attraction value of fulfillment option k in nest $n = 1, 2$ is given by v_{nk} . The customer will choose an option k from nest $n = 1, 2$ with probability

$$\pi_k(S) = Q_n(S_1, S_2) q_{k|n}(S_n) = \frac{V_n(S_n)^{\gamma_n}}{v_0 + \sum_{m=1,2} V_m(S_m)^{\gamma_m}} \frac{v_{kn}}{V_n(S_n)}$$

where $V_n(S_n) = \sum_{k \in S_n} v_{kn}$, and the parameter γ_n is a measure of how easily the products in nest n substitute for each other. To generate each instance, we randomly assign each fulfillment option k to one of the two nests, with equal probability. The nesting parameters γ_1 and γ_2 are independently drawn from an exponential distribution with parameter 1. For attractions, we generate them using a standard lognormal distribution, as in MNL.

MMNL. We consider an MMNL with two types of customers, which is defined as follows:

$$\pi_k(S) = \sum_{g \in \{1,2\}} \alpha^g \frac{v_k^g}{v_0^g + \sum_{k' \in S} v_{k'}^g}$$

where α^g is the proportion of customer being type g and v_k^g is the attraction of option k ($k = 0$ indicates the outside option) for g -type customer. To generate each instance, we draw α^1 from a uniform distribution $\mathcal{U}(0,1)$ and compute $\alpha^2 = 1 - \alpha^1$. The attraction values v_k^g for each k and g are generated using the same method as in the MNL choice model.

As shown in Table EC.1, the number of efficient option lists grows approximately linearly in K – not only under the MNL model, but also under the more complex MMNL and NL models. In addition, Figure EC.1 reports the average CPU runtime required to identify all efficient option lists using Algorithm 5. When the MNL model is used, the algorithm runs efficiently, consistent with the guarantee provided by Proposition 3. Even though the maximum number of efficient options grows exponentially in K under the MMNL and NL models, our algorithm remains highly efficient in practice. As shown in Fig EC.1, the time complexity of our algorithm increases linearly in K . These results demonstrate the algorithm’s practical applicability for problems of realistic size.

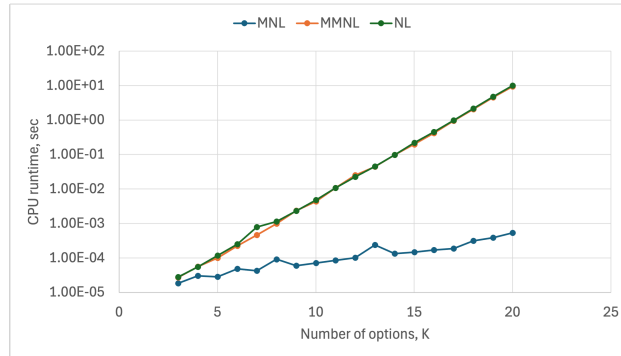


Figure EC.1 CPU runtime for identifying all efficient option lists using Algorithm 5 under various choice models.

EC.8. Benchmark Policies

The first benchmark presents a pre-determined and fixed option list to customers throughout the horizon and determines the fulfillment decisions *myopically* by solving the following problem. It is referred to as the static-myopic policy (**SM**).

$$\begin{aligned} \min_u \quad & \sum_i C_i(u_i) + \sum_j b_j \left(y_{k,j} - \sum_i u_{k,j,i} \right) \\ \text{s.t.} \quad & \sum_i u_{k,j,i} \leq y_{k,j}, \quad u_{k,j,i,\tau} \geq 0, \quad u_{k,j,i,\tau} \in \mathbb{Z} \end{aligned}$$

The fixed list strategy is selected as follows: We evaluate all possible option lists for each simulation case and choose the one with the best performance as the fixed list for that particular case. This policy mimics a seller’s operations when they know customer preferences but does not optimize the assortment or fulfillment decisions.

The second benchmark policy only optimizes its option lists but not the fulfillment decisions. It solves the following assortment optimization problem under capacity constraints (specified in §6.2.2) to determine the option lists, while fulfillment decisions are made in the same manner as in **SM**.

$$\begin{aligned}
& \max_{\gamma} \quad \sum_{\tau} \sum_j \sum_k \sum_S r_{k,j} \Lambda_{j,\tau} \gamma_{j,\tau}(S) \pi_k(S) \\
& \text{s.t.} \quad \sum_j \sum_k \sum_S \Lambda_{j,\tau} \gamma_{j,\tau}(S) \pi_k(S) \leq \Xi, \quad \forall \tau \\
& \quad \sum_S \gamma_{j,\tau}(S) \leq 1, \quad \forall j, \tau \\
& \quad \gamma_{j,\tau}(S) \geq 0, \quad \forall j, \tau, S \in \mathcal{N}
\end{aligned}$$

Ξ specifies the fulfillment capacity for each cycle. We refer to this as the Optimized List policy (**OL**). It is important to note that, for each fulfillment cycle, **ERT** *dynamically* adjusts assortment decisions based on the remaining fulfillment capacity, whereas **OL** remains *stationary*.

In contrast to the second policy, the last one only optimizes the fulfillment decisions. It fixes an option list and solves **DET**, then determines the actual fulfillment in the same manner as **ERT**. It may adopt a different, but best, fixed list for different cases. We refer to it as the Optimized Fulfillment policy (**OF**).

EC.8.1. Supplementary Policies

The first supplementary policy is a randomized rounding fulfillment policy based on the **HYP** policy (**r-HYP**); see Algorithm 6. Lines 5–8 of the algorithm ensure its feasibility. The randomized rounding function $h(\cdot)$ is defined in equation (7). Notably, the **r-HYP** policy does not remove economically nonviable options, nor does it guarantee that the total fulfillment volume respects the implicit fulfillment capacity.

The second supplementary policy is the Economically Randomized Rounding policy (**er-HYP**), which extends **r-HYP** by removing economically nonviable options prior to showing to customers. This allows us to isolate and evaluate the impact of excluding nonviable options by comparing the performance of **r-HYP** and **er-HYP**. Furthermore, by comparing **er-HYP** to **ERT**, we can assess the additional value of imposing threshold-based fulfillment decisions.

EC.9. Additional Computational Experiments

EC.9.1. Varying the Length of Horizon

In this section, we conduct a numerical study to illustrate how the performance of the proposed policy varies with respect to the time horizon T . We vary the horizon from 30 to 70 weeks in increments of 10 weeks. Our results from this numerical study are provided in Figure EC.2. The figure shows that the profit loss $V^D(x) - V^{\text{ERT}}(x)$ grows linearly in T , which aligns with our theoretical performance guarantee.

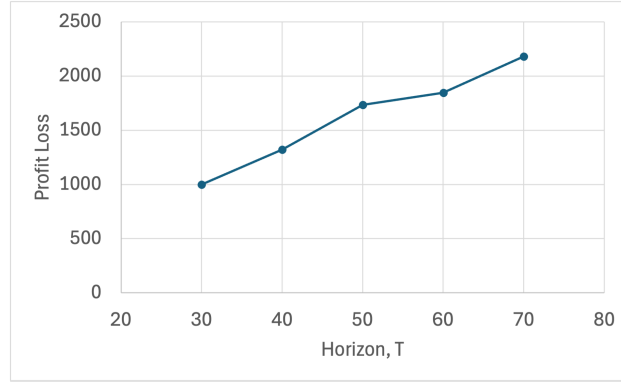


Figure EC.2 Changes of the profit loss with respect to the horizon length T .

EC.9.2. Varying the number of warehouses

We vary the number of warehouses while maintaining a constant demand-to-fulfillment-capacity ratio to investigate the impact of M on policy performance. As shown in Figure EC.3, we change M from 1 to 4 and investigate the changes in the profit loss $V^D(x) - V^{\text{ERT}}(x)$. Unlike the theoretical results, which suggest that the profit loss increases sub-linearly in M , the simulation results indicate that performance is barely affected. These results highlight that the theoretical bound is not tight in M .

We attribute the remaining slack to two algorithmic features of the **ERT** algorithm that the worst-case analysis does not fully exploit. First, the threshold fulfillment rule (namely, “T” in the **ERT** algorithm) caps each warehouse’s contribution by stopping fulfillment once the marginal cost exceeds the penalty b_j , thereby limiting the realized variance below the partition bound. Second, the dynamic elimination of nonviable options (namely, “E” in **ERT** algorithm) removes high-cost states before they can occur, so the variance of **ERT** is in fact strictly smaller than that of **HYP**, which is the policy through which our worst-case bound is derived.

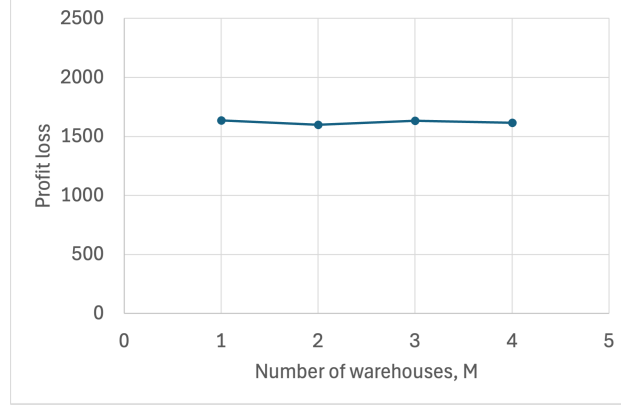


Figure EC.3 Changes of the profit loss with respect to the number of warehouses M .

EC.9.3. Varying Fulfillment Window

As discussed in the paper, the impact of the maximum length of fulfillment window K depends on customer preferences and the structure of the fulfillment surcharges. To numerically verify this theoretical insight, we construct two illustrative cases using an MNL choice model. In Case 1, the set of efficient option lists does not depend on K ; In Case 2, it does. The parameters for each case are defined as follows. For case 1,

$$r_k^1 = 11 - k, \quad v_k^1 = 100$$

For case 2,

$$r_k^2 = 10 + K - k, \quad v_k^2 = 100 + 50 \times (K - k)$$

In both cases, the no-purchase option has fixed utility: $v_0^1 = v_0^2 = 1$. In other words, in case 1, each option has equal choice probability, while faster options are charged with higher prices. In contrast, in case 2, slower options are charged higher prices and are less attractive to customers.

We then change the value of K from 2 to 6 and report the resulting profit loss in Figure EC.4. As shown, the profit loss remains approximately constant in Case 1, confirming that the structure of efficient option lists is unaffected by K . In contrast, in Case 2, the profit loss increases sublinearly with K , indicating that the **ERT** policy does not fully exploit the extended fulfillment flexibility available in the optimal policy. These results empirically support our theoretical analysis of the impact of K .

EC.9.4. Supplementary Benchmark Policies

We compare the **ERT** policy to two supplementary policies based on the **HYP** policy: the randomized rounding HYP policy (**r-HYP**), and the economic option randomized rounding HYP policy (**er-HYP**) to evaluate the importance of removing economically nonviable options and ensuring

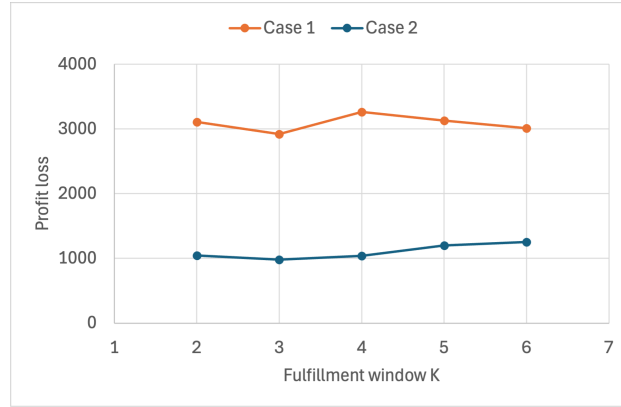


Figure EC.4 Changes of the profit loss with respect to the maximum length of fulfillment window K in various cases.

that fulfillment respects the implicit capacity constraints. This comparison is conducted using real data in §6.2.3, and the results are summarized in Table EC.2.

Our findings show that it is crucial to both remove nonviable options and guarantee that fulfillment decisions satisfy implicit capacity limits, as the **ERT** policy consistently achieves the best performance across all cases. However, the relative importance of these two components depends on customer preferences. Specifically, when customers prefer fast fulfillment options (Cases 1, 2, 4, and 5), removing nonviable options has a greater impact on performance. This reinforces the central insight that demand management is particularly important in speed-sensitive environments. In contrast, when customers are more cost-sensitive (Cases 3 and 6), the seller has greater flexibility to optimize fulfillment assignments, and the value of threshold-based control becomes more pronounced.

Algorithm 4: Upper-envelope construction under nested logit

Input: Nested-logit parameters $(v_{n,k}, \gamma_n)_{n,k}$, revenues $(r_{n,k})_{n,k}$, and outside-option attraction v_0 .

Output: The set of all efficient option lists $\mathcal{E}$.

```

1 Sort the options within each nest in decreasing order of revenue
2 Compute the prefix sums  $A_n(i)$  and  $B_n(i)$  for all  $n$  and  $i$ 
3 Initialize an empty list of line records  $\mathcal{G}$ 
4 foreach  $\mathbf{i} \in \mathcal{I} = \prod_{n=1}^H \{0, \dots, K_n\}$  do
5   | Compute  $Q_{\mathbf{i}} := Q(S(\mathbf{i}))$  and  $R_{\mathbf{i}} := R(S(\mathbf{i}))$ 
6   | Append the record  $(Q_{\mathbf{i}}, R_{\mathbf{i}}, \{S(\mathbf{i})\})$  to  $\mathcal{G}$ 
7 Sort the records in  $\mathcal{G}$  by decreasing  $Q$  and, for ties, decreasing  $R$ 
   // Remove same-slope dominated lines and merge identical lines
8 Initialize an empty list of deduplicated records  $\mathcal{U}$ 
9 foreach  $(Q, R, \mathcal{L}) \in \mathcal{G}$  do
10  | if  $\mathcal{U} = \emptyset$  then
11  |   | Append  $(Q, R, \mathcal{L})$  to  $\mathcal{U}$ 
12  | else
13  |   | Let  $(\bar{Q}, \bar{R}, \bar{\mathcal{L}})$  be the last record in  $\mathcal{U}$ 
14  |   | if  $Q \neq \bar{Q}$  then
15  |   |   | Append  $(Q, R, \mathcal{L})$  to  $\mathcal{U}$ 
16  |   | else if  $R = \bar{R}$  then
17  |   |   | Replace the last record in  $\mathcal{U}$  by  $(\bar{Q}, \bar{R}, \bar{\mathcal{L}} \cup \mathcal{L})$ 
   // Upper-envelope scan
18 Initialize empty stacks  $\mathcal{H}$  and  $\Theta$ 
19 foreach  $(Q, R, \mathcal{L}) \in \mathcal{U}$  do
20  | while  $\mathcal{H} \neq \emptyset$  do
21  |   | Let  $(Q', R', \mathcal{L}')$  be the last record in  $\mathcal{H}$ , and  $\theta^* \leftarrow \frac{R' - R}{Q' - Q}$ 
22  |   | if  $\theta^* < \text{last}(\Theta)$  then
23  |   |   | Remove the last record from  $\mathcal{H}$  and from  $\Theta$ 
24  |   | else
25  |   |   | break
26  | if  $\mathcal{H} = \emptyset$  then
27  |   | Append  $(Q, R, \mathcal{L})$  to  $\mathcal{H}$  and append 0 to  $\Theta$ 
28  | else
29  |   | Append  $(Q, R, \mathcal{L})$  to  $\mathcal{H}$  and append  $\theta^*$  to  $\Theta$ 
30  $\mathcal{E} \leftarrow \bigcup_{(Q, R, \mathcal{L}) \in \mathcal{H}} \mathcal{L}$ 
31 Return  $\mathcal{E}$ 

```

Algorithm 5: Largest Marginal Revenue (LMR) Algorithm (Talluri and Van Ryzin 2004)**Input:** Set of fulfillment options $\mathcal{S}$ and the choice model $\pi_k^j(\cdot)$ for customer j .**Output:** The set of efficient option lists $\mathcal{E}_j$.

- 1 Define $R_j(S) = \sum_k r_{k,j} \pi_k^j(S)$ and $Q_j(S) = \sum_k \pi_k^j(S)$.
- 2 Initialize $S^e = \emptyset$ and $\mathcal{E}_j = \{\emptyset\}$.
- 3 Let set $\Omega = \{S \subseteq \mathcal{S} : R_j(S) \geq R_j(S^e), Q_j(S) \geq Q_j(S^e), S \notin \mathcal{E}_j\}$.
- 4 **while** Ω is not empty **do**
- 5 $S^e \leftarrow \arg \max_{S \in \Omega} \frac{R(S) - R(S^e)}{Q(S) - Q(S^e)}$, break a tie with smallest $Q(S)$.
- 6 Add S^e to $\mathcal{E}_j$, update set Ω .
- 7 **end**

K	3	4	5	6	7	8	9	10	12	14	16	18	20
MNL	4	4	5	6	5	5	7	6	7	9	9	8	8
MMNL	4	5	6	5	6	7	7	7	8	7	8	10	9
NL	4	4	5	5	5	7	6	6	8	9	9	8	9

Table EC.1 Maximum number of efficient option lists with various choice models.**Algorithm 6:** Randomized Rounding Policy**Input:** Customer arrival rate vector $\Lambda_{j,\tau}$, $\forall j \in [N], \tau \in [T]$, Fulfillment cost vector C_i , $\forall i \in [M]$, penalty vector b_j , $\forall j \in [N]$ **Output:** Fulfillment option display decision $S^{\text{r-HYP}} = \{S_{j,\tau}^{\text{HYP}}\}$ and fulfillment decision

$$u^{\text{HYP}} = \{u_{k,j,i,t}^{\text{HYP}}\}$$

- 1 Construct the set of all *efficient assortments* $\mathcal{E}_j$ for all $j \in [N]$.
- 2 Solve **DET** to obtain $\gamma_{j,\tau}^*(S)$ for all $S \in \mathcal{E}_j$ and $\bar{u}_{k,j,i,\tau}^*$.
- 3 **for** $\tau \in [T]$ **do**
- 4 At any time $t \in [0, 1]$, if there is a customer from location j , offer an assortment $S \in \mathcal{E}$ with probability $\gamma_{j,\tau}^*(S)$.
- 5 Let $u_{k,j,i,t}^{\text{r-HYP}} = h\left(\frac{y_{k,j,\tau}}{\bar{y}_{k,j,\tau}^*} \cdot \bar{u}_{k,j,i,\tau}^*\right)$.
- 6 **while** $\sum_i u_{k,j,i,t}^{\text{RR}} > y_{k,j,\tau}$ **do**
- 7 Choose $i' \in \arg \max_i C_i'(w^\top u_{i,\tau}^{\text{r-HYP}})$ and let $u_{k,j,i',t}^{\text{r-HYP}} = u_{k,j,i,t}^{\text{r-HYP}} - 1$.
- 8 **end**
- 9 **end**

Case	(v_0, v_1, v_2, v_3)	ERT	er-HYP	r-HYP
1	(1.00, 3.47, 10.48, 0.12)	2.6%	2.7%	4.0%
2	(1.00, 10.48, 3.47, 0.12)	2.7%	2.8%	3.9%
3	(1.00, 0.12, 3.47, 10.48)	1.5%	4.7%	4.9%
4	(3.00, 3.47, 10.48, 0.12)	2.6%	2.6%	4.1%
5	(3.00, 10.48, 3.47, 0.12)	2.8%	2.8%	3.9%
6	(3.00, 0.12, 3.47, 10.48)	1.3%	5.2%	5.2%

Table EC.2 Comparison of ERT and supplementary benchmark policies under various customer preferences.